

Product Document



User Guide

UG001039

AS7056/57

Evaluation Kit

User Guide

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1 Introduction

The AS7056/57 Evaluation Kit (EVK) allows for the evaluation of all the functions on the AS7056/57 Biosensors and tests them in various applications. The Evaluation Kit is complemented with a graphical user interface (GUI) and optimized firmware, which enables the user to perform measurements, change AS7056/57 register settings, and many more. All these are provided, to prepare and run the AS7056/57 Vital Signs Analog Front End for several healthcare applications like HRM, SpO₂, etc.

This GUI user guide provides an overview of the EVK hardware, installation of the software, description of the GUI sub-menu and their related registers, data logging, and firmware update.

1.1 Kit Content

Figure 1:
AS7056/57 Evaluation Kit



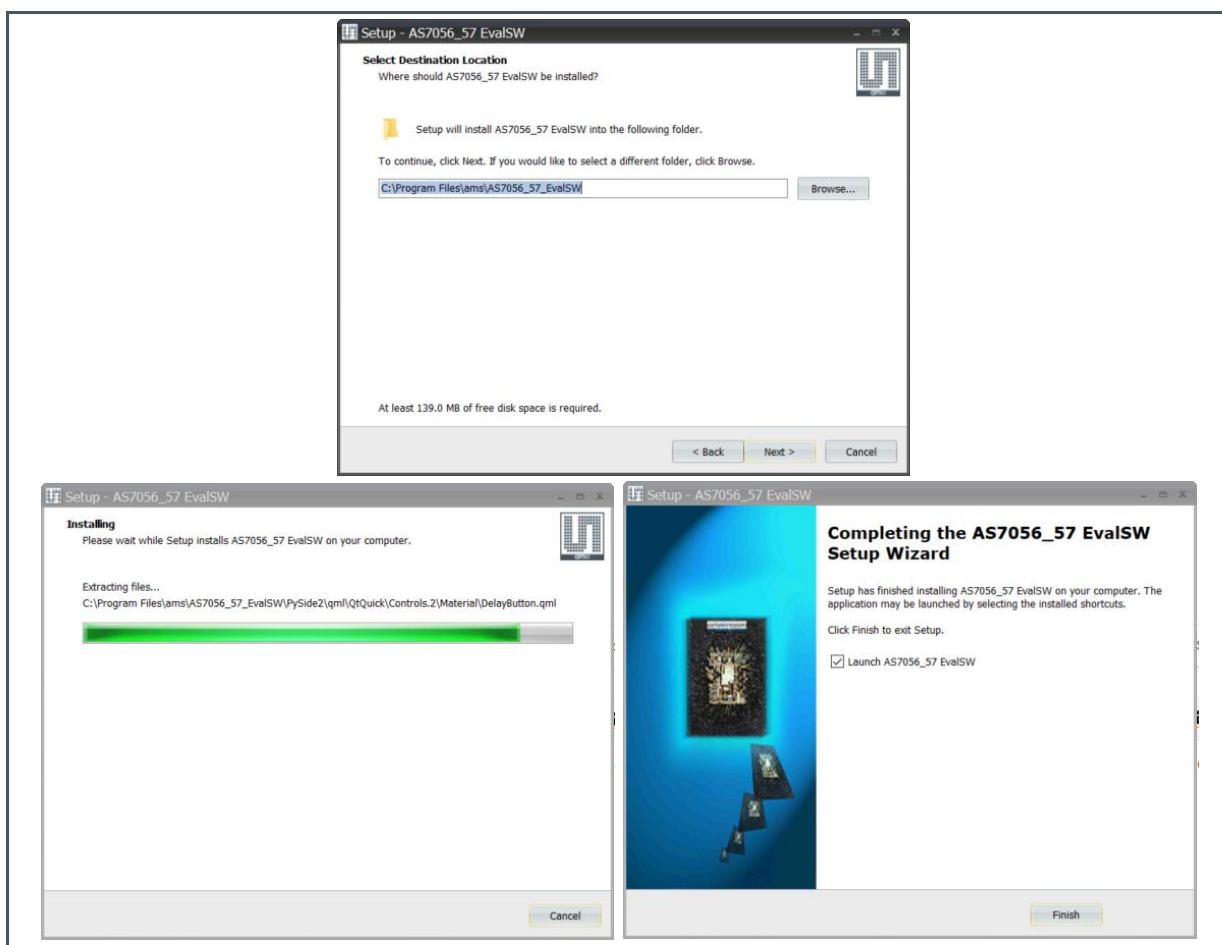
1.2 Ordering Information

Ordering Code	Description
990601215	AS7056/57 Evaluation Kit

2 Getting Started

The latest version of the EVK software is available for download (please contact the ams OSRAM support team for the download link). To install, start the installer executable and follow the instructions as shown in Figure 2 (going from left to right, top to bottom).

Figure 2:
AS7056/57 Vital Signs Sensor Installation



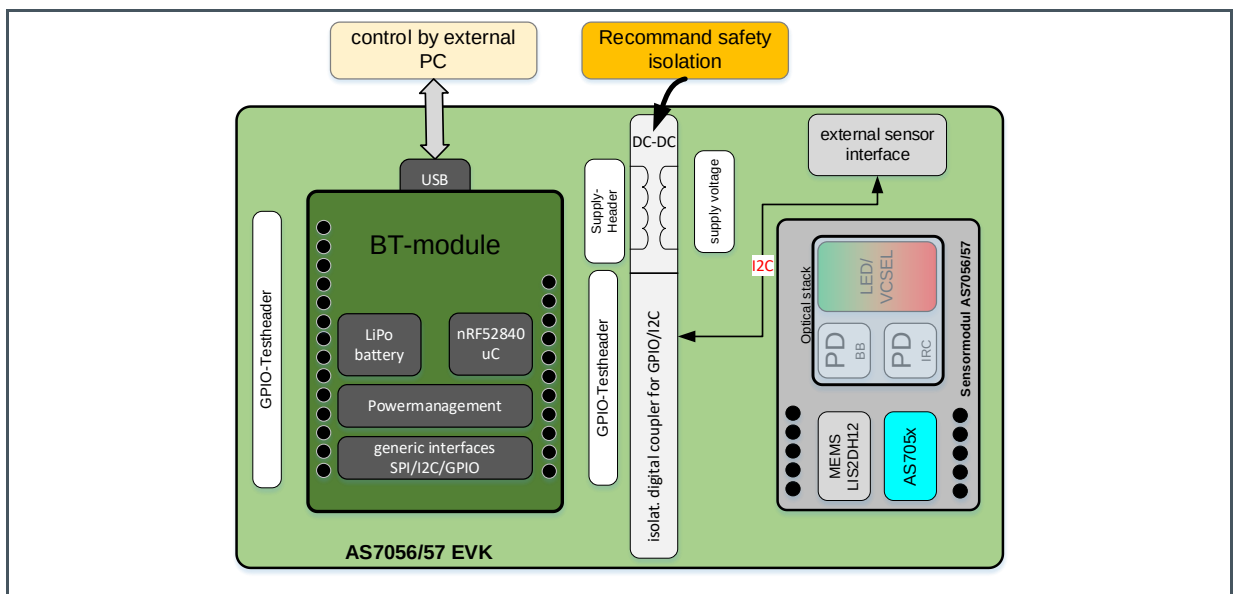
3 Hardware Description

For full flexibility, the Vital Sign optical PPG signal detection needs to be designed on a PCB with 2 blocks. The first block should work as a mainboard and the second block as a sensor board. The Mainboard is used for power management, connectivity, and signal post-processing. Additional to this the Sensor board should be exchangeable for different optical LED/VCSEL/PD set-ups. The evaluation kit (EVK) has to be developed in a way that at least one measurement concept can be used. In the first stage, the measurement in reflective mode has to be preferred for the finger.

3.1 Hardware Architecture

The Hardware architecture overview visualize the electrical main components of the EVK Platform.

Figure 3:
AS7056/57 EVK Architecture



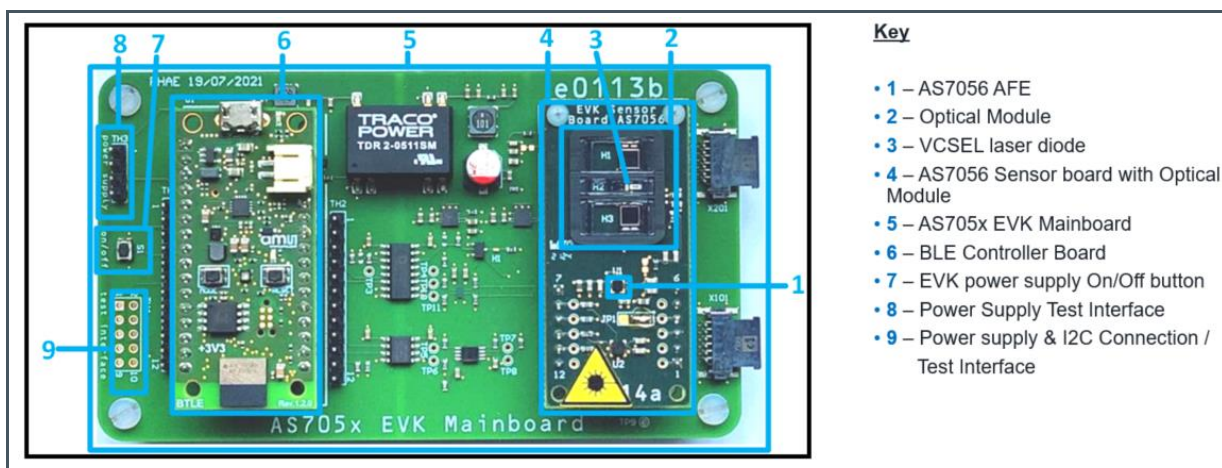
The hardware architecture of the AS7056/57 consists of:

- A BT-module with Insight "ISP1807" (3V3-Version).
- An electrical isolation is used.
- A level shifter for the I²C interface (3V3 to 1V8) is used.
- A "Sensorboard" has a complete unit of AS705x and optical parts.
- Short and directed tracks without additional connectors, to optimize the PCB design.
- Consideration of all the critical parts, to optimize the EMC behavior and improvement in SNR.
- A defined standard interface for the EVK controller board with power supply/I²C.

3.2 AS7056/57 Evaluation Kit (EVK)

The AS7056/57 EVK, as shown in Figure 4, is an evaluation platform for the AS7056/57 Biosensor Converting Unit. The board can be connected through Bluetooth or micro USB. The evaluation kit consists of three printed boards: the Bluetooth module with a micro USB connector, the AS705x mainboard, and the AS7056/57 sensor board. Figure 6 shows the assembled EVK with the mounted Bluetooth module and sensor board.

Figure 4:
AS7056 EVK Top View



The Mainboard, with the Bluetooth module, is used for power management, connectivity, and signal post-processing. The Sensor board is designed with the LED/VCSEL/PD module. The EVK is such that the measurement in reflective mode should be performed on the finger.

Figure 5:
AS7056 & AS7057 Sensor Boards

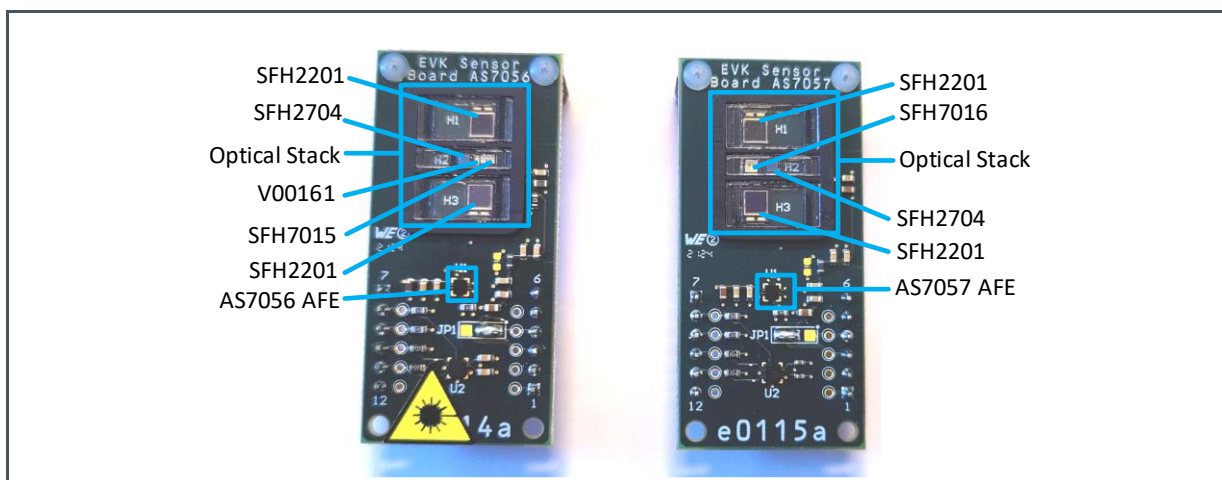
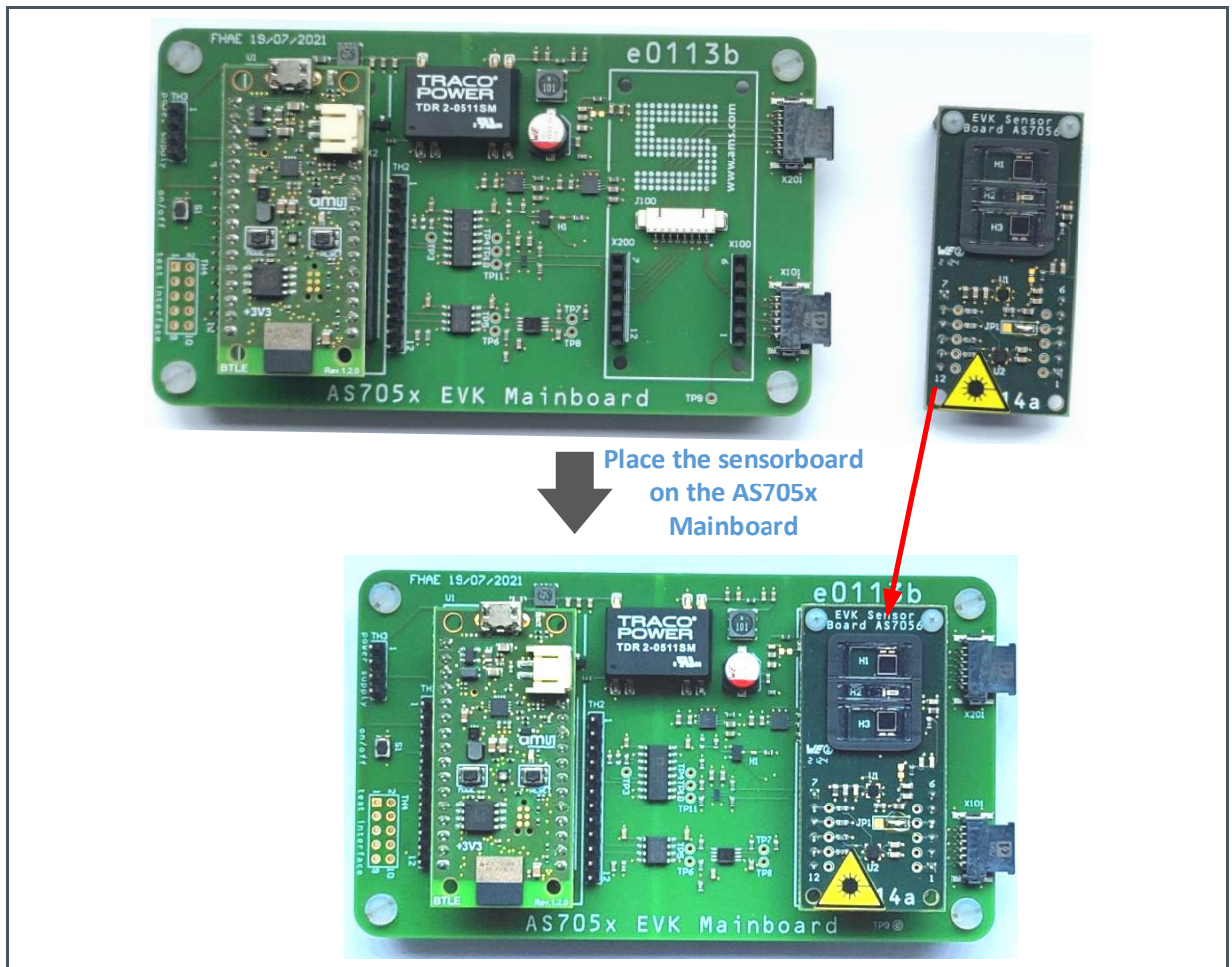


Figure 6:
Fitting the Sensor Board with the AS705x Mainboard



- The AS705x Mainboard supports the AS7056 & AS7057 Sensor boards.
- The Sensor board should be mounted very carefully, as shown in Figure 6.
- During the mounting or changing of the Sensor board, the AS705x Mainboard should be turned off.
- Always mount the optical adapter on the top of the optical module.

3.3 Power Supply

3.3.1 USB Connector

The source of power for the AS7056/57 Evaluation Kit is a USB connector on the Bluetooth module. To avoid electrical interferences on the board: an IEC 60601-1 compliant Traco Power DC/DC converter is assembled on the board, as well as isolator ICs, for all signals passing to the sensor board. Therefore, there is no galvanic connection between the sensor board and the power grid.

4 AS7056/57 Overview

The AS7056/57 is a Biosignal Analog Frontend (AFE) that enables the measurement of photoplethysmography (PPG), pulse transit time (PPT) signals, as well as proximity. PPG is the most used method for heart rate measurement (HRM). It is an optical technique that measures the heart rate by shining light on the skin, and sampling light, modulated by the blood vessels, which expand and contract as blood pulses through them. Algorithms that convert the PPG signals into digital HRM and SpO₂ support the AS7056/57. These can be further utilized to estimate other vital sign parameters, such as heart/pulse rate variability, blood pressure, etc. All the functions of the AS7056/57 are controlled via an I²C/SPI interface.

The AS7056/57 is optimized for biosensing products with the following strengths:

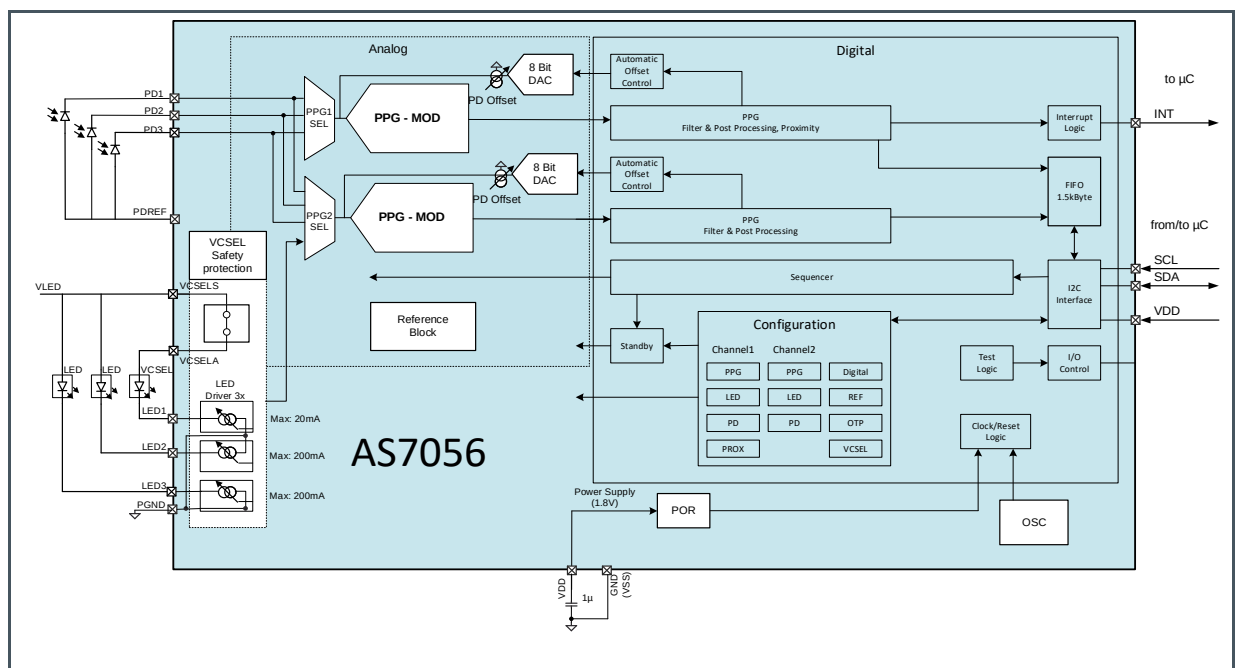
- Low noise, high dynamic range PPG acquisition.
- Simultaneous sampling of two PPG channels.
- Small die size.
- LED/VCSEL drivers.
- Proximity detection.
- Offset cancellation.
- Temperature and power supply monitoring.
- Built-in power control with FIFO, which can optimize power consumption for heart rate monitoring.

4.1 AS7056 Block Diagram

The AS7056 Biosignal Sensor AFE contains two main blocks as shown in Figure 7. An analog front-end for LED driving, signal acquisition, photodiode selection, and signal preconditioning. Moreover, a digital backend for signal filtering, balancing, and sampling. Furthermore, the digital block will handle the sensor configuration, control, and communication to the external MCU.

- PPG acquisition channels
 - 2 independent acquisition channels (2 ADC)
 - Up to 8+4 time interleaved samples per ADC channel
 - Each sample could use a different measurement/sampling mode
 - 3 photodiodes
 - 1 **VCSEL** driver for external VCSEL
 - 2 drivers for external LEDs
- Proximity
- Ultra short conversion time
- HRM/HRV, SpO₂, Blood pressure

Figure 7:
AS7056 Block Diagram

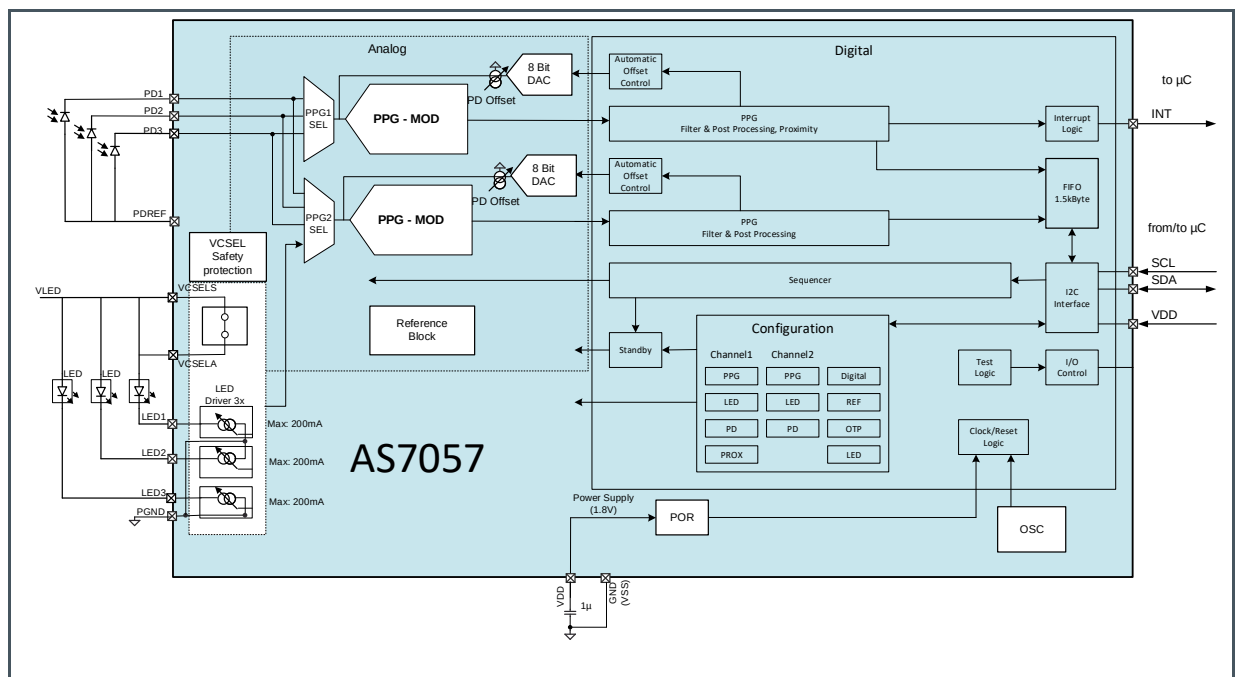


4.2 AS7057 Block Diagram

The AS7057 Biosignal Sensor AFE contains two main blocks as shown in Figure 8. An analog front-end for LED driving, signal acquisition, photodiode selection, and signal preconditioning. Moreover, a digital backend for signal filtering, balancing, and sampling. Furthermore, the digital block will handle the sensor configuration, control, and communication to the external MCU.

- PPG acquisition channels
 - 2 independent acquisition channels (2 ADC)
 - Up to 8+4 time interleaved samples per ADC channel
 - Each sample could use a different measurement/sampling mode
 - 3 photodiodes
 - 3 drivers for external LEDs
- Proximity
- Ultra short conversion time
- HRM/HRV, SpO₂, Blood pressure

Figure 8:
AS7057 Block Diagram



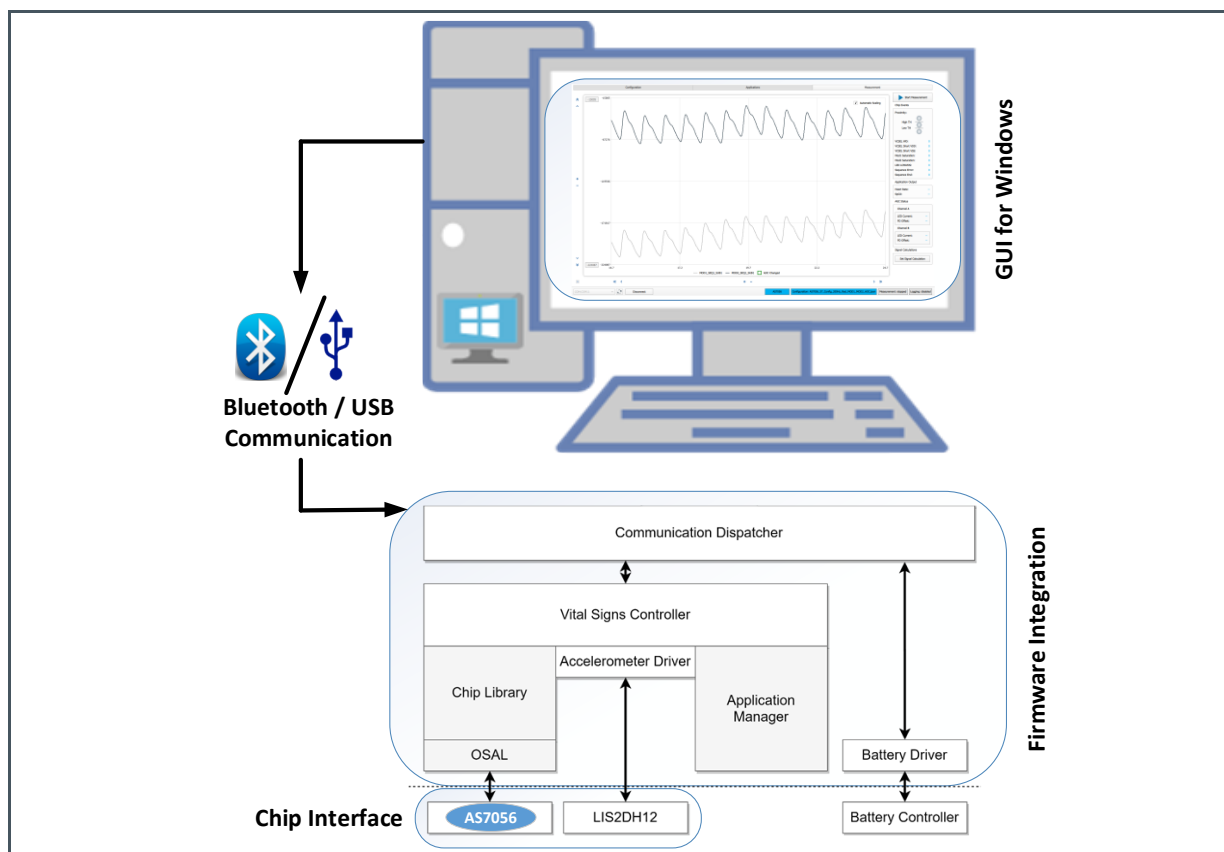
5 Software Description

The aim is to describe the essential workability of the AS7056/57 AFE. The software module provides software components, which enable a user to rapidly prototype their solutions.

5.1 Software Architecture

As the schematic show in Figure 9, there is a window PC needed to run the GUI application which is able to connect to the Microcontroller board / Firmware over USB or BT. The main components of the Firmware Integration are also shown.

Figure 9:
SW Architecture for Windows GUI



5.2 Graphical User Interface

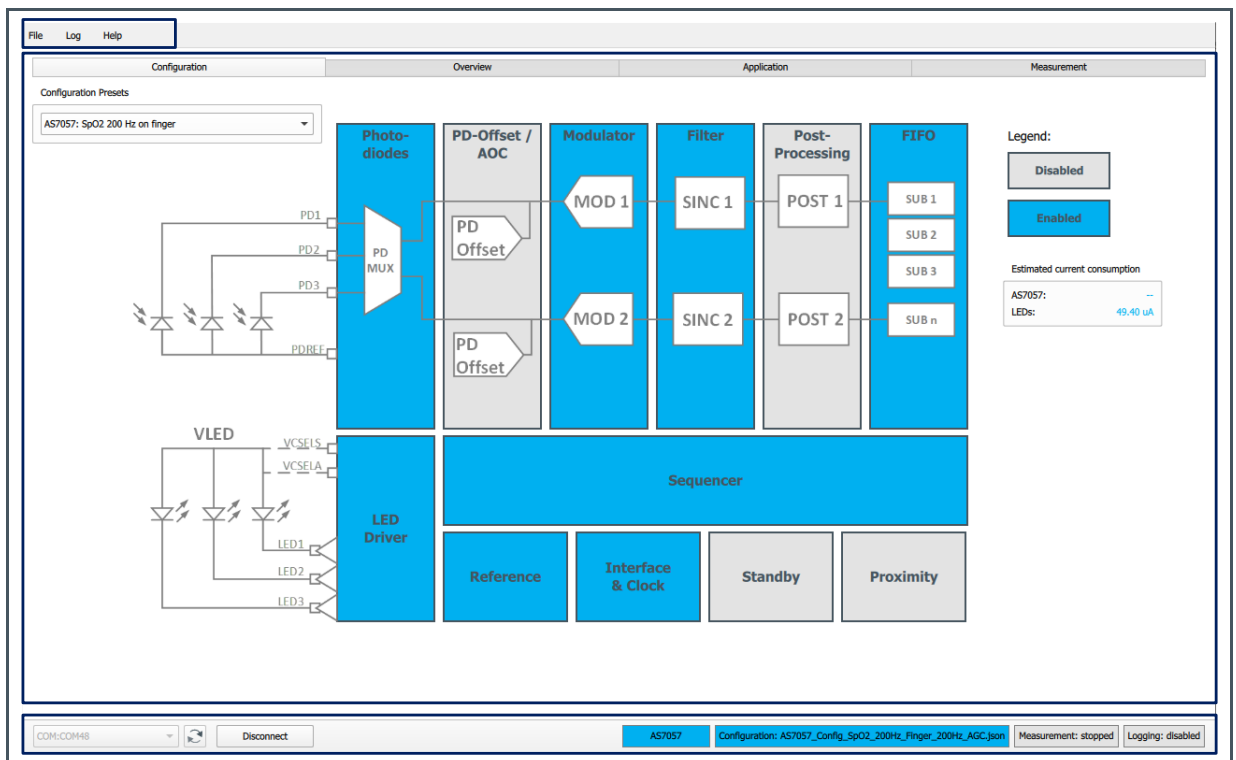
This section describes the Graphical User Interface (GUI) of the AS7056/57 Vital Sign Sensor application. The application is designed to be used with the AS7056/57 sensor series evaluation kits.

5.2.1 Overview

The GUI consists of three main segments highlighted in Figure 10.

- The section at the top contains options to the register map list to load and save a configuration file, save the log file with the timestamp, and update the firmware.
- The middle section contains the control functions with four tabs -> Configuration, Overview, Application, and Measurement.
- The status bar at the bottom contains the connection information, configuration file information, measurement condition, and logging status.

Figure 10:
Overview of the GUI



When the GUI starts, the status bar at the bottom of the window is grey. While the GUI establishes communication with the USB port of the EVK, the green LED on the Bluetooth module will turn on and the status bar at the bottom of the window will turn blue.

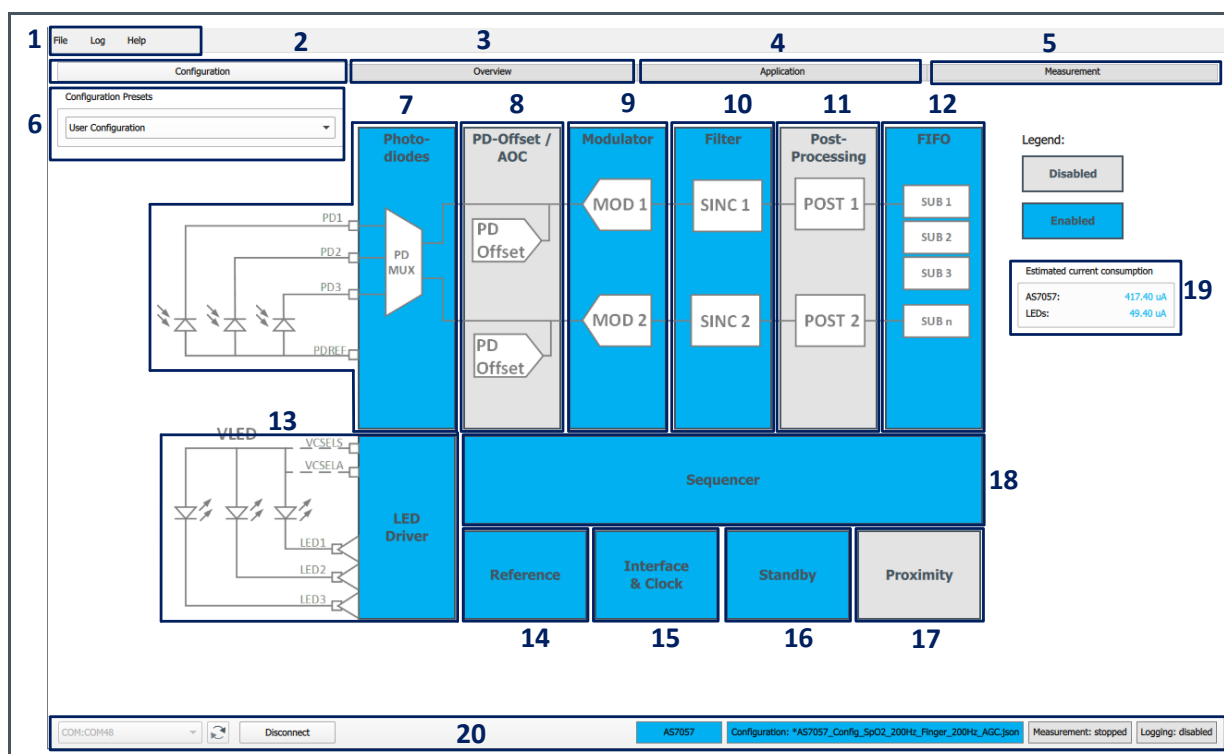
**Information**

- Valid for the following software version - AS7056/57 Vital Signs Sensor
 - Supported hardware - Evalboard AS7056/57 Rev.1.0
 - Download – Go <https://ams.com/download-center> to download the latest version.
-

6 GUI Block Description

This section describes each block of the AS7056/57 GUI and their relation with the register map if needed.

Figure 11:
AS7056/57 GUI Blocks



1 Menu bar to open the Register map, load and save the configuration file, save the log file with the timestamp, enable a moving average filter (if needed), and update the firmware.

2 Configuration Tab

3 Overview Tab

4 Application Tab

5 Measurement Tab

6 Configuration Presets

7 Photodiode Configuration

8 PD-Offset Configuration

9 Modulator Configuration

10 Filter Configuration

11 Post-Processing Configuration

12 FIFO Configuration

13 LED/VCSEL Driver Configuration

14 Reference Configuration

15 Interface & Clock Configuration

16 Standby Configuration

17 Proximity Configuration

18 Sample Rate & Measurement Configuration

19 Estimated Current Consumption

20 Status Bar

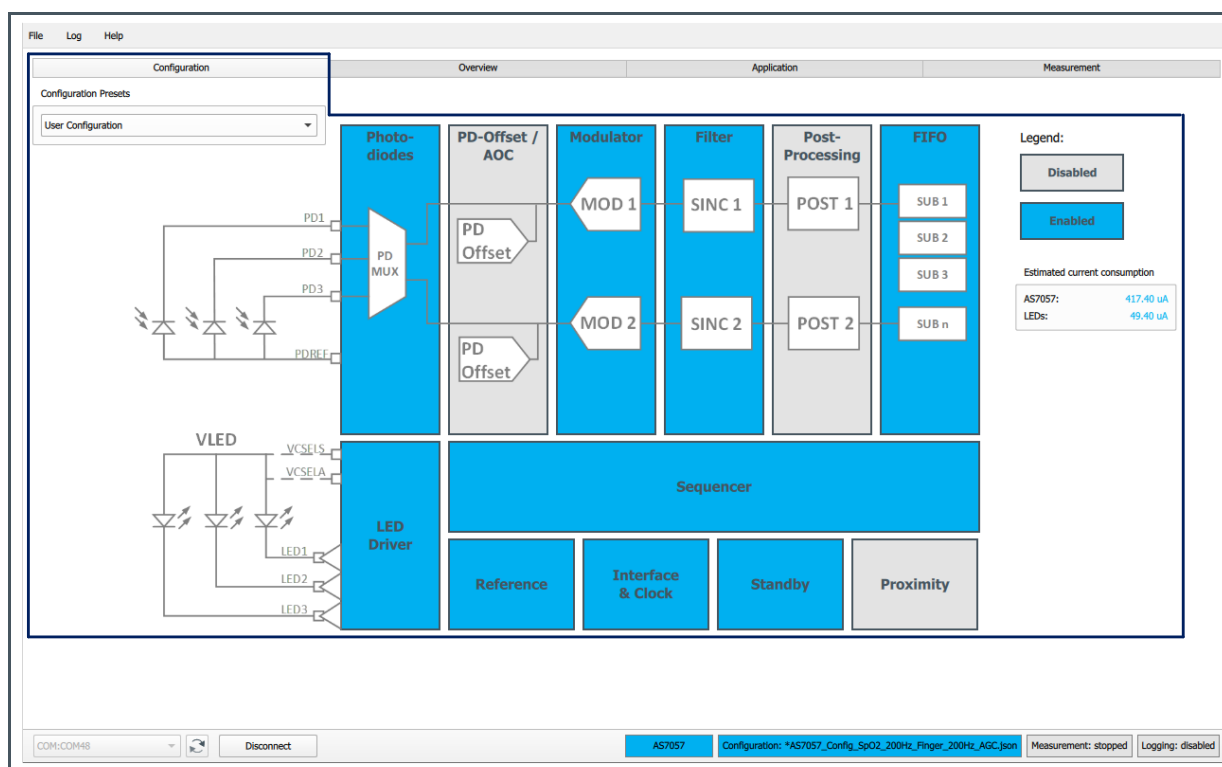
6.1 Configuration Tab

- After launching the EVK GUI, the “Configuration” tab is displayed.
- A few configurations presets are provided for the user to quickly start using the device. These can be chosen from the “Configuration Presets” Tab. Each file comes with a description, which gives the user a brief idea of which measurements could be done using these.

From the block diagram of the AS7056/57 (Figure 12), the sections in the blue show which sub-blocks are enabled, and those in grey shown which sub-blocks are disabled in a particular configuration file.

Each sub-block can be accessed by clicking on them.

Figure 12:
Configuration Tab Overview



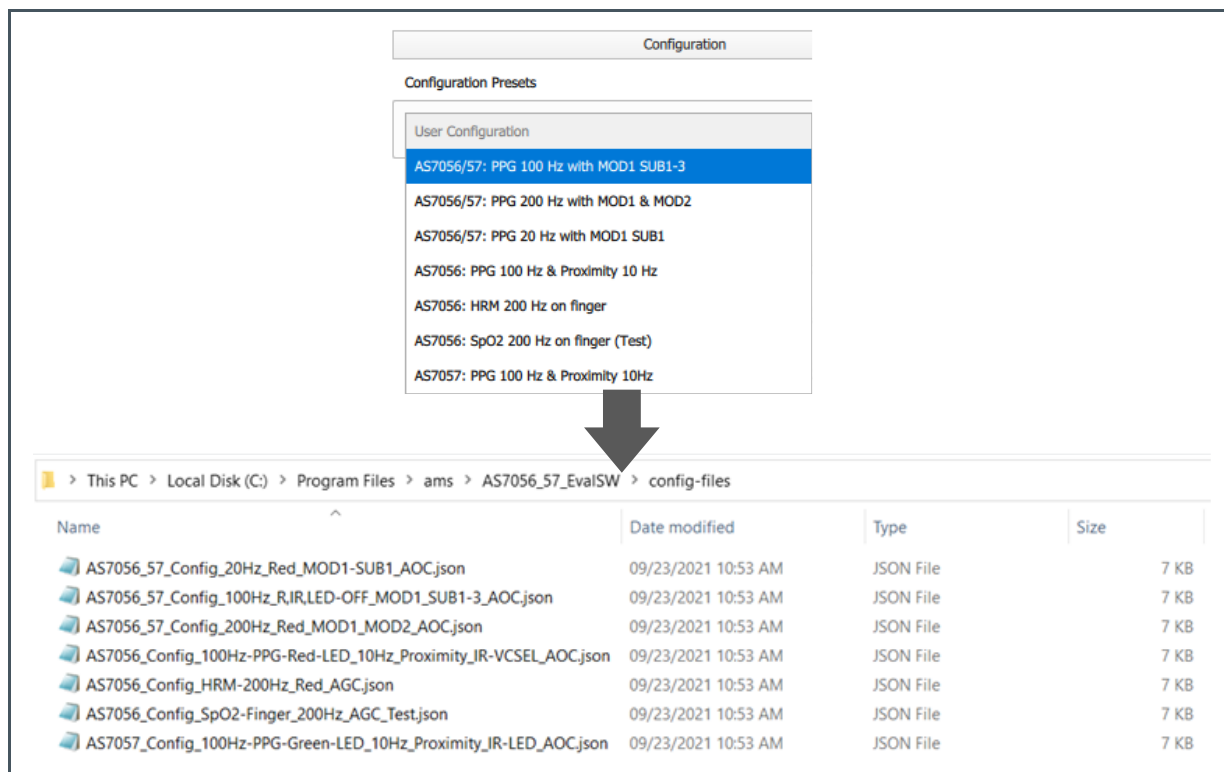
6.1.1 AS7056/57 Configuration Preset Settings

The AS7056/57 configuration preset settings are located in the configuration JSON files of the latest release.

The individual settings of each of the AS7056/57 blocks can be found in a JSON file. Moreover, new settings can be saved in the JSON file.

By default, the configuration presets are located in the folder “C:\Program Files\ams\AS7056_57_EvalSW\config-files”, after the AS7056/57 software is installed, or the user directory depends on the installation folder.

Figure 13:
Default Preset Configuration Files



Application-specific default preset configuration JSON files store almost all the register information for the AS7056/57 AFE that are easy to read, as shown in Figure 14.



Information

Please note that the configuration preset files can be changed with each new GUI version.

Figure 14:
AS7056/57 Configuration File

```

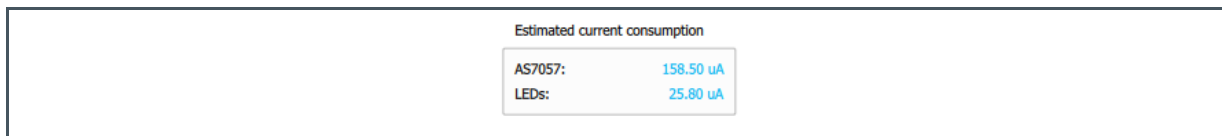
"gui_as7056-57": {
  "filter": {
    "moving_average": {
      "average_seq1": 0,
      "average_seq2": 0
    }
  },
  "signal_calculation": {
    "formula": ""
  }
},
"applications": {
  "enabled_apps": 1,
  "hrm": {
    "enable_prv": 0
  },
  "spo2": {
    "cal_coeff_a": 0,
    "cal_coeff_b": 0,
    "cal_coeff_c": 0,
    "dc_comp_red": 0,
    "dc_comp_ir": 0
  },
  "raw_data": {
    "enable_acc_data": 0
  }
},
"signal_routing_as7056-57": {
  "hrm": {
    "ppg": 1
  },
  "spo2": {
    "red": 1,
    "ir": 2,
    "ambient": 3
  }
},
"as7056-57": {
  "ctrl": {
    "CONTROL": "0x00",

```

- (1) All the register values displayed here are in hexadecimal. However, decimals are also supported (0-255), without quotation marks.

6.1.2 Estimated Current Consumption

Figure 15:
Estimated Current Consumption



The estimated current consumption of the AS7056/57 and the LEDs during measurement are displayed as shown in Figure 15. Based on the register configuration and LED configuration, the library calculates the estimated current of the chip and the LEDs. To estimate the current, the standby mode is activated to put the analog blocks to sleep after the end of the measurement.

6.1.3 Sequencer Configuration

To synchronize the LED currents, the integration time, and the ADC sampling time, a built-in sampling sequencer can be used. All settings could be changed in that sequencer configuration.

Figure 16:
Sequencer Configuration

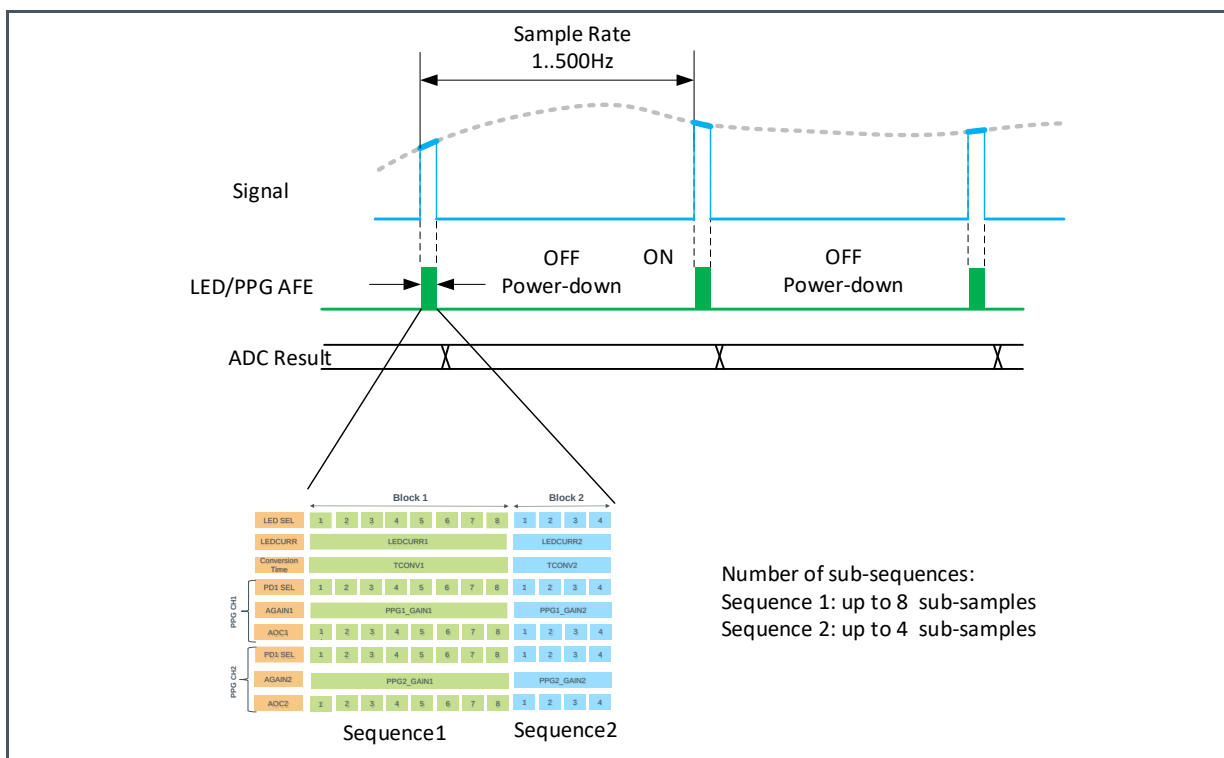


The sequencer generates the 8-bit-timings based on a 1 μ s clock. The PPG acquisition channels run in parallel and operate with different sampling frequencies. For this, the sampling frequency and the two frequency dividers for the partial sequences must be programmed.

Each data sample is constructed from a set of individual measurements, called sub-sequences. These sub-sequences are grouped in two sequences with separate sample rates, LED current, and PPG Gain control settings. The number of measurements can be configured from one to eight for

Sequence 1 and one to four for Sequence 2. Each sub-sequence has its settings for operation mode, photodiode-, LED- and AOC configurations.

Figure 17:
Sequencer



IRQ

A high active interrupt can be programmed for continuous data acquisition. With IRQ_ENABLE, the sources for generating an interrupt are enabled.

Figure 18:
IRQ Registers

0x3F	SEQ_LED_INIT			
ID	Bit Name	Default	Access	Description
7	IR_EN_PROX_ON	0	RW	Proximity rising edge interrupt enable
6	IRQ_EN_PROX_OFF	0	RW	Proximity falling edge interrupt enable
5	IRQ_EN_VCSEL	0	RW	VCSEL short to VDD/VSS interrupt enable
4	IRQ_EN_ASAT	0	RW	Analog saturation interrupt enable
3	IRQ_EN_LED_LOWVDS	0	RW	LED low voltage interrupt enable
2	IRQ_EN_FIFOOVERFLOW	0	RW	FIFO overflow interrupt enable

0x3F SEQ_LED_INIT				
ID	Bit Name	Default	Access	Description
1	IRQ_EN_FIFOTHRESHOLD	0	RW	FIFO threshold interrupt enable
0	IRQ_EN_SEQUENCER	0	RW	Sequencer interrupt enable

LED Settings

Figure 19:
Time After Switching on the LEDs

0x45 SEQ_LED_INIT				
ID	Bit Name	Default	Access	Description
7:0	SEQ_LED_INIT	0	RW	Setup time after turn on LEDs

➤ LED setup time = $N * 1\mu s$ | where $N = 0 \dots 255$

Sequencer

Sequencer 1 and sequencer 2 use the same parameters in this block.

SEQ_MODCONF:

- MOD_CLK (Modulator clock frequency) – The frequencies 10 MHz, 5 MHz, 2.5 MHz, and 1.25MHz can be set. With a reduction of the Modulator clock frequency, the converting time will be increased, and the signal-to-noise ratio will improve. Also, the power consumption of the AS7056/57 will increase.
- MOD_RESET_DELAY (Reset ADC) - Reset time for all modulators.
Internal reset process time = $N * MOD_CLK$

Figure 20:
SEQ_MODCONF Register

0x42 SEQ_MODCONF				
ID	Bit Name	Default	Access	Description
4:2	MOD_RESET_DELAY	0	RW	Internal reset process time = $N * MOD_CLK$ Number 0 - 4 Number 1 - 8 Number 2 - 16 Number 3 - 32 Number 4 - 64 Number 5 - 128 Number > 5 - 256

0x42		SEQ_MODCONF		
ID	Bit Name	Default	Access	Description
1:0	LED_SEQ1_SUB2	0	RW	Modulator clock frequency Number 0 – 10 MHz Number 1 – 5 MHz Number 2 – 2.5 MHz Number 3 – 1.25 MHz

SEQ_SAMPLE:

The number of samples for the activated channel.

Figure 21:
SEQ_SAMPLE Register

0x40		SEQ_SAMPLE		
ID	Bit Name	Default	Access	Description
7:0	SEQ_SAMPLE	0	RW	Number of samples for the activated channel

If the seq_sample = 0, the sequencer runs continuously.

SEQ_SUB_WAIT (Address 0x41):

The distance between the subsamples.

Figure 22:
SEQ_SUB_WAIT Register

0x41		SEQ_SUB_WAIT		
ID	Bit Name	Default	Access	Description
7:0	SUB_WAIT	0	RW	Setup time after turn on LEDs

➤ Wait time = N * 1μs

SEQ_SAR_WAIT (Address 0x44):

The distance between the SAR measurements.

Figure 23:
SEQ_SAR_WAIT Register

0x44		SEQ_SAR_WAIT		
ID	Bit Name	Default	Access	Description
7:0	SAR_WAIT	0	RW	Distance between the SAR measurements

➤ $\text{Time} = N * 1\mu\text{s}$

SEQ_FREQL & SEQ_FREQH:

Configuration of the sequence sample period. The Divider for calculating the sample rate (Hz) for the sequence configuration is shown below:

- Sample Rate = $1 / (\text{Divider} * 31.25\mu\text{s})$ with divider = $[1 \dots 2^{16}]$
- The divider value is configured as a 16-bit value $[0 \dots (2^{16}-1)]$.

Figure 24:
SEQ_FREQL Register

0x46		SEQ_FREQL		
ID	Bit Name	Default	Access	Description
7:0	SEQ_FREQ [7:0]	0	RW	Configuration of the sequence sample period

Figure 25:
SEQ_FREQH Register

0x47		SEQ_FREQH		
ID	Bit Name	Default	Access	Description
7:0	SEQ_FREQ [15:8]	0	RW	Configuration of the sequence sample period

There are already some pre-defined base sample rates [Hz] for the sequence configuration in the Base Sample Rate Preset [Hz] option.

After selecting the Base Sample Rate Preset [Hz] or selecting the Base Sample Rate Divider, the exact sample rate can be seen on the Base Sample Rate [Hz] option.

Sequencer 1 & Sequencer 2

The partial sequences, Sequence 1 and Sequence 2, of a sample, can be operated with different frequencies. For this, the sampling frequency, and the two frequency dividers for the partial sequences, must be programmed.

Each sample is made up of two sequences with several sub-samples: sequence 1 with a maximum of eight and sequence 2 with a maximum of four sub-samples. Several parameters for the sequences and modulators, as well as sub-samples, must be set within a sample.

SEQ_CONFIG (Address 0x43):

Figure 26:
SEQ_CONFIG Register

0x43		SEQ_CONFIG		
ID	Bit Name	Default	Access	Description
7	SEQ2_EN	0	RW	Enable the Sequence 2
5:4	SEQ2_SUB_SAMPLE	0	RW	Number of sub samples for Sequence 2 (1 to 4)
3	SEQ1_EN	0	RW	Enable the Sequence 1
2:0	SEQ1_SUB_SAMPLE	0	RW	Number of sub samples for Sequence 1 (1 to 8)

SEQ_FREQDIVL & SEQ1_FREQDIVH:

The configuration of sequence 1 sample period. The Divider for calculating the sample rate (Hz) for the sequence 1 configuration is shown below:

- Sample Rate = Base Sample Rate / Divider with (divider = [1... 2^{16}])
- The divider value is configured as a 16-bit value [0... ($2^{16}-1$)]

Figure 27:
SEQ1_FREQDIVL Register

0x48		SEQ1_FREQDIVL		
ID	Bit Name	Default	Access	Description
7:0	SEQ1_FREQDIV [7:0]	0	RW	Configuration of the sequence1 sample period

Figure 28:
SEQ1_FREQDIVH Register

0x49		SEQ1_FREQDIVH		
ID	Bit Name	Default	Access	Description
7:0	SEQ1_FREQDIV [15:8]	0	RW	Configuration of the sequence1 sample period

SEQ2_FREQDIVL & SEQ2_FREQDIVH:

The configuration of sequence 2 sample period. The Divider for calculating the sample rate (Hz) for the sequence 2 configuration is shown below:

- Sample Rate = Base Sample Rate / Divider with (divider = [1... 2¹⁶])
- The divider value is configured as a 16-bit value [0... (2¹⁶-1)]

Figure 29:
SEQ2_FREQDIVL Register

0x4A		SEQ2_FREQDIVL		
ID	Bit Name	Default	Access	Description
7:0	SEQ2_FREQDIV [7:0]	0	RW	Configuration of the sequence2 sample period

Figure 30:
SEQ2_FREQDIVH Register

0x4B		SEQ2_FREQDIVH		
ID	Bit Name	Default	Access	Description
7:0	SEQ2_FREQDIV [15:8]	0	RW	Configuration of the sequence2 sample period

MOD1_SEQ1_SUB_EN & MOD2_SEQ1_SUB_EN:

This enables sub-samples for Sequence 1 in Modulator 1 & Modulator 2.

Figure 31:
MOD1_SEQ1_SUB_EN Register

0x4C		MOD1_SEQ1_SUB_EN		
ID	Bit Name	Default	Access	Description
7:0	MOD1_SEQ1_SUB_EN [7:0]	0	RW	Enable Sub samples for Sequence 1 in Modulator 1

Figure 32:
MOD2_SEQ1_SUB_EN Register

0x4E		MOD2_SEQ1_SUB_EN		
ID	Bit Name	Default	Access	Description
7:0	MOD2_SEQ1_SUB_EN [7:0]	0	RW	Enable Sub samples for Sequence 1 in Modulator 2

MOD1_SEQ2_SUB_EN (Address 0x4D) & MOD2_SEQ2_SUB_EN (Address 0x4F):

This enables sub-samples for Sequence 2 in Modulator 1 & Modulator 2.

Figure 33:
MOD1_SEQ2_SUB_EN Register

0x4D		MOD1_SEQ2_SUB_EN		
ID	Bit Name	Default	Access	Description
7:0	MOD1_SEQ2_SUB_EN [7:0]	0	RW	Enable Sub samples for Sequence 2 in Modulator 1

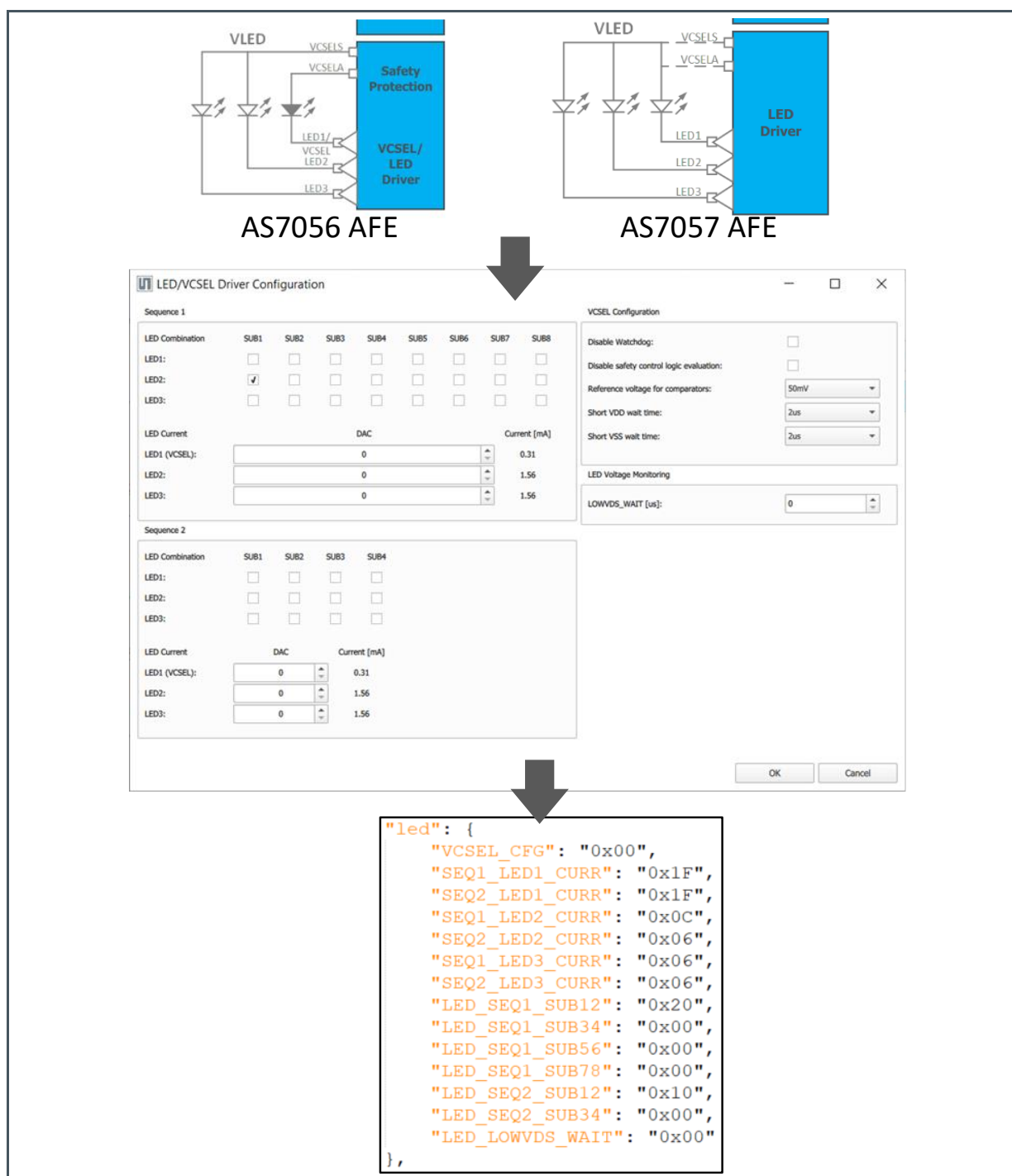
Figure 34:
MOD2_SEQ2_SUB_EN Register

0x4F		MOD2_SEQ2_SUB_EN		
ID	Bit Name	Default	Access	Description
7:0	MOD2_SEQ2_SUB_EN [7:0]	0	RW	Enable Sub samples for Sequence 2 in Modulator 2

6.1.4 LED Driver Configuration

The AS7056 contains 2 LED plus 1 VCSEL/LED drivers and the AS7057 3 LED drivers, which can be configured in the LED driver configuration.

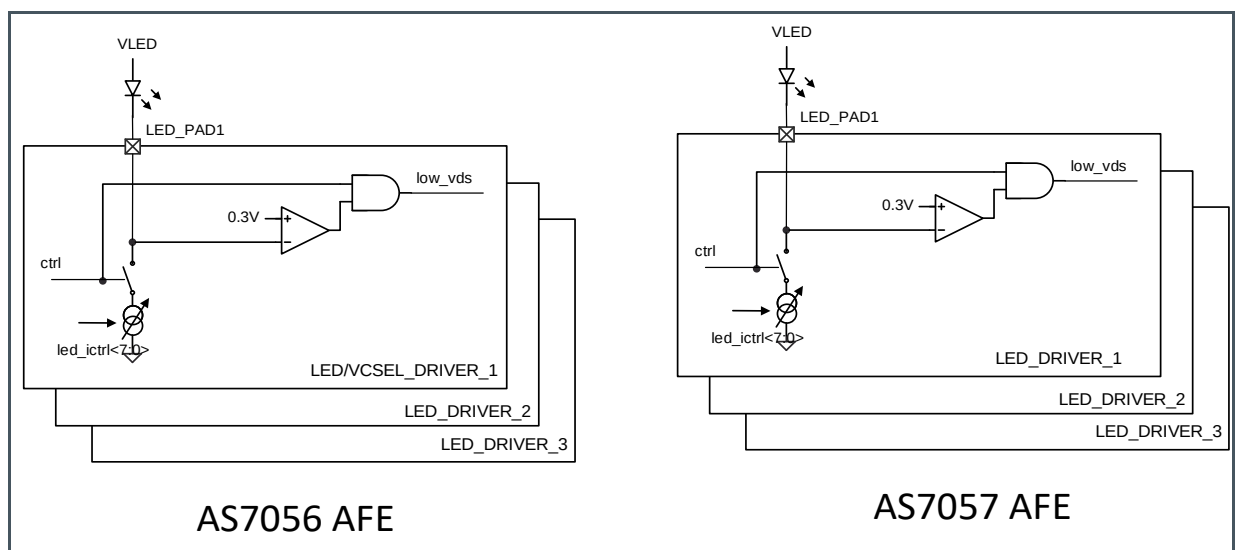
Figure 35:
LED Driver Configuration



The AS7056/57 contains three LED drivers which are implemented as a regulated current sink. The VCSEL (for the AS7056)/LED (for the AS7057) driver 1 can be configured for a maximum current of 20 mA (for the AS7056)/200 mA (for the AS7057). The other two drivers have a maximum current of 200 mA. The 200 mA LED drivers can be adjusted in 128 steps, and the 20 mA VCSEL driver in 64 steps.

The driver contains a low output voltage detection. If the LED is on and the voltage at the LEDn pad is below the DROP voltage (0.3 V), then the output low is set, and it is assumed that the expected output current is not reached. In the digital part, the output signal is synchronized with the current status of the LED on signals to avoid false detections.

Figure 36:
LED/VCSEL Drivers



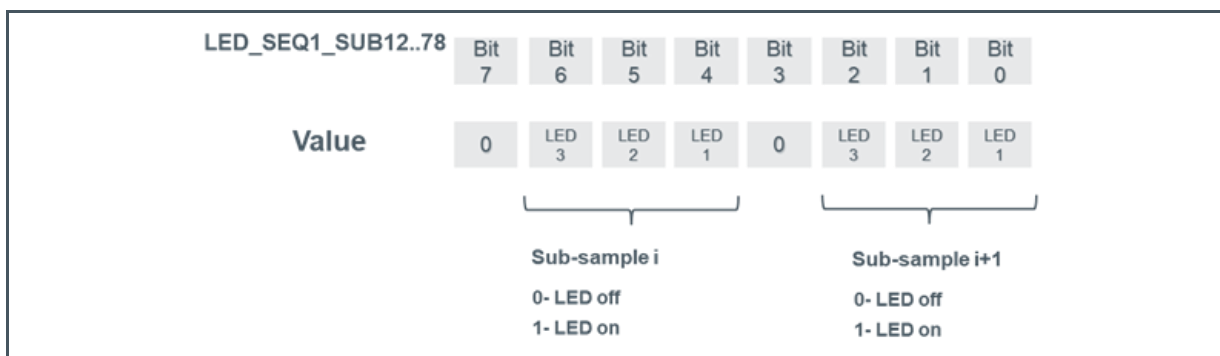
Two register sets can be used to configure the LEDs selection in the subsamples of measurement sequences and the current through them.

Sequence 1

LED Selection:

Four registers are used to select which LED, out of the available three, should be connected to which subsample and subsequence. Each LED could be connected to any subsample inside sequence 1.

Figure 37:
LED/VCSEL Selection for Sequence 2



LED_SEQ1_SUB (12/34/56/78): Select LED1 (for the AS7056)/VESEL1 (for the AS7057), LED2, and LED3 for subsample (1/3/5/7) and subsample (2/4/6/8) in Sequence 1. Bit 6:4 corresponds to subsample (1/3/5/7) and bit 2:0 to the subsample (2/4/6/8). Bit 7 and 3 remain unused and set to zero.

Figure 38:
LED_SEQ1_SUB12 Register

0x2F	LED_SEQ1_SUB12			
ID	Bit Name	Default	Access	Description
6:4	LED_SEQ1_SUB1	0	RW	Select LED for SubSample1 in Sequence1. Bit 0 – Off Bit 1 – On
2:0	LED_SEQ1_SUB2	0	RW	Select LED for SubSample2 in Sequence1. Bit 0 – Off Bit 1 – On

Figure 39:
LED_SEQ1_SUB34 Register

0x30	LED_SEQ1_SUB34			
ID	Bit Name	Default	Access	Description
6:4	LED_SEQ1_SUB3	0	RW	Select LED for SubSample3 in Sequence1. Bit 0 – Off Bit 1 – On
2:0	LED_SEQ1_SUB4	0	RW	Select LED for SubSample4 in Sequence1. Bit 0 – Off Bit 1 – On

Figure 40:
LED_SEQ1_SUB56 Register

0x31		LED_SEQ1_SUB56		
ID	Bit Name	Default	Access	Description
6:4	LED_SEQ1_SUB5	0	RW	Select LED for SubSample7 in Sequence1. Bit 0 – Off Bit 1 – On
2:0	LED_SEQ1_SUB6	0	RW	Select LED for SubSample8 in Sequence1. Bit 0 – Off Bit 1 – On

Figure 41:
LED_SEQ1_SUB78 Register

0x32		LED_SEQ1_SUB78		
ID	Bit Name	Default	Access	Description
6:4	LED_SEQ1_SUB7	0	RW	Select LED for SubSample1 in Sequence1. Bit 0 – Off Bit 1 – On
2:0	LED_SEQ1_SUB8	0	RW	Select LED for SubSample2 in Sequence1. Bit 0 – Off Bit 1 – On

LED/VCSEL Current:

The LED/VCSEL current of each LED can also be programmed. Note that the LED current can be programmed for each sequence but not for each subsample. The output current for the different LEDs and the VCSEL (for the AS7056) can be given as:

Equation 1:

$$I_{OUT} = SEQ_X - LED_Y - CURR|_{dec} \times \frac{I_{OUT_MAX}}{64} \Big|_{Y=1}$$

$$I_{OUT} = SEQ_X - LED_Y - CURR|_{dec} \times \frac{I_{OUT_MAX}}{128} \Big|_{Y=2,3}$$

In the above equation, I_{OUT_MAX} is 200 mA for LED 1, LED2, and LED3. For AS7056, the I_{OUT_MAX} is given as 20 mA for LED1/VCSEL1.

For AS7056: The current for VCSEL1 in Sequence 1 is shown in the table below.

Figure 42:
SEQ1_LED1_CURR Register (AS7056)

0x29		SEQ1_LED1_CURR		
ID	Bit Name	Default	Access	Description
6:0	SEQ1_LED1_CURR	0	RW	VCSEL minimum current is 0.31 mA when SEQ1_LED1_CURR = 0 VCSEL maximum current is 20 mA when SEQ1_LED1_CURR = 63

For AS7057: The current for LED1 in Sequence 1 is shown in the table below.

Figure 43:
SEQ1_LED1_CURR Register (AS7057)

0x29		SEQ1_LED1_CURR		
ID	Bit Name	Default	Access	Description
6:0	SEQ1_LED1_CURR	0	RW	LED1 minimum current is 1.56 mA when SEQ1_LED1_CURR = 0 LED1 maximum current is 200 mA when SEQ1_LED1_CURR = 127

For AS7056/57: The current for LED2 in Sequence 1 is shown below.

Figure 44:
SEQ1_LED2_CURR Register (AS7056/57)

0x2B		SEQ1_LED2_CURR		
ID	Bit Name	Default	Access	Description
6:0	SEQ1_LED2_CURR	0	RW	LED2 minimum current is 1.56 mA when SEQ1_LED2_CURR = 0 LED2 maximum current is 200 mA when SEQ1_LED2_CURR = 127

For AS7056/57: The current for LED3 in Sequence 1 is shown below.

Figure 45:
SEQ1_LED3_CURR Register (AS7056/57)

0x2D		SEQ1_LED3_CURR		
ID	Bit Name	Default	Access	Description
6:0	SEQ1_LED3_CURR	0	RW	LED3 minimum current is 1.56 mA when SEQ1_LED3_CURR = 0

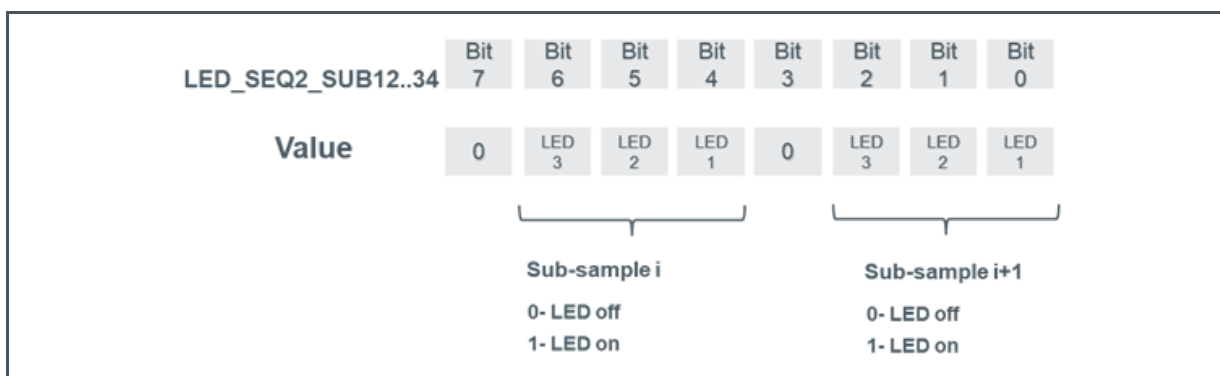
0x2D	SEQ1_LED3_CURR			
ID	Bit Name	Default	Access	Description
				LED3 maximum current is 200 mA when SEQ1_LED3_CURR = 127

Sequence 2

LED Selection:

Two registers are used to select which LED, out of the available three, should be connected to which subsample and subsequence. Each LED could be connected to any subsample inside sequence 2.

Figure 46:
LED/VCSEL Selection for Sequence 2



LED_SEQ2_SUB(12/34): Select LED1 (for the AS7056)/VESEL1 (for the AS7057), LED2, and LED3 for subsample (1/3), and subsample (2/4) in Sequence 1. Bit 6:4 corresponds to subsample (1/3) and bit 2:0 to the subsample (2/4). Bit 7 and 3 remain unused and set to zero.

Figure 47:
LED_SEQ2_SUB12 Register

0x33	LED_SEQ2_SUB12			
ID	Bit Name	Default	Access	Description
6:4	LED_SEQ2_SUB1	0	RW	Select LED for SubSample1 in Sequence2. Bit 0 – Off Bit 1 – On
2:0	LED_SEQ2_SUB2	0	RW	Select LED for SubSample2 in Sequence2. Bit 0 – Off Bit 1 – On

Figure 48:
LED_SEQ1_SUB34 Register

0x34	LED_SEQ1_SUB34			
ID	Bit Name	Default	Access	Description
6:4	LED_SEQ2_SUB3	0	RW	Select LED for SubSample3 in Sequence2. Bit 0 – Off Bit 1 – On
2:0	LED_SEQ2_SUB4	0	RW	Select LED for SubSample4 in Sequence2. Bit 0 – Off Bit 1 – On

LED/VCSEL Current:

The LED/VCSEL current of each LED can also be programmed. Note that the LED current can be programmed for each sequence but not for each subsample. The output current for the different LEDs and the VCSEL (for AS7056) can be given as:

Equation 2:

$$I_{OUT} = SEQ_X - LED_Y - CURR|_{dec} \times \frac{I_{OUT_MAX}}{64} \Big|_{Y=1}$$

$$I_{OUT} = SEQ_X - LED_Y - CURR|_{dec} \times \frac{I_{OUT_MAX}}{128} \Big|_{Y=2,3}$$

In the above equation, I_{OUT_MAX} is 200 mA for LED 1, LED2, and LED3. For AS7056, the I_{OUT_MAX} is given as 20 mA for LED1/VCSEL1.

For AS7056: The current for VCSEL1 in Sequence 2 is shown in the table below.

Figure 49:
SEQ2_LED1_CURR Register (AS7056)

0x2A	SEQ2_LED1_CURR			
ID	Bit Name	Default	Access	Description
6:0	SEQ2_LED1_CURR	0	RW	VCSEL minimum current is 0.31 mA when SEQ2_LED1_CURR = 0 VCSEL maximum current is 20 mA when SEQ21_LED1_CURR = 63

For AS7057: The current for LED1 in Sequence 2 is given in the table below.

Figure 50:
SEQ2_LED1_CURR Register (AS7057)

0x2A	SEQ2_LED1_CURR			
ID	Bit Name	Default	Access	Description
6:0	SEQ2_LED1_CURR	0	RW	LED1 minimum current is 1.56 mA when SEQ2_LED1_CURR = 0 LED1 maximum current is 200 mA when SEQ2_LED1_CURR = 127

For AS7056/57: The current for LED2 in Sequence 2 is shown in the table below.

Figure 51:
SEQ2_LED2_CURR Register (AS7056/57)

0x2C	SEQ2_LED2_CURR			
ID	Bit Name	Default	Access	Description
6:0	SEQ2_LED2_CURR	0	RW	LED2 minimum current is 1.56 mA when SEQ2_LED2_CURR = 0 LED2 maximum current is 200 mA when SEQ2_LED2_CURR = 127

For AS7056/57: The current for LED3 in Sequence 2 is shown in the table below.

Figure 52:
SEQ2_LED3_CURR Register (AS7056/57)

0x2E	SEQ2_LED3_CURR			
ID	Bit Name	Default	Access	Description
6:0	SEQ2_LED3_CURR	0	RW	LED3 minimum current is 1.56 mA when SEQ2_LED3_CURR = 0 LED3 maximum current is 200 mA when SEQ2_LED3_CURR = 127

VCSEL Configuration for Safety Protection

The VCSEL safety protection provides the functional monitoring of the VCSEL/LED1 driver. The VCSEL safety protection controller generates the control signals for the VCSEL short circuit. The watchdog automatically starts, if the VCSEL/LED1 switches on. If a watchdog event occurs, the VCSEL/LED1 and power switch are turned off.

Figure 53:
SEQ2_LED1_CURR Register

0x28		SEQ2_LED1_CURR		
ID	Bit Name	Default	Access	Description
7	VCSEL_WD_DISABLE	0	RW	Disable VCSEL/LED1 watchdog.
6	VCSEL_SAFETY_DISABLE	0	RW	Disable safety control logic evaluation of the short to VSS/VDD signals.
5:4	VCSEL_VRSEL	0	RW	If the voltage at VCSEL/LED1 pin is below some level, a short circuit is detected. 50 mV, 100 mV, 150 mV & 200 mV can be chosen as a reference voltage for comparators.
3:2	VCSEL_SHORT_VDD_WAIT	0	RW	The time between switching on the short circuit detection and the valid result. 2 μ s, 4 μ s, 8 μ s, and 12 μ s can be chosen as a wait time.
1:0	VCSEL_SHORT_VSS_WAIT	0	RW	The time between the switching on short circuit detection and the valid result. 2 μ s, 4 μ s, 8 μ s, and 12 μ s can be chosen as a wait time.

LED Voltage Monitoring

LOWVDS_WAIT: The time between switching on an LED and the start of voltage monitoring. All LEDs use the same time.

➤ Time = lowvds_wait * 1 μ s | where lowvds_wait = 0...255

Figure 54:
LED_LOWVDS_WAIT Register

0x35		LED_LOWVDS_WAIT		
ID	Bit Name	Default	Access	Description
7:0	LOWVDS_WAIT	0	RW	The LOWVDS_WAIT defines the time between switching on an LED and the start of voltage monitoring.

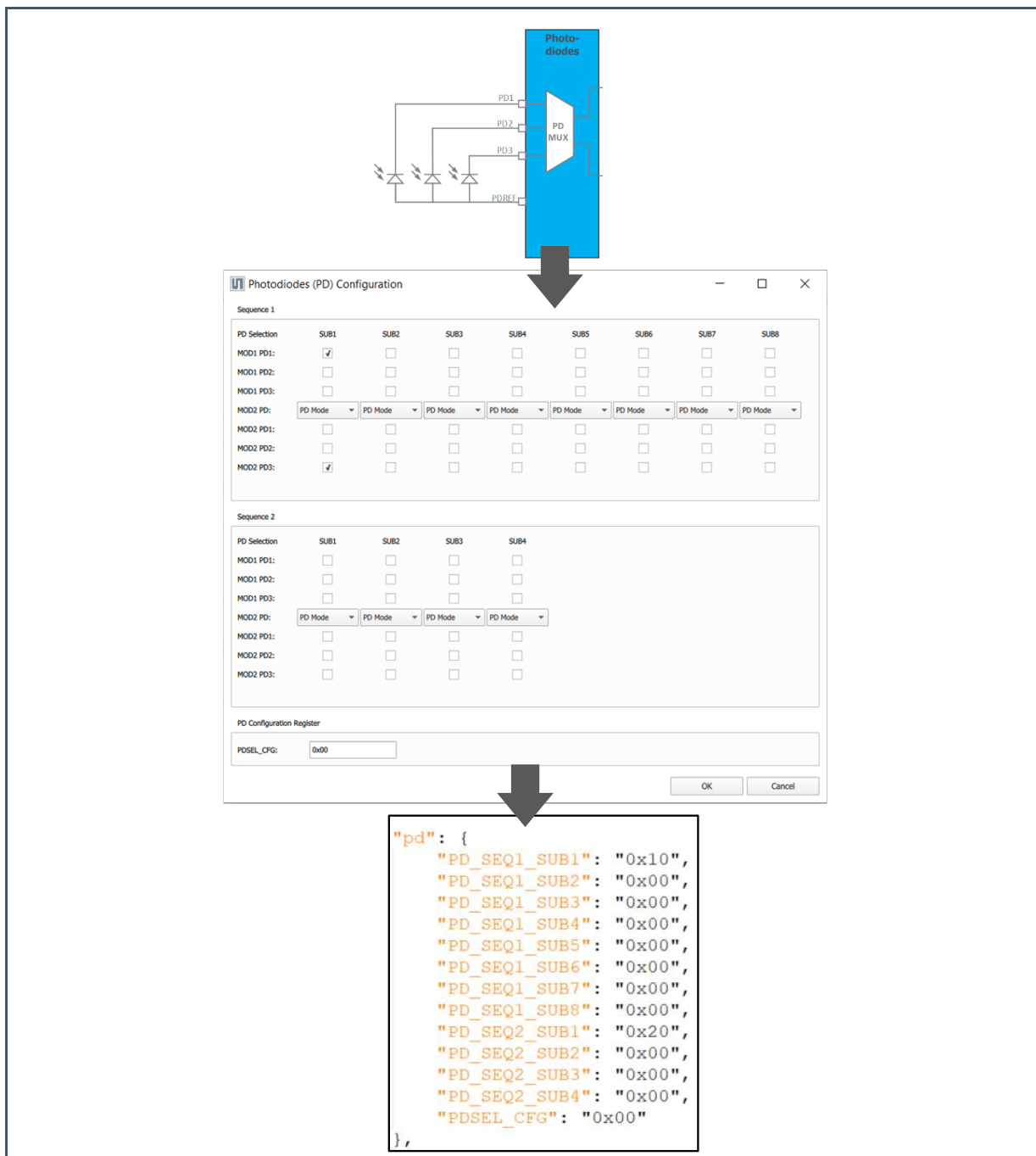
Permissible Skin Exposure to VCSEL

The AS7056 VCSEL can be used for PPG measurements and during that process the VCSEL touches the skin more or less directly and the power per area can be high for a divergent laser. This makes it possible to damage a small area on the skin even with small absolute laser powers. Therefore, it is important to understand the skin compatibility with the eye safety of the VCSEL.

6.1.5 Photodiodes (PD) Configuration

The Configuration of the three usable photodiodes can be done in this section.

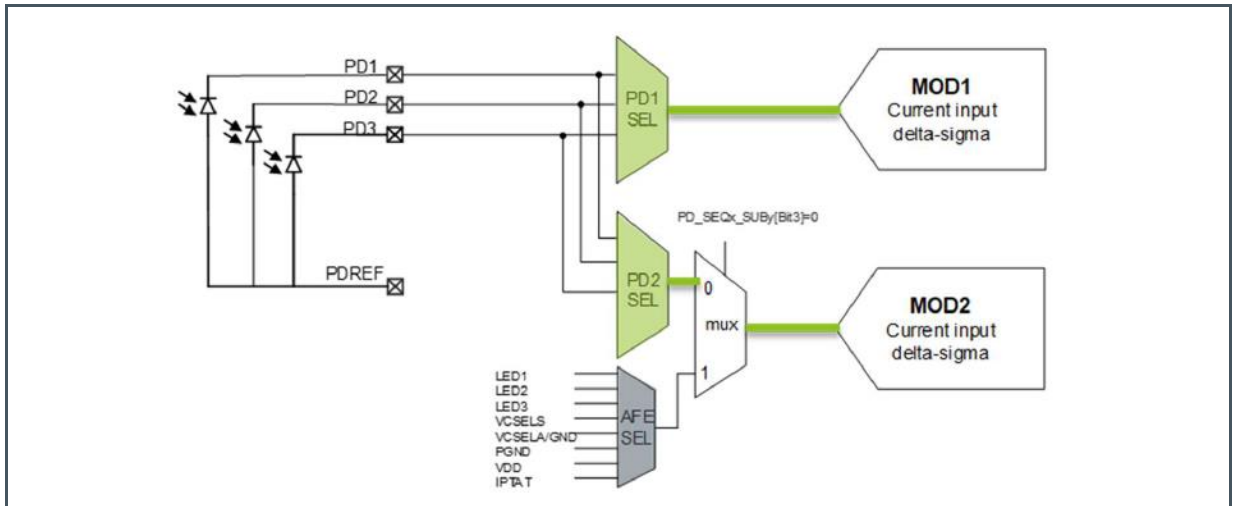
Figure 55:
Photodiodes Configuration



The PPG front-ends consist of three main functional blocks: photodiode (PD) selection, PD offset, and the PPG ADC. The PD selection supports up to three different photodiodes. The photodiodes are

connected via a multiplexer to the PPG ADC. To minimize daylight impacts, the photodiode input currents can be compensated by the PD offset. After amplifying and converting the PD currents, the digital PD output signals pass to the digital PPG filter inside the digital block of the AS7056/57.

Figure 56:
PPG Selection



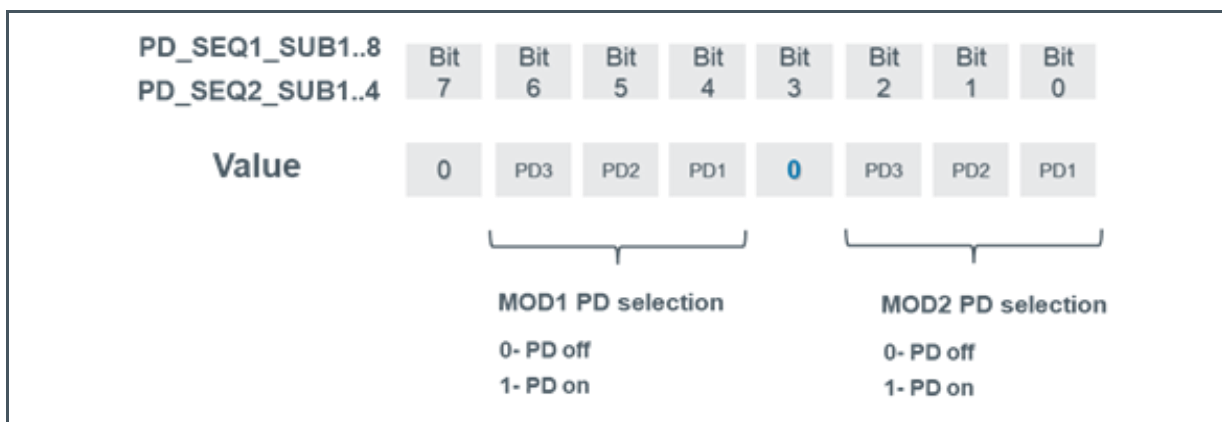
Each of the three photodiodes can be connected to PPG1-MOD or PPG2-MOD.

Sequence 1 & Sequence 2

PD Selection:

Eight registers are used to select which PD, out of the available three, should be connected to which subsample and subsequence. Each PD could be connected to any subsample inside sequence 1. The generic form of the register can be written as PD_SEQx_SUBy, where “x” shows the sequence number for each channel, and “y” shows the subsample number inside each sequence. In the default mode, bit 7 and bit 3 remain set to zero. Bits 6:4 are used to select PDs for MOD1, while bits 2:0 are used to select PDs for MOD2. For MOD1, bit 6 corresponds to PD3, bit 5 corresponds to PD2, and bit 4 to PD1. Similarly, bits 2:0 are used for MOD2, as shown in Figure 57 below. PD3, PD2, and PD1 are different photodiodes and occupy different positions on the evaluation board.

Figure 57:
PD Selection for Sequence 1



The table below (Figure 58) shows the connection of the PD3, PD2, and PD1 to MOD1 and MOD2 and the corresponding binary and hexadecimal code. If two or all three PDs are connected to MOD1 or MOD2, they must be interleaved in time, since at one time, only one MOD can be connected to a PD. However, two different PDs can be simultaneously connected to different MODs. For instance, PD1 is connected to MOD2 and simultaneously PD2 can be applied to MOD1.

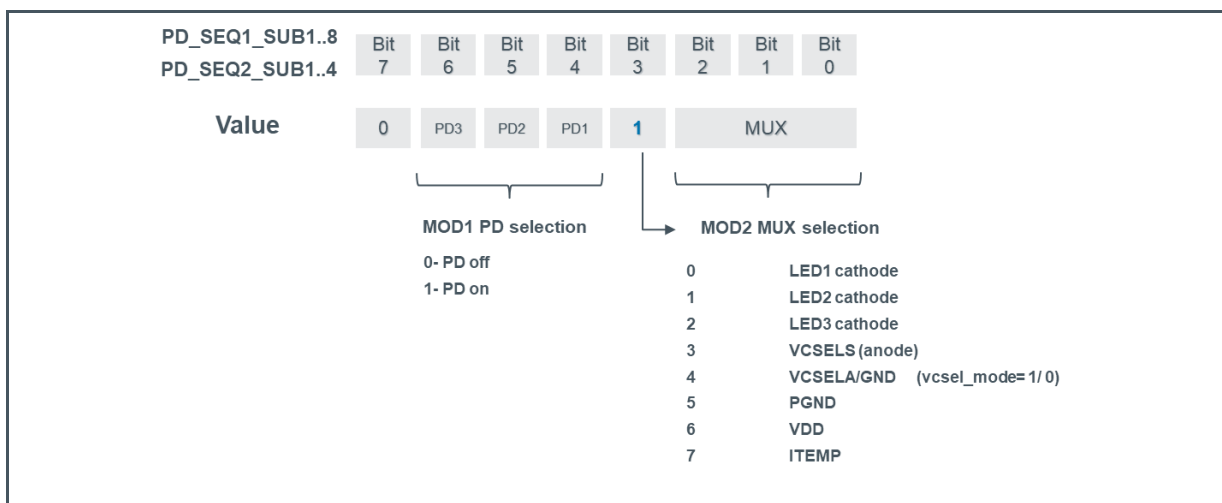
Figure 58:
PD Selection

Synchronous/ Time Interleaved	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	PD_SEQx_SUBy Photodiode Selection
		MOD1				MOD2			
T/S		PD3	PD2	PD1		PD3	PD2	PD1	Hex
T	0	0	0	0	0	0	0	0	0x00
T	0	0	0	1	0	0	0	0	0x10
T	0	0	0	0	0	0	0	1	0x01
T	0	0	1	0	0	0	0	0	0x20
T	0	0	1	1	0	0	0	0	0x30
S	0	0	1	0	0	0	0	1	0x21
T	0	0	0	0	0	0	1	0	0x02
S	0	0	0	1	0	0	1	0	0x12
T	0	0	0	0	0	0	1	1	0x03
T	0	1	0	0	0	0	0	0	0x40
T	0	1	0	1	0	0	0	0	0x50

Synchronous/ Time Interleaved	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	PD_SEQx_SUBy Photodiode Selection
		MOD1				MOD2			
T/S		PD3	PD2	PD1		PD3	PD2	PD1	Hex
S	0	1	0	0	0	0	0	1	0x41
T	0	1	1	0	0	0	0	0	0x60
T	0	1	1	1	0	0	0	0	0x70
S	0	1	1	0	0	0	0	1	0x61
S	0	1	0	0	0	0	1	0	0x42
S	0	1	0	1	0	0	1	0	0x52
S	0	1	0	0	0	0	1	1	0x43
T	0	0	0	0	0	1	0	0	0x04
S	0	0	0	1	0	1	0	0	0x14
T	0	0	0	0	0	1	0	1	0x05
S	0	0	1	0	0	1	0	0	0x24
S	0	0	1	1	0	1	0	0	0x34
S	0	0	1	0	0	1	0	1	0x25
T	0	0	0	0	0	1	1	0	0x06
S	0	0	0	1	0	1	1	0	0x16
T	0	0	0	0	0	1	1	1	0x07

PPG2/MOD2 can be connected to external voltages instead of the PDs. This is achieved by setting bit 3 inside the PD_SEQx_SUBy register to one. Note that MOD1 is configured to be connected only to PDs and not any external voltage, while MOD2 can be configured to be connected to either PDs or the external voltages.

Figure 59:
MOD2 – Internal MUX Selection



After setting bit 3 to high, bit 2:0 decides which of the external voltages will be connected to MOD2. The PDs current and the external voltages can be simultaneously measured via MOD1 and MOD2 respectively.

Figure 60:
MOD1 – PD Selection and MOD2 – Internal MUX Selection

PD_SEQx_SUBy Photodiode Selection			Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
PD3	PD2	PD1								
-	-	-	0	0	0	0	1	0..7		
-	-	MOD1	0	0	0	1	1	0..7		
-	MOD1	-	0	0	1	0	1	0..7		
-	MOD1	MOD1	0	0	1	1	1	0..7		
MOD1	-	-	0	1	0	0	1	0..7		
MOD1	-	MOD1	0	1	0	1	1	0..7		
MOD1	MOD1	-	0	1	1	0	1	0..7		
MOD1	MOD1	MOD1	0	1	1	1	1	0..7		

PD Configuration Register

PDSEL_CFG (Address 0x5F):

The PDREF (photodiode reference voltage) can be selected between VSSA (Analog ground) and VCM_PPG (Common mode voltage for PPG channel). When PDREF_SEL (Selection of PDREF voltage) is logical '1' then PDREF is connected to VSSA. When PDREF_SEL is logical '0', then

PDREF is connected to VCM_PPG if the signals PPG1_EN (Enable PPG1 channel) or PPG2_EN (Enable PPG2 channel) is digital high (otherwise PDREF is connected to VSSA).

6.1.6 PD-Offset/Automatic Offset Control (AOC) Configuration

The AS7056/57 offers an automatic offset control to balance and reduce ambient light currents automatically. This feature can be configured in this section.

Figure 61:
AOC Configuration

PD-Offset / Automatic Offset Control (AOC) Configuration

Sequence 1

Enable AOC	SUB1	SUB2	SUB3	SUB4	SUB5	SUB6	SUB7	SUB8
MOD1:	☒	☒	☐	☐	☐	☐	☐	☐
MOD2:	☐	☐	☐	☐	☐	☐	☐	☐

PD Offset

SUB1	SUB2	SUB3	SUB4	SUB5	SUB6	SUB7	SUB8
MOD1:	0	0	0	0	0	0	0
MOD2:	0	0	0	0	0	0	0

Sequence 2

SUB1	SUB2	SUB3	SUB4	
MOD1:	0	0	0	0

AOC Threshold

	ADC Value	Register Value
Minimum Threshold MOD1:	397312	0x61
Maximum Threshold MOD1:	-176128	0xAB
Minimum Threshold MOD2:	208896	0x33
Maximum Threshold MOD2:	-311296	0xCC
Threshold for SAR:	0	0x80

Oversampling

Oversampling: 8

LED off

Disable PD Offset for LED off: ☐

PD Offset for LED off: 0

OK Cancel

```

"aoc": {
  "AOC_MOD1_SEQ1_SUB1": "0x00",
  "AOC_MOD1_SEQ1_SUB2": "0x00",
  "AOC_MOD1_SEQ1_SUB3": "0x00",
  "AOC_MOD1_SEQ1_SUB4": "0x00",
  "AOC_MOD1_SEQ1_SUB5": "0x00",
  "AOC_MOD1_SEQ1_SUB6": "0x00",
  "AOC_MOD1_SEQ1_SUB7": "0x00",
  "AOC_MOD1_SEQ1_SUB8": "0x00",
  "AOC_MOD1_SEQ2_SUB1": "0x00",
  "AOC_MOD1_SEQ2_SUB2": "0x00",
  "AOC_MOD1_SEQ2_SUB3": "0x00",
  "AOC_MOD1_SEQ2_SUB4": "0x00",
  "AOC_MOD2_SEQ1_SUB1": "0x00",
  "AOC_MOD2_SEQ1_SUB2": "0x00",
  "AOC_MOD2_SEQ1_SUB3": "0x00",
  "AOC_MOD2_SEQ1_SUB4": "0x00",
  "AOC_MOD2_SEQ1_SUB5": "0x00",
  "AOC_MOD2_SEQ1_SUB6": "0x00",
  "AOC_MOD2_SEQ1_SUB7": "0x00",
  "AOC_MOD2_SEQ1_SUB8": "0x00",
  "AOC_LED_OFF": "0x00",
  "AOC_CFG": "0x00",
  "AOC_MOD1_THH": "0xAB",
  "AOC_MOD1_THL": "0x61",
  "AOC_MOD2_THH": "0xCC",
  "AOC_MOD2_THL": "0x33",
  "AOC_SAR_THRES": "0x80",
  "MOD1_SEQ1_AOC_EN": "0x03",
  "MOD2_SEQ1_AOC_EN": "0x00"
}

```

Apart from the configuration before or after a measurement cycle, it is also possible to adjust the AOC registers during a measurement by an external microcontroller.

The PD offset registers can be overwritten if the AOC is active and take effect immediately. The control of the AOC continues as usual. A separate PD offset current can be programmed for twenty sub-samples in a sample, as shown in the tables below.

Sequence 1

Figure 62:
MOD1_SEQ1_AOC_EN Register

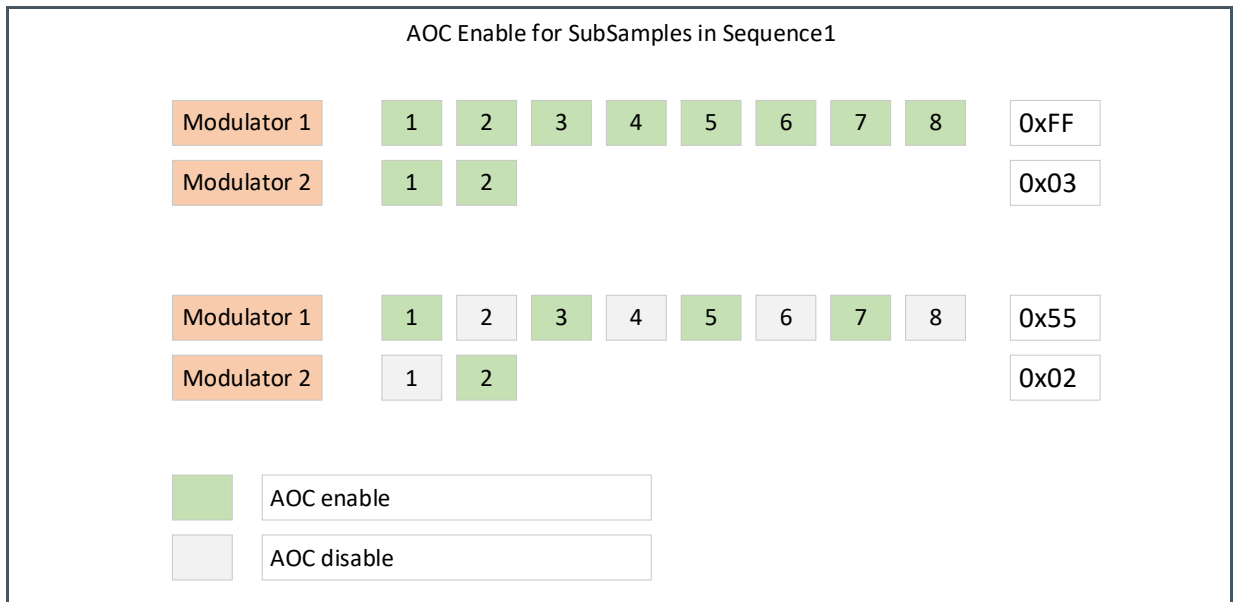
0x8B		MOD1_SEQ1_AOC_EN		
ID	Bit Name	Default	Access	Description
7:0	MOD1_SEQ1_AOC_EN	0	RW	Enable AOC for Subsamples 1 to 8 in Sequence1 for Modulator1. Bit 7 – Subsample 8 Bit 6 - Subsample 7 Bit 5 - Subsample 6 Bit 4 - Subsample 5 Bit 3 - Subsample 4 Bit 2 - Subsample 3 Bit 1 - Subsample 2 Bit 0 - Subsample 1

Figure 63:
MOD2_SEQ1_AOC_EN Register

0x8C		MOD2_SEQ1_AOC_EN		
ID	Bit Name	Default	Access	Description
7:0	MOD2_SEQ1_AOC_EN	0	RW	Enable AOC for Subsamples 1 to 8 in Sequence1 for Modulator2. Bit 1 - Subsample 2 Bit 0 - Subsample 1

The AOC is only available in sequence1 and can be activated for subsamples 1 to 8 for PPG MOD1 and for subsamples 1 to 2, in sequence1 for PPG MOD2, as shown in the figure below.

Figure 64:
AOC Enable for Subsamples in Sequence 1



Subsamples 1 to 4 in Sequence 2 for PPG MOD1 and subsamples 3 to 8 in Sequence 1 for PPG MOD2 only work with the manual programmable PD offset value.

Figure 65:
AOC_MOD1_SEQ1_SUB1 Register

0x70 AOC_MOD1_SEQ1_SUB1				
ID	Bit Name	Default	Access	Description
7:0	AOC_MOD1_SEQ1_SUB1	0	R_PUSH	PD Offset for Modulator1 in SubSample1/Sequence1.

Figure 66:
AOC_MOD1_SEQ1_SUB2 Register

0x71 AOC_MOD1_SEQ1_SUB2				
ID	Bit Name	Default	Access	Description
7:0	AOC_MOD1_SEQ1_SUB2	0	R_PUSH	PD Offset for Modulator1 in SubSample2/Sequence1.

Figure 67:
AOC_MOD1_SEQ1_SUB3 Register

0x72 AOC_MOD1_SEQ1_SUB3				
ID	Bit Name	Default	Access	Description
7:0	AOC_MOD1_SEQ1_SUB3	0	R_PUSH	PD Offset for Modulator1 in SubSample3/Sequence1.

Figure 68:
AOC_MOD1_SEQ1_SUB4 Register

0x73 AOC_MOD1_SEQ1_SUB4				
ID	Bit Name	Default	Access	Description
7:0	AOC_MOD1_SEQ1_SUB4	0	R_PUSH	PD Offset for Modulator1 in SubSample4/Sequence1.

Figure 69:
AOC_MOD1_SEQ1_SUB5 Register

0x74 AOC_MOD1_SEQ1_SUB5				
ID	Bit Name	Default	Access	Description
7:0	AOC_MOD1_SEQ1_SUB5	0	R_PUSH	PD Offset for Modulator1 in SubSample5/Sequence1.

Figure 70:
AOC_MOD1_SEQ1_SUB6 Register

0x75 AOC_MOD1_SEQ1_SUB6				
ID	Bit Name	Default	Access	Description
7:0	AOC_MOD1_SEQ1_SUB6	0	R_PUSH	PD Offset for Modulator1 in SubSample6/Sequence1.

Figure 71:
AOC_MOD1_SEQ1_SUB7 Register

0x76 AOC_MOD1_SEQ1_SUB7				
ID	Bit Name	Default	Access	Description
7:0	AOC_MOD1_SEQ1_SUB7	0	R_PUSH	PD Offset for Modulator1 in SubSample7/Sequence1.

Figure 72:
AOC_MOD1_SEQ1_SUB8 Register

0x77		AOC_MOD1_SEQ1_SUB8		
ID	Bit Name	Default	Access	Description
7:0	AOC_MOD1_SEQ1_SUB8	0	R_PUSH	PD Offset for Modulator1 in SubSample8/Sequence1.

Figure 73:
AOC_MOD2_SEQ1_SUB1 Register

0x7C		AOC_MOD2_SEQ1_SUB1		
ID	Bit Name	Default	Access	Description
7:0	AOC_MOD2_SEQ1_SUB1	0	R_PUSH	PD Offset for Modulator2 in SubSample1/Sequence1.

Figure 74:
AOC_MOD2_SEQ1_SUB2 Register

0x7D		AOC_MOD2_SEQ1_SUB2		
ID	Bit Name	Default	Access	Description
7:0	AOC_MOD2_SEQ1_SUB2	0	R_PUSH	PD Offset for Modulator2 in SubSample2/Sequence1.

Figure 75:
AOC_MOD2_SEQ1_SUB3 Register

0x7E		AOC_MOD2_SEQ1_SUB3		
ID	Bit Name	Default	Access	Description
7:0	AOC_MOD2_SEQ1_SUB3	0	R_PUSH	PD Offset for Modulator2 in SubSample3/Sequence1.

Figure 76:
AOC_MOD2_SEQ1_SUB4 Register

0x7F		AOC_MOD2_SEQ1_SUB4		
ID	Bit Name	Default	Access	Description
7:0	AOC_MOD2_SEQ1_SUB4	0	R_PUSH	PD Offset for Modulator2 in SubSample4/Sequence1.

Figure 77:
AOC_MOD2_SEQ1_SUB5 Register

0x80		AOC_MOD2_SEQ1_SUB5		
ID	Bit Name	Default	Access	Description
7:0	AOC_MOD2_SEQ1_SUB5	0	R_PUSH	PD Offset for Modulator2 in SubSample5/Sequence1.

Figure 78:
AOC_MOD2_SEQ1_SUB6 Register

0x81		AOC_MOD2_SEQ1_SUB6		
ID	Bit Name	Default	Access	Description
7:0	AOC_MOD2_SEQ1_SUB6	0	R_PUSH	PD Offset for Modulator2 in SubSample6/Sequence1.

Figure 79:
AOC_MOD2_SEQ1_SUB7 Register

0x82		AOC_MOD2_SEQ1_SUB7		
ID	Bit Name	Default	Access	Description
7:0	AOC_MOD2_SEQ1_SUB7	0	R_PUSH	PD Offset for Modulator2 in SubSample7/Sequence1.

Figure 80:
AOC_MOD2_SEQ1_SUB8 Register

0x83		AOC_MOD2_SEQ1_SUB8		
ID	Bit Name	Default	Access	Description
7:0	AOC_MOD2_SEQ1_SUB8	0	R_PUSH	PD Offset for Modulator2 in SubSample8/Sequence1.

$$\triangleright PD_offset_current = (AOC_MODx_SEQy_SUBz) \times (PPG_DAC\ full_{scale}range)/256$$

Sequence 2

Subsamples 1 to 4 in sequence 2 for PPG MOD2 are intended for the analog front end and do not have a programmable PD offset value.

Figure 81:
AOC_MOD1_SEQ2_SUB1 Register

0x78		AOC_MOD1_SEQ2_SUB1		
ID	Bit Name	Default	Access	Description
7:0	AOC_MOD1_SEQ2_SUB1	0	R_PUSH	PD Offset for Modulator1 in SubSample1/Sequence2.

Figure 82:
AOC_MOD1_SEQ2_SUB2 Register

0x79		AOC_MOD1_SEQ2_SUB2		
ID	Bit Name	Default	Access	Description
7:0	AOC_MOD1_SEQ2_SUB2	0	R_PUSH	PD Offset for Modulator1 in SubSample2/Sequence2.

Figure 83:
AOC_MOD1_SEQ2_SUB3 Register

0x7A		AOC_MOD1_SEQ2_SUB3		
ID	Bit Name	Default	Access	Description
7:0	AOC_MOD1_SEQ2_SUB3	0	R_PUSH	PD Offset for Modulator1 in SubSample3/Sequence2.

Figure 84:
AOC_MOD1_SEQ2_SUB4 Register

0x7B		AOC_MOD1_SEQ2_SUB4		
ID	Bit Name	Default	Access	Description
7:0	AOC_MOD1_SEQ2_SUB4	0	R_PUSH	PD Offset for Modulator1 in SubSample4/Sequence2.

AOC Threshold

Modulator 1 and Modulator 2 have their parameters for setting the automatic offset control AOC. The threshold for increasing or decreasing the PD_OFFSET for Modulator 1, Modulator 2, and the threshold for SAR for all subsamples, are described below.

Figure 85:
AOC_MOD1_THH Register

0x86	AOC_MOD1_THH			
ID	Bit Name	Default	Access	Description
7:0	AOC_MOD1_THH	255	RW	Threshold for decreasing the PD_OFFSET for all Subsamples for Modulator1.

Figure 86:
AOC_MOD1_THL Register

0x87	AOC_MOD1_THL			
ID	Bit Name	Default	Access	Description
7:0	AOC_MOD1_THL	0	RW	Threshold for increasing the PD_OFFSET for all Subsamples for Modulator1.

Figure 87:
AOC_MOD2_THH Register

0x88	AOC_MOD2_THH			
ID	Bit Name	Default	Access	Description
7:0	AOC_MOD2_THH	255	RW	Threshold for decreasing the PD_OFFSET for all Subsamples for Modulator2.

Figure 88:
AOC_MOD2_THL Register

0x89	AOC_MOD2_THL			
ID	Bit Name	Default	Access	Description
7:0	AOC_MOD2_THL	0	RW	Threshold for increasing the PD_OFFSET for all Subsamples for Modulator2.

Figure 89:
AOC_SAR_TRES Register

0x8A	AOC_SAR_TRES			
ID	Bit Name	Default	Access	Description
7:0	AOC_SAR_TRES	0	RW	Threshold for SAR for all Subsamples.

Oversampling

Figure 90:
AOC_CFG Register (Oversampling)

0x85	AOC_CFG			
ID	Bit Name	Default	Access	Description
2:0	AOC_OVS	0	RW	Value for oversampling AOC. Oversampling = $2^{(aoc_ovs+3)}$

LED Off

Figure 91:
AOC_CFG Register (LED Off)

0x85	AOC_CFG			
ID	Bit Name	Default	Access	Description
4	DIS_LED OFF	0	RW	Disable PD Offset for LED off (Double and Triple Sampling) and Use PD Offset for SubSample.

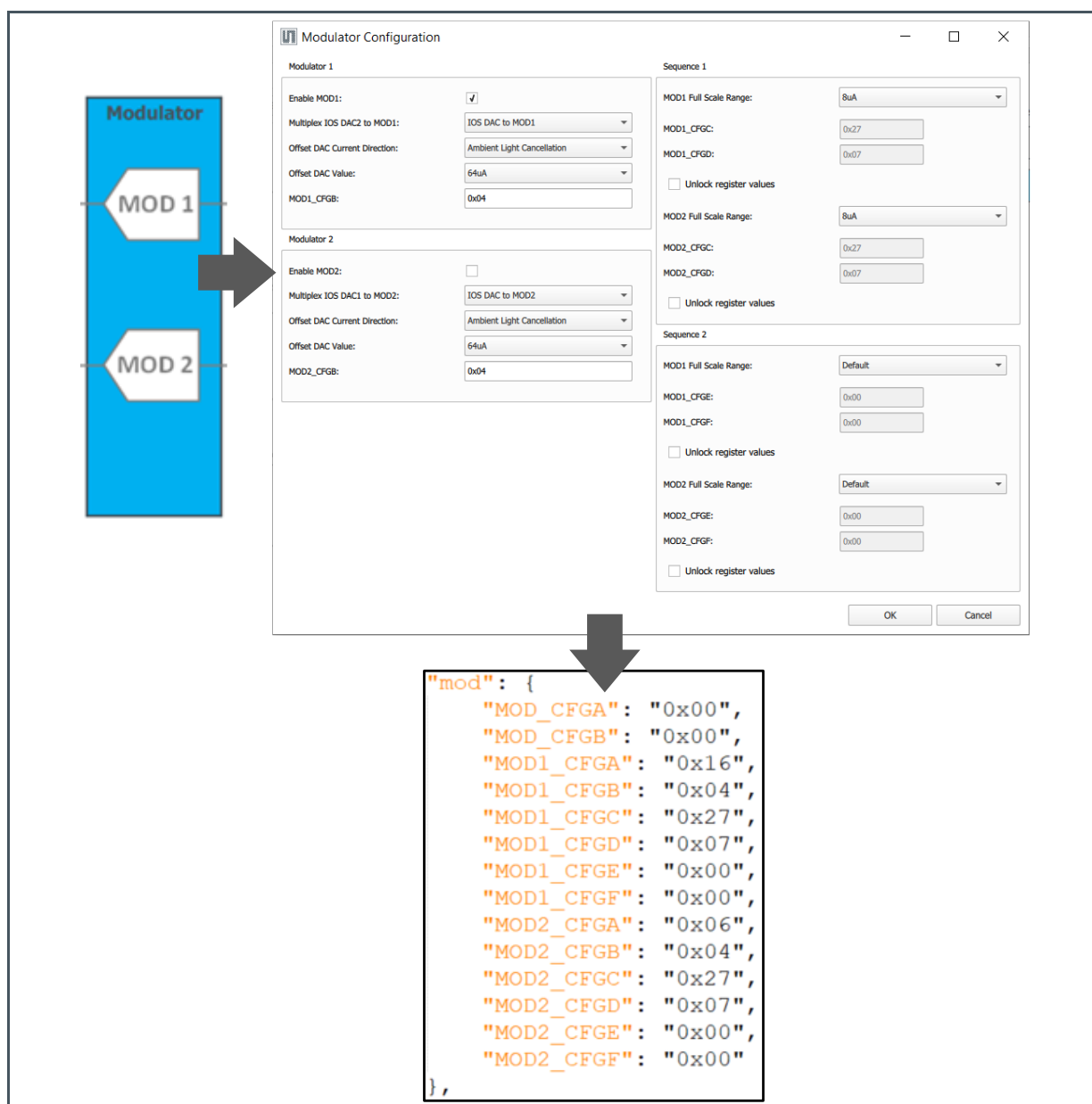
Figure 92:
AOC_LED OFF Register

0x85	AOC_LED OFF			
ID	Bit Name	Default	Access	Description
7:0	AOC_LED OFF	0	RW	PD Offset for LED off (Double and Triple Sampling)

6.1.7 Modulator Configuration

The PPG ADCs, also known as Modulators, are current input ADCs and could be configured in the Modulator configuration.

Figure 93:
Modulator Configuration



The PPG ADCs are current input ADCs. The sequencer for each time slot controls it. The ADC Input Full-Scale Range from PDs to PPG ADC has been programmed by the reference DAC full-scale range. A stable operation is enabled by proper operation conditions with a DAC with bipolar references. The modulator is used to scale the reference DAC current.

Modulator 1

MOD1_CFGA [2:0] is used to set the IOS DAC Full-Scale current independently for MOD1, as shown below. Setting the three bits to 111 enables the IOS DAC current up to 128 μ A.

Figure 94:
MOD1_CFGA Register

0x1B	MOD1_CFGA			
ID	Bit Name	Default	Access	Description
5	MOD1_IOS_MUX	0	RW	Multiplex IOS DAC2 to modulator 1 for calibration. Bit 0 – ios DAC to MOD1 Bit 1 – ios DAC to MOD2, if MOD1_OIS_DIR = 1
4	MOD1_EN	0	RW	Enable modulator 1. Bit 0 – Off Bit 1 – On
3	MOD1_IOS_DIR	0	RW	Offset DAC current direction. Bit 0 – Ambient light cancelation is enabled.
2:0	MOD1_IOS_FS	0	RW	Offset DAC full scale current. Number 0 – 1 μ A Number 1 – 2 μ A Number 2 – 4 μ A Number 3 – 8 μ A Number 4 – 16 μ A Number 5 – 32 μ A Number 6 – 64 μ A Number 7 – 128 μ A

MOD1_CFGB (Address 0x1C): Current reference DAC IRN (Negative reference current)/IRNmax scale factor. The scale factor is constant, which is 0.625.

Modulator 2

MOD2_CFGA [2:0] is used to set the IOS DAC Full-Scale current independently for MOD1, as shown below. Setting the three bits to 111 enables the IOS DAC current up to 128 μ A.

Figure 95:
MOD2_CFGA Register

0x21	MOD2_CFGA			
ID	Bit Name	Default	Access	Description
5	MOD2_IOS_MUX	0	RW	Multiplex IOS DAC1 to modulator 2 for calibration. Bit 0 – ios DAC to MOD1 Bit 1 – ios DAC to MOD2,

0x21 MOD2_CFGA				
ID	Bit Name	Default	Access	Description
				if MOD1_OIS_DIR = 1
4	MOD2_EN	0	RW	Enable modulator 2. Bit 0 – Off Bit 1 – On
3	MOD2_IOS_DIR	0	RW	Offset DAC current direction. Bit 0 – Ambient light cancelation is enabled.
2:0	MOD2_IOS_FS	0	RW	Offset DAC full scale current. Number 0 – 1 μ A Number 1 – 2 μ A Number 2 – 4 μ A Number 3 – 8 μ A Number 4 – 16 μ A Number 5 – 32 μ A Number 6 – 64 μ A Number 7 – 128 μ A

MOD2_CFGB (Address 0x22): Current reference DAC IRN (Negative reference current)/IRNmax scale factor.

Figure 96:
SEQ_FREQ[7:0] Register

0x46 SEQ_FREQ[7:0]				
ID	Bit Name	Default	Access	Description
3:0	SEQ_FREQ [7:0]	0	RW	Current reference DAC IRP/IRN scale factor. Number 0 – 0.125 Number 1 – 0.250 Number 2 – 0.375 Number 3 – 0.500 Number 4 – 0.625 Number 5 – 0.750 Number 6 – 0.875 Number 7 – 1.000

Sequence 1 & Sequence 2

The configuration registers MODx_CFGC and MODx_CFGD contain parameters that vary with the AGAIN. Each AGAIN and the corresponding configuration registers address a different input full-scale range, as shown in Figure 97 below. The configuration registers, C and D, vary AGAIN parameters for sequence 1, and registers E and F, vary parameters for sequence 2. The default values of these registers for each AGAIN setting are shown in Figure 97 below.

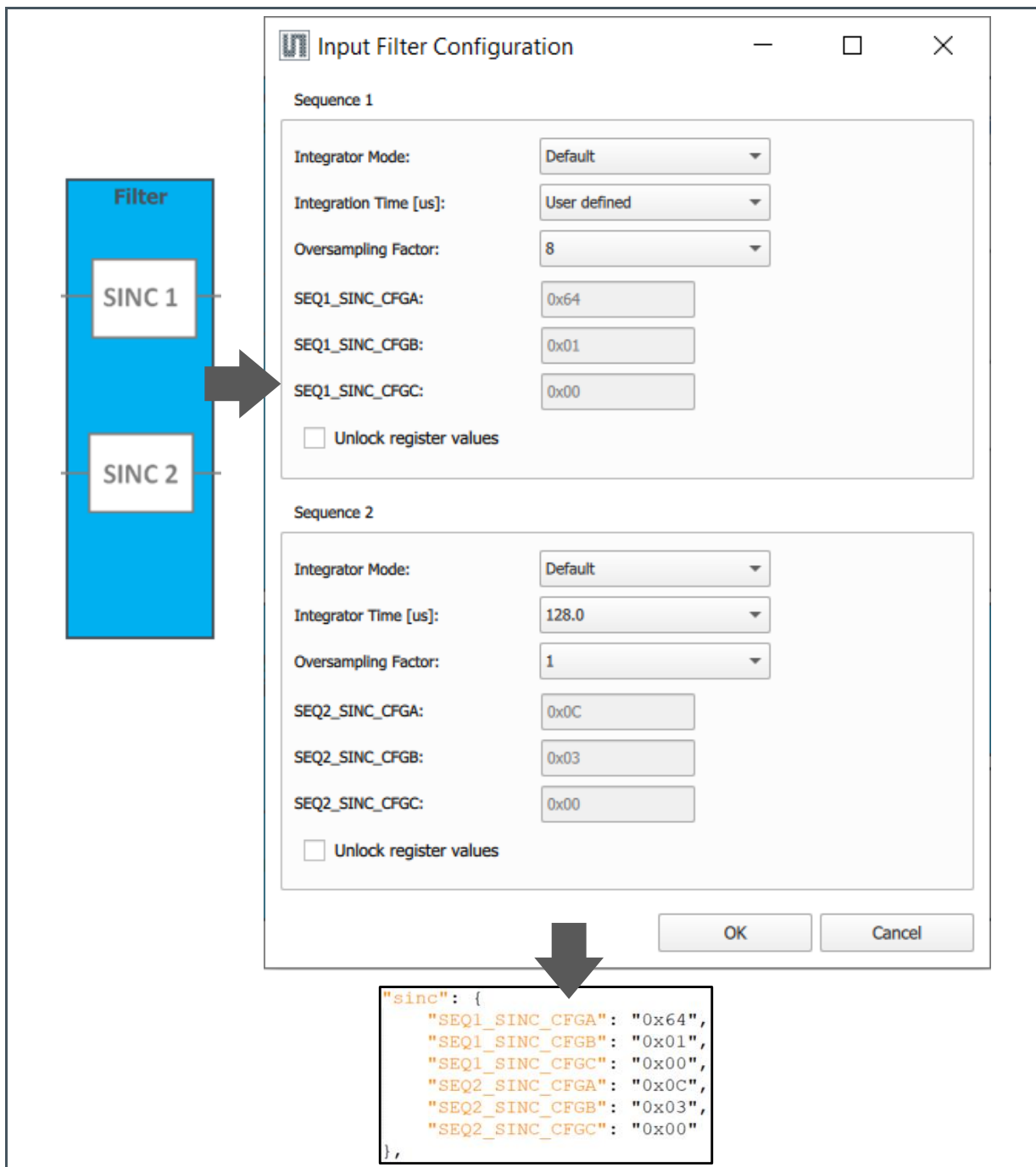
Figure 97:
MOD1 and MOD2 FSR Settings

			IFS	1 μA	2 μA	4 μA	8 μA	16 μA	32 μA	64μA
Register	Address	MOD	AGAIN	0	1	2	3	4	5	6
MOD1_CFGC	0x1D	MOD1	SEQ1	0x27	0x27	0x27	0x27	0x2F	0x1F	0x1F
MOD1_CFGD	0x1E			0x00	0x01	0x03	0x07	0x0F	0x1F	0x3F
MOD1_CFGE	0x1F		SEQ2	0x27	0x27	0x27	0x27	0x2F	0x1F	0x1F
MOD1_CFGF	0x20			0x00	0x01	0x03	0x07	0x0F	0x1F	0x3F
MOD2_CFGC	0x23	MOD2	SEQ1	0x27	0x27	0x27	0x27	0x2F	0x1F	0x1F
MOD2_CFGD	0x24			0x00	0x01	0x03	0x07	0x0F	0x1F	0x3F
MOD2_CFGE	0x25		SEQ2	0x27	0x27	0x27	0x27	0x2F	0x1F	0x1F
MOD2_CFGF	0x26			0x00	0x01	0x03	0x07	0x0F	0x1F	0x3F

6.1.8 Input Filter Configuration

The AS7056/57 offers different filter modes, which are accessible on that setting page.

Figure 98:
Input Filter Configuration



Sequence 1 and 2 have their parameters for setting the SINC filter. All subsamples of a sequence use the same parameters for both modulators.

Sequence 1

Figure 99:
SEQ1_SINC_CFGA Register

0x61	SEQ1_SINC_CFGA			
ID	Bit Name	Default	Access	Description
7:5	SEQ1_SINC_OVS	0	RW	Value for oversampling SINC filter. Oversampling = $2^{\text{seq1_sinc_ovs}}$
4:2	SEQ1_SINC_DEC	0	RW	Decimation Number 0 – 16 Number 1 – 32 Number 2 – 64 Number 3 – 128 Number 4 – 256

Figure 100:
SEQ1_SINC_CFGB Register

0x62	SEQ1_SINC_CFGB			
ID	Bit Name	Default	Access	Description
6:3	SEQ1_OS_DELAY	0	RW	Delay after which data is valid for the average calculation:
1	SEQ1_SEL_ORDER	0	RW	Select filter order 4/5. Bit 0 – 4 Bit 1 – 5
0	SEQ1_FILTER_MODE	1	RW	Select mode of filter for Sequence1. Bit 0 – Integrator Bit 1 – CIC Mode

Figure 101:
SEQ1_SINC_CFGC Register

0x63	SEQ1_SINC_CFGC			
ID	Bit Name	Default	Access	Description
7:0	SEQ1_START_DELAY	0	RW	Start delay

Sequence 2

Figure 102:
SEQ2_SINC_CFGA Register

0x64	SEQ2_SINC_CFGA			
ID	Bit Name	Default	Access	Description
7:5	SEQ2_SINC_OVS	0	RW	Value for oversampling SINC filter. Oversampling = $2^{\text{seq2_sinc_ovs}}$
4:2	SEQ2_SINC_DEC	0	RW	Decimation Number 0 – 16 Number 1 – 32 Number 2 – 64 Number 3 – 128 Number 4 – 256

Figure 103:
SEQ2_SINC_CFGB Register

0x65	SEQ2_SINC_CFGB			
ID	Bit Name	Default	Access	Description
6:3	SEQ2_OS_DELAY	0	RW	Delay after which data is valid for the average calculation:
1	SEQ2_SEL_ORDER	0	RW	Select filter order 4/5. Bit 0 – 4 Bit 1 – 5
0	SEQ2_FILTER_MODE	1	RW	Select mode of filter for Sequence2. Bit 0 – Integrator Bit 1 – CIC Mode

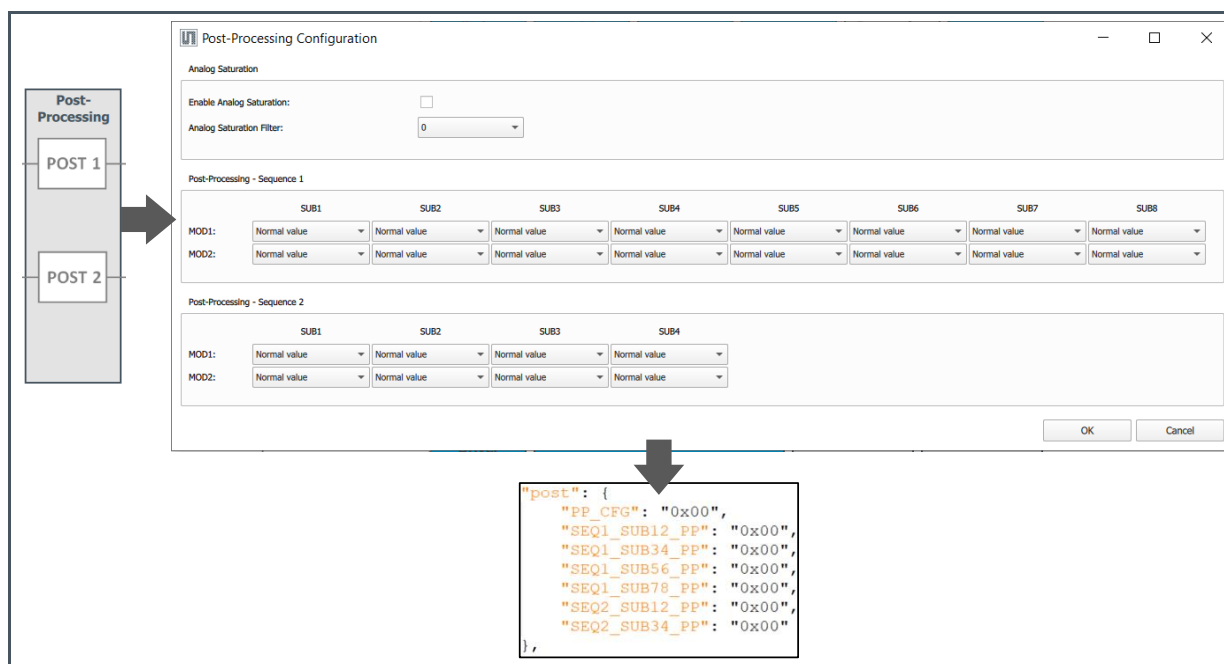
Figure 104:
SEQ2_SINC_CFGC Register

0x66	SEQ2_SINC_CFGC			
ID	Bit Name	Default	Access	Description
7:0	SEQ2_START_DELAY	0	RW	Start delay

6.1.9 Post-Processing Configuration

The configuration of the post-processing behavior can be adjusted in this section.

Figure 105:
Post-Processing Configuration



Analog Saturation

If analog saturation occurs in the modulator during a measurement, the measured value can be set to the maximum value according to the sign.

- Negative measurement values from 0x80000 to 0xFFFFF. In this case, the maximum value is 0x80000.
- Positive measurement values from 0x00000 to 0x7FFFF. In this case, the maximum value is 0x7FFFF.

Enable analog saturation: This means enabling the change in the measured signed value when an analog saturation is detected.

Figure 106:
PP_CFG Register

0x37		PP_CFG		
ID	Bit Name	Default	Access	Description
7	ASAT_ON]	0	RW	Enable Analog Saturation; enabling the change in the measured signed value when an analog saturation is detected. Bit 0 – The measured signed value has not changed. Bit 1 – The measured signed value has changed.
3:0	ASAT_FIL	0	RW	Digital Filter for the analog input signal for saturation; the pulse for the detection of the analog saturation must be longer than $n * MOD_CLK$.

Post-Processing - Sequence 1

Each subsample uses, independently of one another, one of the following functions for post-processing:

- 2'b00 – Transmission of the measured value -> Normal value
- 2'b01 – Transmission of the inverted measured value -> Invert value
- 2'b10 – Transfer of the result = measured value - PP offset -> Value – pp_offset
- 2'b11 – Writing the measured value in the PD offset and transmission it -> Value to pp_offset

Modulator 1 and Modulator 2 use the same setting for post-processing.

Figure 107:
SEQ1_SUB12_PP Register

0x38		SEQ1_SUB12_PP		
ID	Bit Name	Default	Access	Description
7:6	MOD1_SEQ1_SUB1_PP	0	RW	Post Processing in Modulator1 for SubSample1 in Sequence1. Number 0 – Normal value Number 1 – Inverted value Number 2 – Value pp_offset Number 3 - Write a value to pp_offset
5:4	MOD2_SEQ1_SUB1_PP	0	RW	Post Processing in Modulator2 for SubSample1 in Sequence1. Number 0 – Normal value Number 1 – Inverted value Number 2 – Value pp_offset Number 3 - Write a value to pp_offset
3:2	MOD1_SEQ1_SUB2_PP	0	RW	Post Processing in Modulator1 for SubSample2 in Sequence1. Number 0 – Normal value Number 1 – Inverted value

0x38	SEQ1_SUB12_PP			
ID	Bit Name	Default	Access	Description
				Number 2 – Value pp_offset Number 3 - Write value to pp_offset
1:0	MOD2_SEQ1_SUB2_PP	0	RW	Post Processing in Modulator2 for SubSample2 in Sequence1. Number 0 – Normal value Number 1 – Inverted value Number 2 – Value pp_offset Number 3 - Write a value to pp_offset

Figure 108:
SEQ1_SUB34_PP Register

0x39	SEQ1_SUB34_PP			
ID	Bit Name	Default	Access	Description
7:6	MOD1_SEQ1_SUB3_PP	0	RW	Post Processing in Modulator1 for SubSample3 in Sequence1. Number 0 – Normal value Number 1 – Inverted value Number 2 – Value pp_offset Number 3 - Write a value to pp_offset
5:4	MOD2_SEQ1_SUB3_PP	0	RW	Post Processing in Modulator2 for SubSample3 in Sequence1. Number 0 – Normal value Number 1 – Inverted value Number 2 – Value pp_offset Number 3 - Write a value to pp_offset
3:2	MOD1_SEQ1_SUB4_PP	0	RW	Post Processing in Modulator1 for SubSample4 in Sequence1. Number 0 – Normal value Number 1 – Inverted value Number 2 – Value pp_offset Number 3 - Write a value to pp_offset
1:0	MOD2_SEQ1_SUB4_PP	0	RW	Post Processing in Modulator2 for SubSample4 in Sequence1. Number 0 – Normal value Number 1 – Inverted value Number 2 – Value pp_offset Number 3 - Write a value to pp_offset

Figure 109:
SEQ1_SUB56_PP Register

0x3A	SEQ1_SUB56_PP			
ID	Bit Name	Default	Access	Description
7:6	MOD1_SEQ1_SUB5_PP	0	RW	Post Processing in Modulator1 for SubSample5 in Sequence1.

0x3A	SEQ1_SUB56_PP			
ID	Bit Name	Default	Access	Description
				Number 0 – Normal value Number 1 – Inverted value Number 2 – Value pp_offset Number 3 - Write a value to pp_offset
5:4	MOD2_SEQ1_SUB5_PP	0	RW	Post Processing in Modulator2 for SubSample5 in Sequence1. Number 0 – Normal value Number 1 – Inverted value Number 2 – Value pp_offset Number 3 - Write a value to pp_offset
3:2	MOD1_SEQ1_SUB6_PP	0	RW	Post Processing in Modulator1 for SubSample6 in Sequence1. Number 0 – Normal value Number 1 – Inverted value Number 2 – Value pp_offset Number 3 - Write a value to pp_offset
1:0	MOD2_SEQ1_SUB6_PP	0	RW	Post Processing in Modulator2 for SubSample6 in Sequence1. Number 0 – Normal value Number 1 – Inverted value Number 2 – Value pp_offset Number 3 - Write a value to pp_offset

Figure 110:
SEQ1_SUB78_PP Register

0x3B	SEQ1_SUB78_PP			
ID	Bit Name	Default	Access	Description
7:6	MOD1_SEQ1_SUB7_PP	0	RW	Post Processing in Modulator1 for SubSample7 in Sequence1. Number 0 – Normal value Number 1 – Inverted value Number 2 – Value pp_offset Number 3 - Write a value to pp_offset
5:4	MOD2_SEQ1_SUB7_PP	0	RW	Post Processing in Modulator2 for SubSample7 in Sequence1. Number 0 – Normal value Number 1 – Inverted value Number 2 – Value pp_offset Number 3 - Write a value to pp_offset
3:2	MOD1_SEQ1_SUB8_PP	0	RW	Post Processing in Modulator1 for SubSample8 in Sequence1. Number 0 – Normal value Number 1 – Inverted value Number 2 – Value pp_offset Number 3 - Write value to pp_offset
1:0	MOD2_SEQ1_SUB8_PP	0	RW	Post Processing in Modulator2 for SubSample8 in Sequence1.

0x3B	SEQ1_SUB78_PP			
ID	Bit Name	Default	Access	Description
				Number 0 – Normal value
				Number 1 – Inverted value
				Number 2 – Value pp_offset
				Number 3 - Write a value to pp_offset

Post-Processing - Sequence 2

Each subsample uses, independently of one another, one of the following functions for post-processing:

- 2'b00 – Transmission of the measured value -> Normal value
- 2'b01 – Transmission of the inverted measured value -> Invert value
- 2'b10 – Transfer of the result = measured value - PP offset -> Value – pp_offset
- 2'b11 – Writing the measured value in the PD offset and transmission it -> Value to pp_offset

Modulator 1 and Modulator 2 use the same setting for post-processing.

Figure 111:
SEQ2_SUB12_PP Register

0x3C	SEQ2_SUB12_PP			
ID	Bit Name	Default	Access	Description
7:6	MOD1_SEQ2_SUB1_PP	0	RW	Post Processing in Modulator1 for SubSample1 in Sequence2. Number 0 – Normal value Number 1 – Inverted value Number 2 – Value pp_offset Number 3 - Write a value to pp_offset
5:4	MOD2_SEQ2_SUB1_PP	0	RW	Post Processing in Modulator2 for SubSample1 in Sequence2. Number 0 – Normal value Number 1 – Inverted value Number 2 – Value pp_offset Number 3 - Write a value to pp_offset
3:2	MOD1_SEQ2_SUB2_PP	0	RW	Post Processing in Modulator1 for SubSample2 in Sequence2. Number 0 – Normal value Number 1 – Inverted value Number 2 – Value pp_offset Number 3 - Write value to pp_offset
1:0	MOD2_SEQ2_SUB2_PP	0	RW	Post Processing in Modulator2 for SubSample2 in Sequence2. Number 0 – Normal value Number 1 – Inverted value Number 2 – Value pp_offset Number 3 - Write a value to pp_offset

Figure 112:
SEQ2_SUB34_PP Register

0x3D		SEQ2_SUB34_PP		
ID	Bit Name	Default	Access	Description
7:6	MOD1_SEQ2_SUB3_PP	0	RW	Post Processing in Modulator1 for SubSample3 in Sequence2. Number 0 – Normal value Number 1 – Inverted value Number 2 – Value pp_offset Number 3 – Write a value to pp_offset
5:4	MOD2_SEQ2_SUB3_PP	0	RW	Post Processing in Modulator2 for SubSample3 in Sequence2. Number 0 – Normal value Number 1 – Inverted value Number 2 – Value pp_offset Number 3 – Write a value to pp_offset
3:2	MOD1_SEQ2_SUB4_PP	0	RW	Post Processing in Modulator1 for SubSample4 in Sequence2. Number 0 – Normal value Number 1 – Inverted value Number 2 – Value pp_offset Number 3 – Write a value to pp_offset
1:0	MOD2_SEQ2_SUB4_PP	0	RW	Post Processing in Modulator2 for SubSample4 in Sequence2. Number 0 – Normal value Number 1 – Inverted value Number 2 – Value pp_offset Number 3 – Write a value to pp_offset

6.1.10 Proximity Configuration

The Proximity output is an additional signal to the two PPG signal outputs. The integrated proximity detection, with separate programmable, has high and low threshold values, in a way that the high threshold is more than the low threshold.

Figure 113:
Proximity Configuration

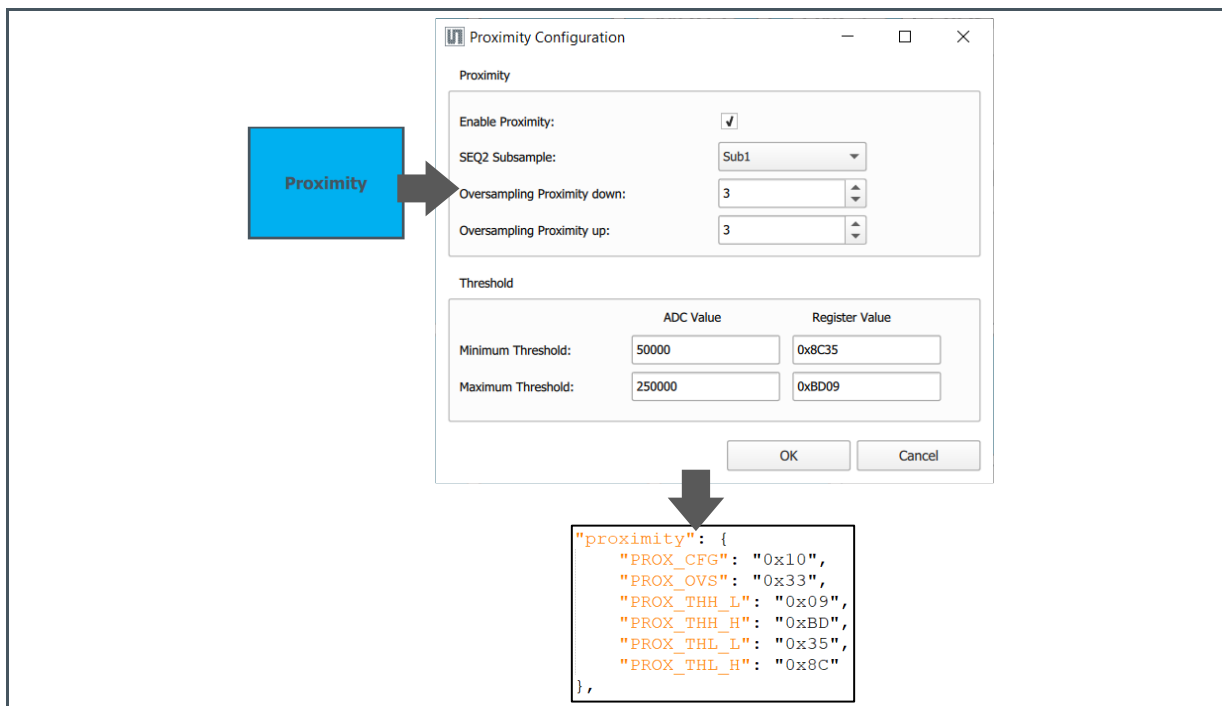
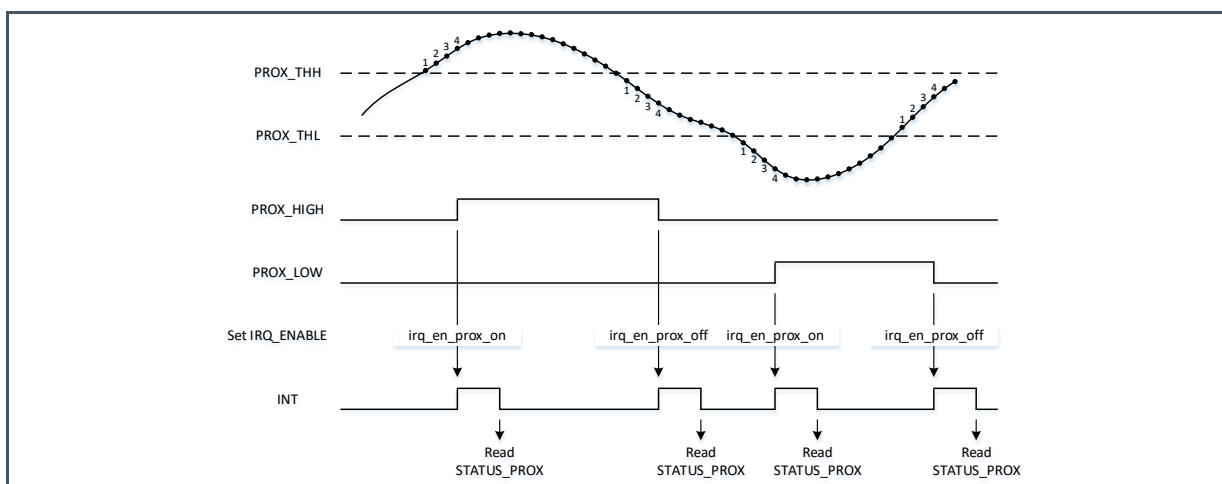


Figure 114:
Proximity Detection



Proximity

A subsample from Sequence2 for Modulator1 can be used for the proximity function.

Figure 115:
PROX_CFG Register

0x98	PROX_CFG			
ID	Bit Name	Default	Access	Description
4	PROX_EN	0	RW	Enable Proximity
1:0	PROX_SUB	0	RW	Select SubSample for Proximity; MOD1_SEQ2: 0-3: SUB1-SUB4

Figure 116:
PROX_OVS Register

0x99	PROX_OVS			
ID	Bit Name	Default	Access	Description
6.4	PROX_ON_OVS	0	RW	Oversampling for Proximity up.
2:0	PROX_OFF_OVS	0	RW	Oversampling for Proximity down.

Threshold

The minimum and maximum threshold for proximity with a value range of $[-2^{19}: 2^{19} - 1]$:

The minimum threshold value will be mapped as a 16-bit unsigned value into the following registers:

[15:8] Reg: 0x9D (PROX_THL_H) | Bit 7:0

[7:0] Reg: 0x9C (PROX_THL_L) | Bit 7:0

The maximum threshold value will be mapped as a 16-bit unsigned value into the following registers:

[15:8] Reg: 0x9B (PROX_THL_H) | Bit 7:0

[7:0] Reg: 0x9A (PROX_THL_L) | Bit 7:0

For calculating the register values,

1. The signed value will first be added with 2^{19} .
2. Secondly, the value will be shifted to 4 bits to the right.
3. The calculated 16-bit value will be mapped into both registers.

Figure 117:
PROX_THH_L Register

0x9A		PROX_THH_L		
ID	Bit Name	Default	Access	Description
7:0	PROX_THH [7:0]	0	RW	Threshold High for Proximity.

Figure 118:
PROX_THH_H Register

0x9B		PROX_THH_H		
ID	Bit Name	Default	Access	Description
7:0	PROX_THH [15:8]	0	RW	Threshold High for Proximity.

Figure 119:
PROX_THH_L Register

0x9C		PROX_THH_L		
ID	Bit Name	Default	Access	Description
7:0	PROX_THL [7:0]	0	RW	Threshold Low for Proximity.

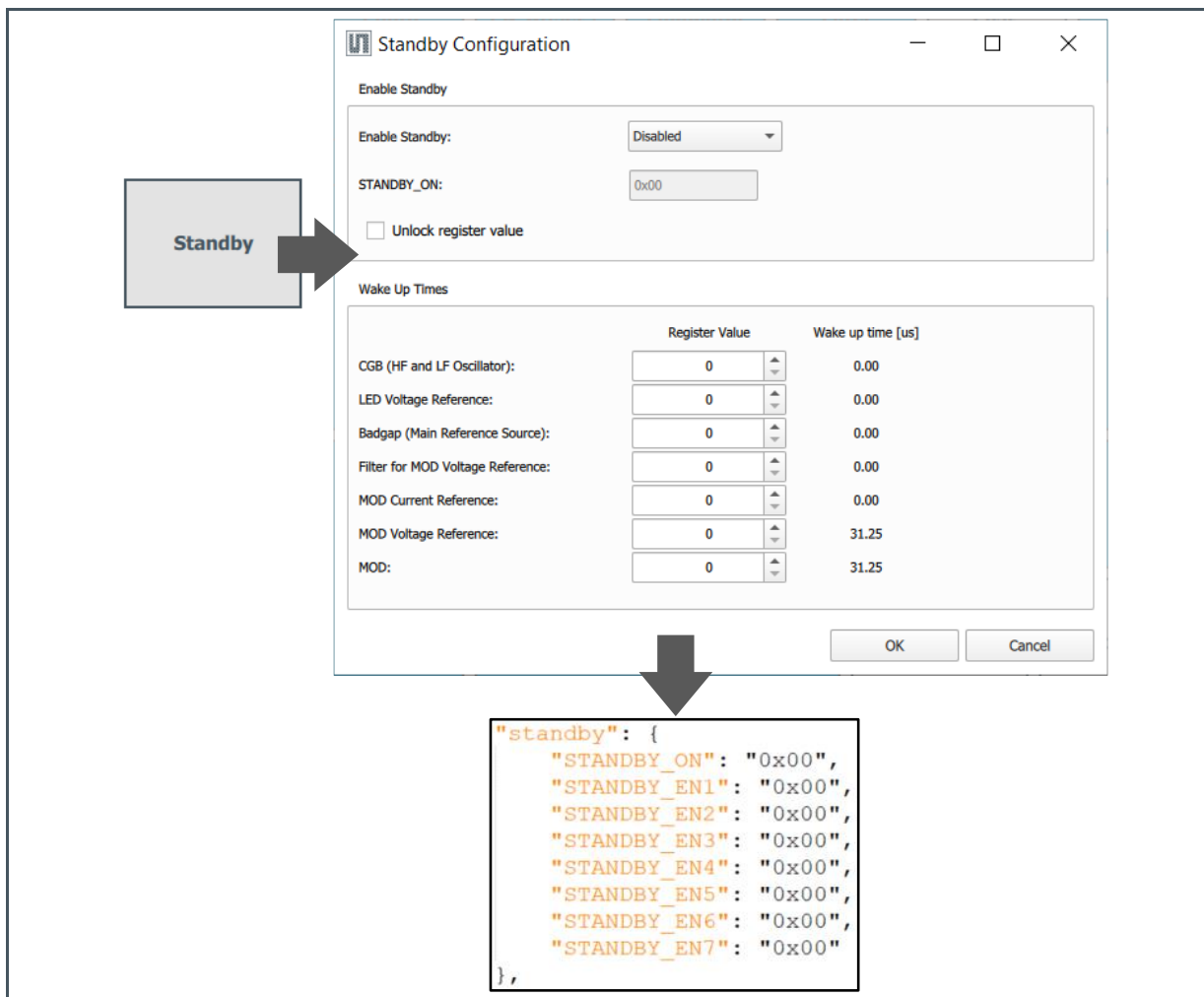
Figure 120:
PROX_THH_H Register

0x9D		PROX_THH_H		
ID	Bit Name	Default	Access	Description
7:0	PROX_THL [15:8]	0	RW	Threshold Low for Proximity.

6.1.11 Standby Configuration

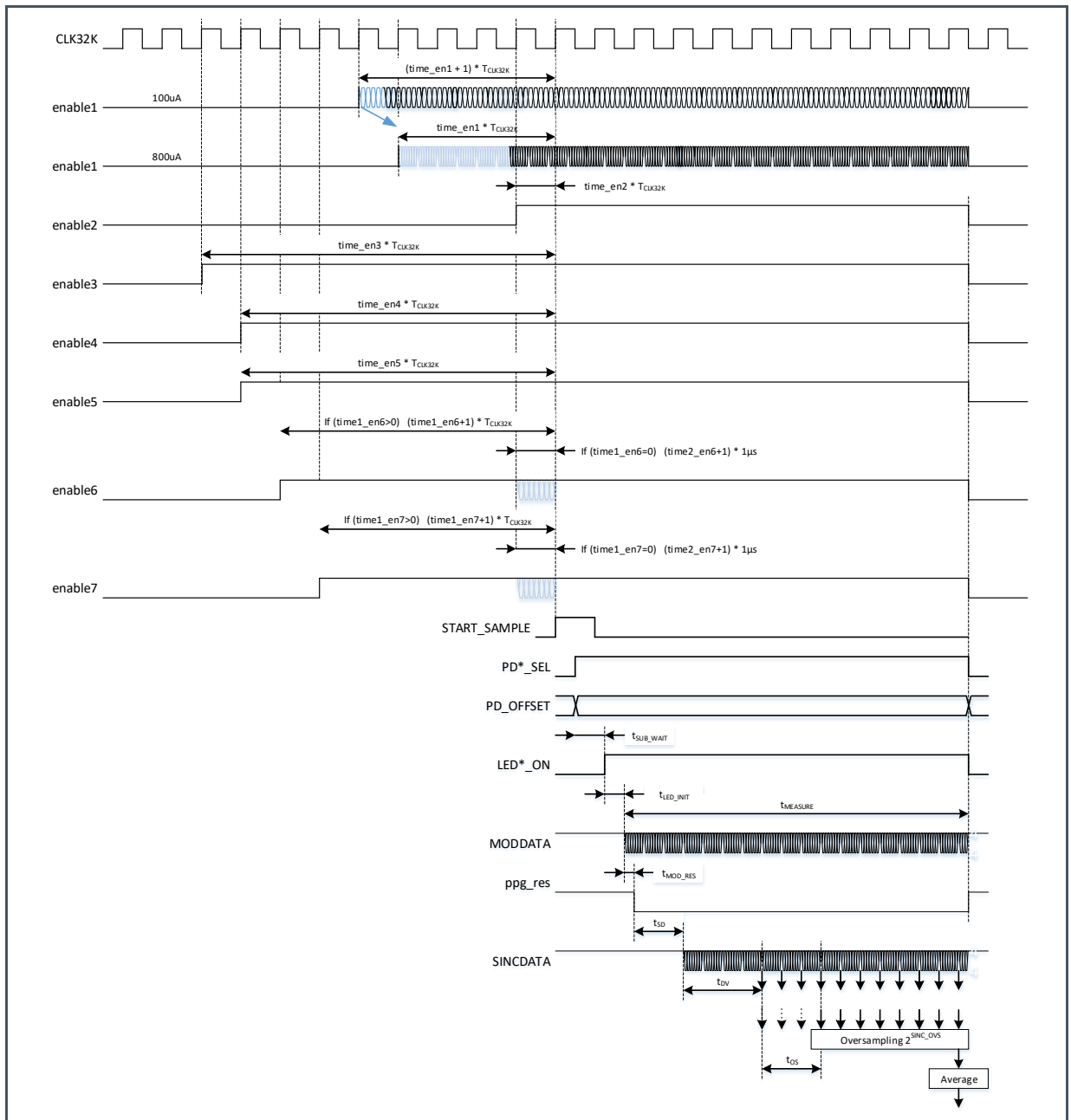
The power management configuration of the AS7056/57 can be set in the standby configuration.

Figure 121:
Standby Configuration



The standby registers are part of the AS7056/57 power management to reduce the power consumption of the AS7056/57 and the measurement system itself. Also, during continuous measurements, blocks such as ADCs, clock generation, or reference block can switch to standby mode between several PPG sample slots if they are inactive. For usage of the standby functionality, the dedicated registers have to be activated. In the standard default setting, the automatic standby is disabled.

Figure 122:
Standby Time Behavior



Enable Standby

When the STANDBY_ON is set to 0x7F, i.e. all to one, all the enabled signals are reset to zero, except byp_ref_lp (REF_CFGA, 0x15), which is set to 1 and can be sequentially turned on, but not all at once. These enabled signals will be turned on with the configured stby_en (standby enable) time.

Figure 123:
STANDBY_ON Register

0xA0	STANDBY_ON			
ID	Bit Name	Default	Access	Description
6:0	STBY_EN_ON	0	RW	Set on Standby Enable 1...7

Wake Up Times

The STANDBY_ENx registers turn on the corresponding enable signal after a configured amount of time. This time is given as the STANDBY_ENx register value times 31.25µs (the low-frequency clock period i.e. 32 kHz). However, to enable STANDBY_EN6 and STANDBY_EN7, 32 kHz or the 1MHz clock, dictates the delay time before the enable signal is turned on. Bit 7:5 for these two registers allow the delay to be set by the 32 kHz clock, while bit 4:0 allows the delay to be set by the 1MHz clock. The pattern in which the delay is set is the same as described above for the other registers, i.e. the decimal value of bits 7:5 times 31.25 µs, or the decimal value of bits 4:0 times 1µs. The default values of STANDBY_ENx registers are shown in the tables below.

Figure 124:
STANDBY_EN1 Register

0xA1	STANDBY_EN1			
ID	Bit Name	Default	Access	Description
7:0	STBY_EN1_TIME	4	RW	Selecting the wake-up time for CGB (HF and LF Oscillator). Time for Enable1: N * 31.25 µs OR N * TCLK_32kHz

Figure 125:
STANDBY_EN2 Register

0xA2	STANDBY_EN2			
ID	Bit Name	Default	Access	Description
7:0	STBY_EN2_TIME	2	RW	Selecting the wake-up time for the LED Voltage Reference. Time for Enable2: N * 31.25 µs OR N * TCLK_32kHz

Figure 126:
STANDBY_EN3 Register

0xA3	STANDBY_EN3			
ID	Bit Name	Default	Access	Description
7:0	STBY_EN3_TIME	4	RW	Selecting the wake-up time for the Bandgap (Main Reference Source). Time for Enable3: $N * 31.25\mu s$ OR $N * TCLK_32kHz$

Figure 127:
STANDBY_EN4 Register

0xA4	STANDBY_EN4			
ID	Bit Name	Default	Access	Description
7:0	STBY_EN4_TIME	0	RW	Selecting the wake-up time for Filter for MOD Voltage Reference. Time for Enable4: $N * 31.25\mu s$ OR $N * TCLK_32kHz$

Figure 128:
STANDBY_EN5 Register

0xA5	STANDBY_EN5			
ID	Bit Name	Default	Access	Description
7:0	STBY_EN5_TIME	3	RW	Selecting the wake-up time for the MOD Current Reference. Time for Enable5: $N * 31.25\mu s$ OR $N * TCLK_32kHz$

Figure 129:
STANDBY_EN6 Register

0xA6	STANDBY_EN6			
ID	Bit Name	Default	Access	Description
7:5	STBY_EN6_TIME1	0	RW	Time1 for Enable6: $N * TCLK_32kHz$
4:0	STBY_EN6_TIME2	16	RW	Time2 for Enable6: $N * TCLK_2MHz$

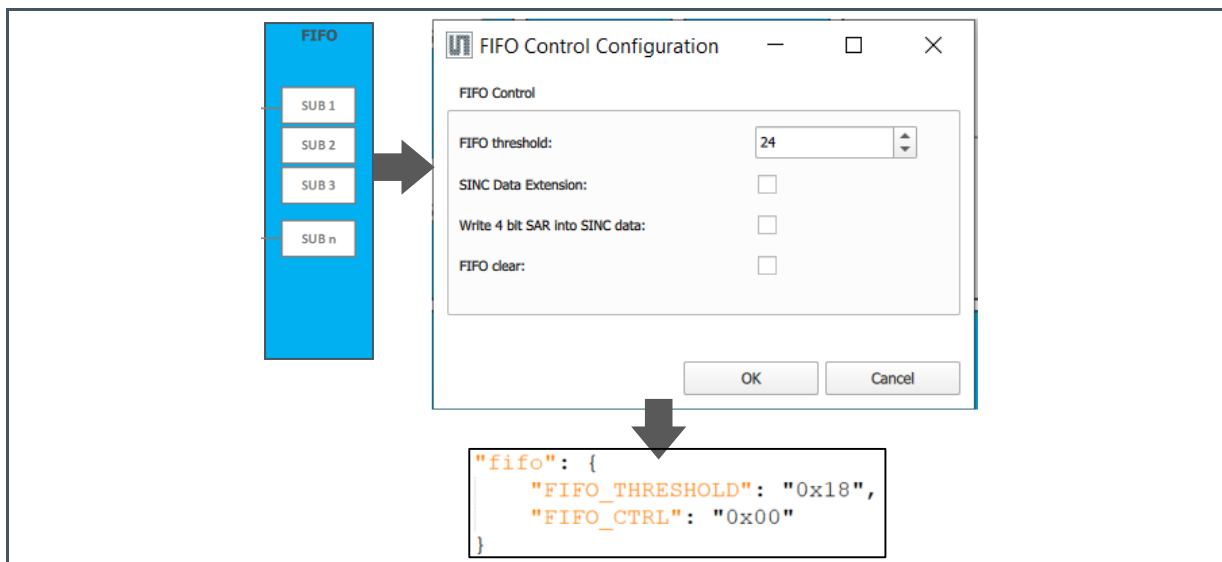
Figure 130:
STANDBY_EN7 Register

0xA7	STANDBY_EN7			
ID	Bit Name	Default	Access	Description
7:5	STBY_EN7_TIME1	0	RW	Time1 for Enable7: $N * TCLK_32kHz$
4:0	STBY_EN7_TIME2	16	RW	Time2 for Enable7: $N * TCLK_2MHz$

6.1.12 FIFO Control Configuration

The AS7056/57 provides a 1.5 kbyte FIFO which configuration could be adjusted in this section.

Figure 131:
FIFO Control Configuration



The measurement data are saved in the FIFO register for the data output to an external microcontroller via the I²C interface. The FIFO buffering allows the external MCU to stay in idle mode during the measurements for power saving.

FIFO Control

Figure 132:
FIFO_THRESHOLD Register

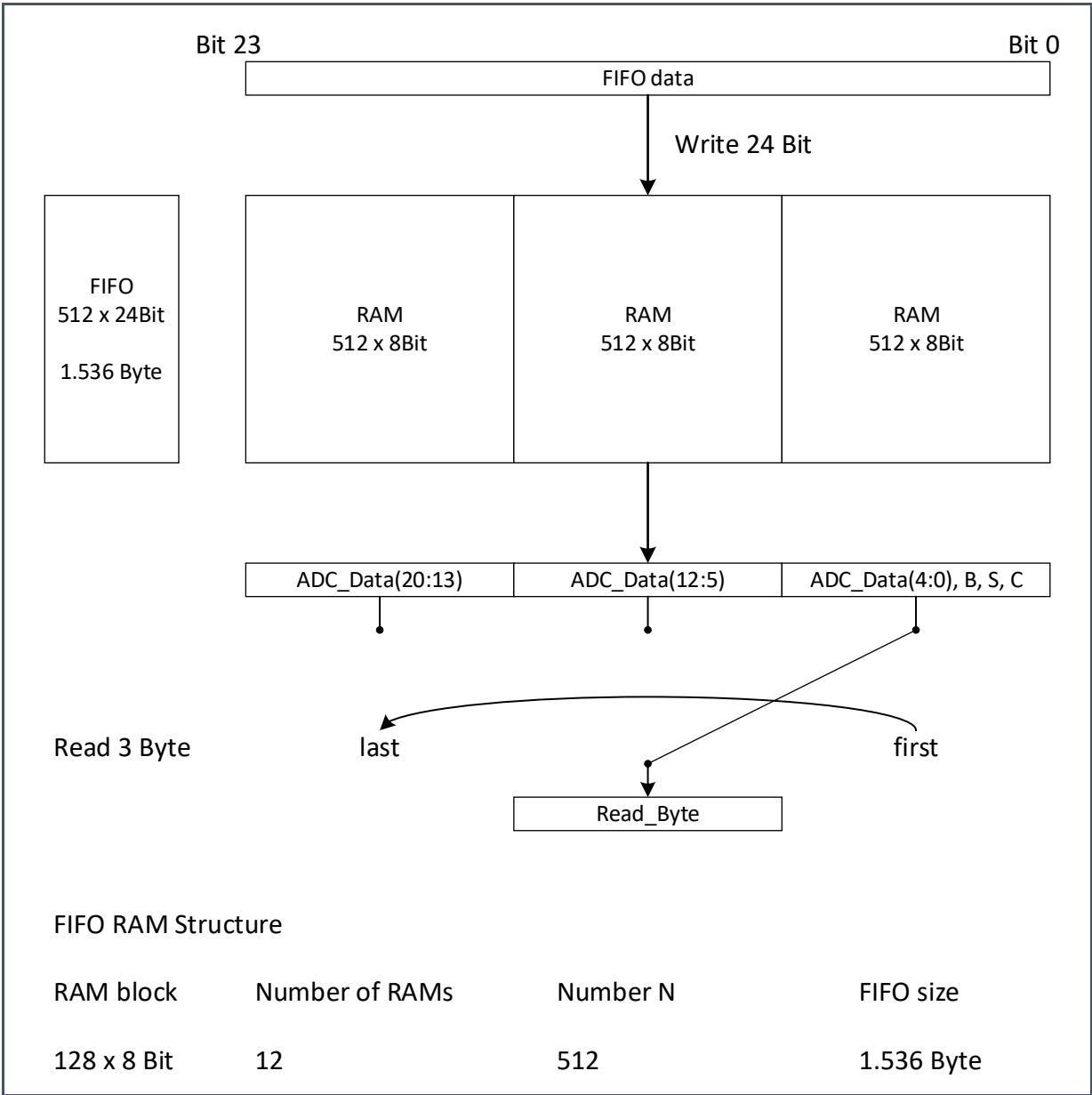
0xD0	FIFO_THRESHOLD			
ID	Bit Name	Default	Access	Description
7:0	FIFO_THRESHOLD [7:0]	64	RW	FIFO threshold Bit 7:0

Figure 133:
FIFO_CTRL Register

0xD1	FIFO_CTRL			
ID	Bit Name	Default	Access	Description
7	FIFO_CLEAR	0	PUSH	Write 1 here to clear the FIFO.
3	SAR_DATA_EN	0	RW	Write 4 Bit SAR into SINC data Bit (3:1).

0xD1 FIFO_CTRL				
ID	Bit Name	Default	Access	Description
1	FIFO_THRESHOLD [8]	0	RW	FIFO threshold Bit 8.

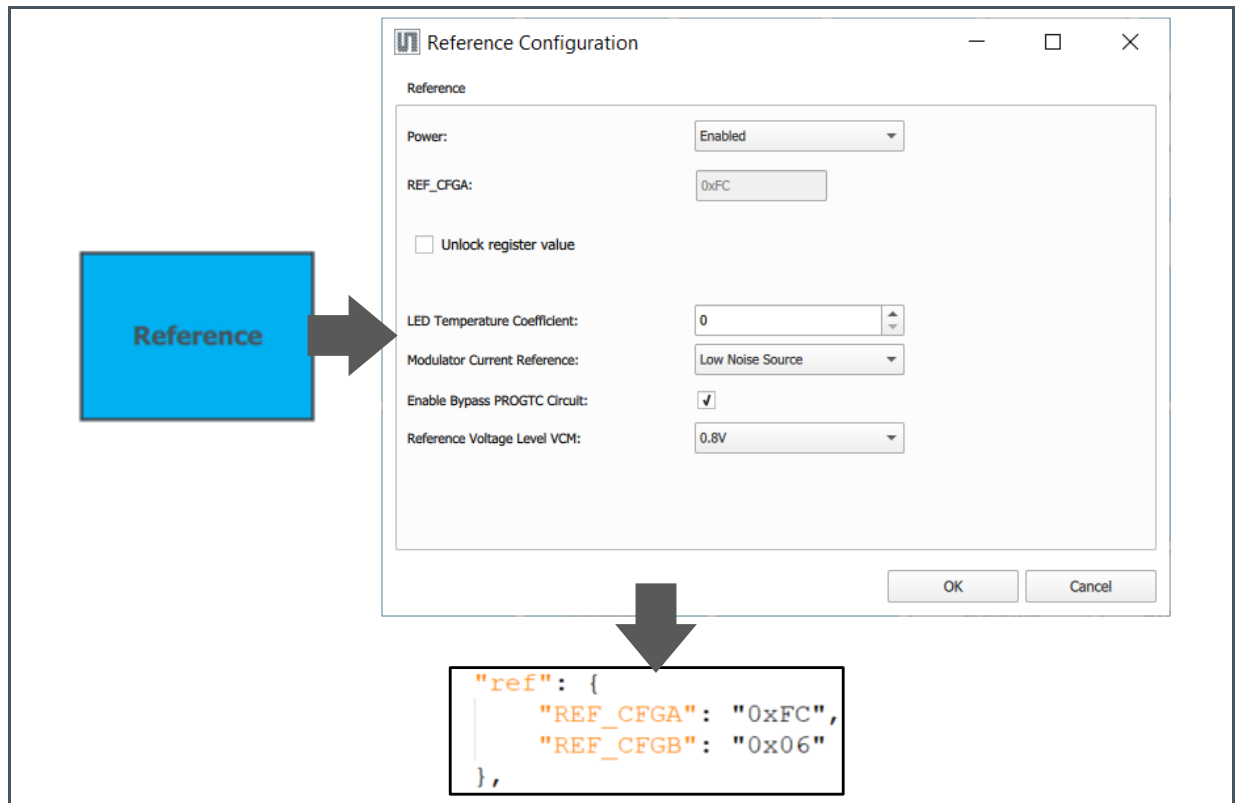
Figure 134:
FIFO Data Structure and Organization



6.1.13 Reference Configuration

The configuration of the reference voltage/current could be set in the reference configuration.

Figure 135:
Reference Configuration



The Reference Block provides voltage and current references for all the blocks inside the analog part and the LED/VCSEL drivers. It consists of a low voltage bandgap reference, which provides trim able 0.9 V and 0.6 V voltages, a set of buffers for delivering references voltages to the PPG channels and LED/VCSEL drivers, and a current reference circuit.

Reference

Figure 136:
REF_CFGA Register

0x15 REF_CFGA				
ID	Bit Name	Default	Access	Description
7	EN_BG	0	RW	Enable Bandgap; The bandgap is the main Voltage reference that is used to generate all internal voltages and currents. It must be enabled for the modulator measurement and disabled in standby mode. Bit 0 – Disable Bit 1 – Enable
6	SEL_IN_iLED	1	RW	Select a low noise source for the LED reference current. Bit 0 – Regular source selected Bit 1 – Low noise source selected
5	EN_VCM_PPG	0	RW	Enable VCM PPG Buffer; Voltage reference for ADC enabled before the measurement and disabled in standby mode. Bit 0 – Disable Bit 1 – Enable
4	SEL_STRTUP	0	RW	Select the Bandgap Start-up Circuit. Bit 0 – New start-up circuit Bit 1 – Old start-up circuit
3	EN_VR_LED	0	RW	Enable the LED Buffer; Voltage reference for the LED driver. Bit 0 - Disable Bit 1 – Enable
2	EN_BIAS	0	RW	Enable IREF_DAC and IOS_DAC; Current bias reference for the MOD operation. Bit 0 - Disable Bit 1 – Enable
1	EN_PTAT	0	RW	Select the tail current of the VCM buffers for power supply optimization. Bit 0 – 2 mA Bit 1 – Reserved
0	BYP_REF_LP	0	RW	The bypass of a low-pass filter of the bandgap voltage for a faster startup. Bit 0 – Not bypassed Bit 1 – Bypassed

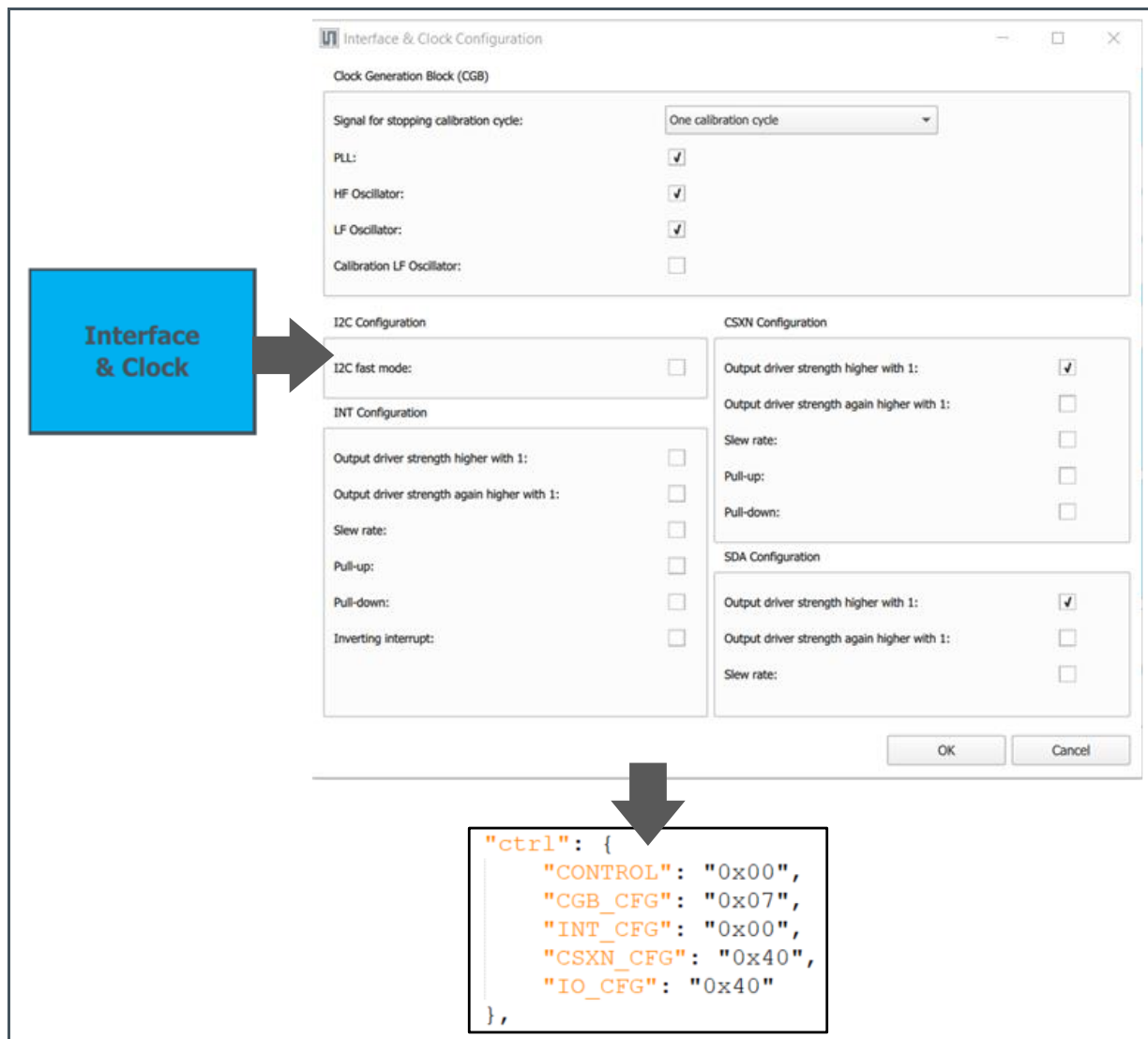
Figure 137:
REF_CFGB Register

0x16		REF_CFGB		
ID	Bit Name	Default	Access	Description
7:3	iLED_TC	0	RW	Temperature coefficient of the LED current.
2	SEL_IREF_IN	1	RW	Select a low noise current reference for PPG IREF_DAC and IOS_DAC. Bit 0 - Regular source selected Bit 1 - Low noise source selected
1	BYP_PROGTC	1	RW	The bypass TC selection circuit for the LEC current. Bit 0 – Not bypassed Bit 1 – Bypassed
0	SEL_REF	0	RW	Select the reference voltage level for VCM; Voltage reference for ADC. All signals are centered around this voltage. Bit 0 - 0.8 V Bit 1 - 0.75 V

6.1.14 Interface & Clock Configuration

All basic clock configurations could be adjusted in the interface & clock configuration section.

Figure 138:
Interface & Clock Configuration



Clock Generation Block

The clock reference block provides three clock frequencies needed for the system: 32 kHz, 2 MHz, and 20 MHz.

The 32 kHz low-frequency clock is always active to enable low power consumption over the whole time. That clock is needed by the digital part to generate the time basis for the sampling windows.

The low-frequency clock is recalibrated from time to time to achieve the needed accuracy. The calibration is done in the clock generation block itself.

The 2 MHz high-frequency oscillator is used as a clock basis for the PLL and recalibrates the low-frequency clock. The activity is controlled by the digital core.

The 20 MHz PLL clock is used for the modulators and is the highest frequency available in the system. It runs together with the high-frequency oscillator and is enabled on-demand by the digital core.

Figure 139:
CGB_CFG Register

0x11	CGB_CFG			
ID	Bit Name	Default	Access	Description
7	BGCAL_DONE_SEL	0	RW	Select which signal is used to stop the calibration. Bit 0 - Run only one calibration cycle Bit 1 - Run calibration until LF oscillator is close to 32 kHz
6	BGCAL_EN	0	PUSH1	Start calibration of the LF oscillator.
2	PLL_ON	0	RW	Turn the pll to 20 MHz; needs to be enabled after hf_osc_en (min. 30 μ s). Bit 1 – On Bit 0 – Off
1	HF_OSC_ON	0	RW	Turn the hf oscillator to 2 MHz. 0: needs to be enabled after lf_osc_en (min. 100 μ s). Bit 1 – On Bit 0 - Off
0	LF_OSC_ON	0	RW	Turn the lf oscillator to 32 kHz. Bit 1 – On Bit 0 – Off

I²C Configuration

I²C Fast Mode: Enable I²C fast mode with 1 MHz. By enabling this function, the internal clock will be switched to 1 MHz from the standard 100 kHz or the fast mode, 400 kHz.

Figure 140:
CONTROL Register

0x10	CONTROL			
ID	Bit Name	Default	Access	Description
0	I2C_FM_PLUS	0	RW	I ² C fastmode plus with 1 MHz. Bit 1 – On Bit 0 – Off

INT Configuration

The interrupt events are processed by the interrupt manager. These interrupt events must be released for processing via an interrupt enable register.

Figure 141:
INT_CFG Register

0x12	INT_CFG			
ID	Bit Name	Default	Access	Description
6	INT_E2	0	RW	Set the output driver strength high with 1.
5	INT_E4	0	RW	Set output driver strength high again with 1.
4	INT_SR	0	RW	Set the slew rate to another value with 1.
3	INT_PU	0	RW	PU=1, pull-up if IO is activated.
2	INT_PD	0	RW	PD=1, pull-down if IO is activated.
0	INT_INV	0	RW	Inverting interrupt.

CSXN Configuration

Figure 142:
CSXN_CFG Register

0x13	CSXN_CFG			
ID	Bit Name	Default	Access	Description
6	CSXN_E2	0	RW	Set the output driver strength high with 1.
5	CSXN_E4	1	RW	Set the output driver strength high again with 1.
4	CSXN_SR	0	RW	Set the slew rate to another value with 1.
3	CSXN_PU	0	RW	PU=1, pull-up if IO is activated.
2	CSXN_PD	0	RW	PD=1, pull-down if IO is activated.

SDA Configuration

Figure 143:
SDA_CFG Register

0x14	SDA_CFG			
ID	Bit Name	Default	Access	Description
6	SDA_E2	0	RW	Set the output driver strength high with 1.
5	SDA_E4	1	RW	Set the output driver strength high again with 1.
4	SDA_SR	0	RW	Set the slew rate to another value with 1.

6.2 Overview Tab

Figure 144:
Overview Tab



The "Overview" tab gives a quick overview of the measurement configuration and shows which modulator is used, which sequencer is used, which subsamples are used, which LED current has been selected, which PDs have been selected, what is ADC full-scale State, which SINC filter is selected, i.e. all necessary parameters are quickly visible. This is equivalent to looking at each part of the "Configuration" tab. All measurement configurations or IC settings can be quickly surveyed with the "Overview" tab.

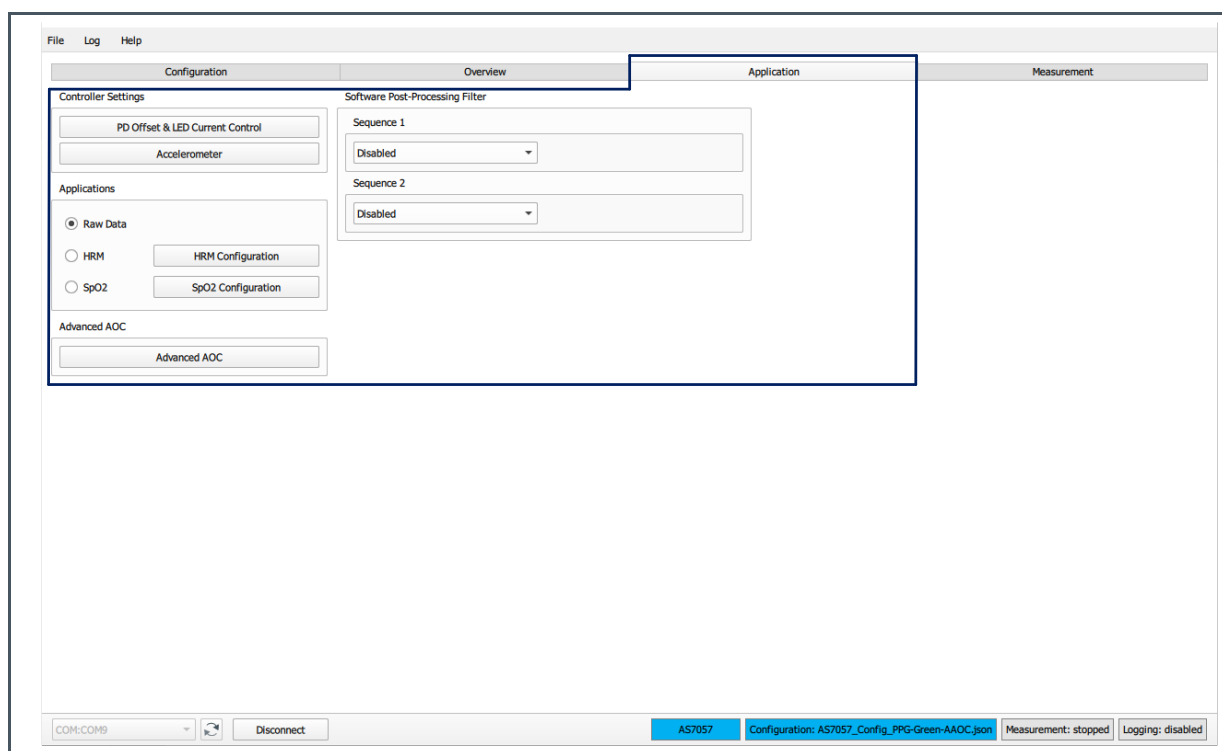
6.3 Application Tab

Automatic gain control (AGC) implementation and vital signs application-related parameters are seen under the “Application” tab.

This block has the AGC Configuration that controls the PD offset and LED current, as well as algorithms such as HRM and SpO₂. The parameters related to these applications can be checked and adjusted individually under these tabs.

Different applications can be enabled, like Raw Data, HRM, or SpO₂. The Accelerometer (ACC) can also be enabled or disabled.

Figure 145:
Application Tab Overview



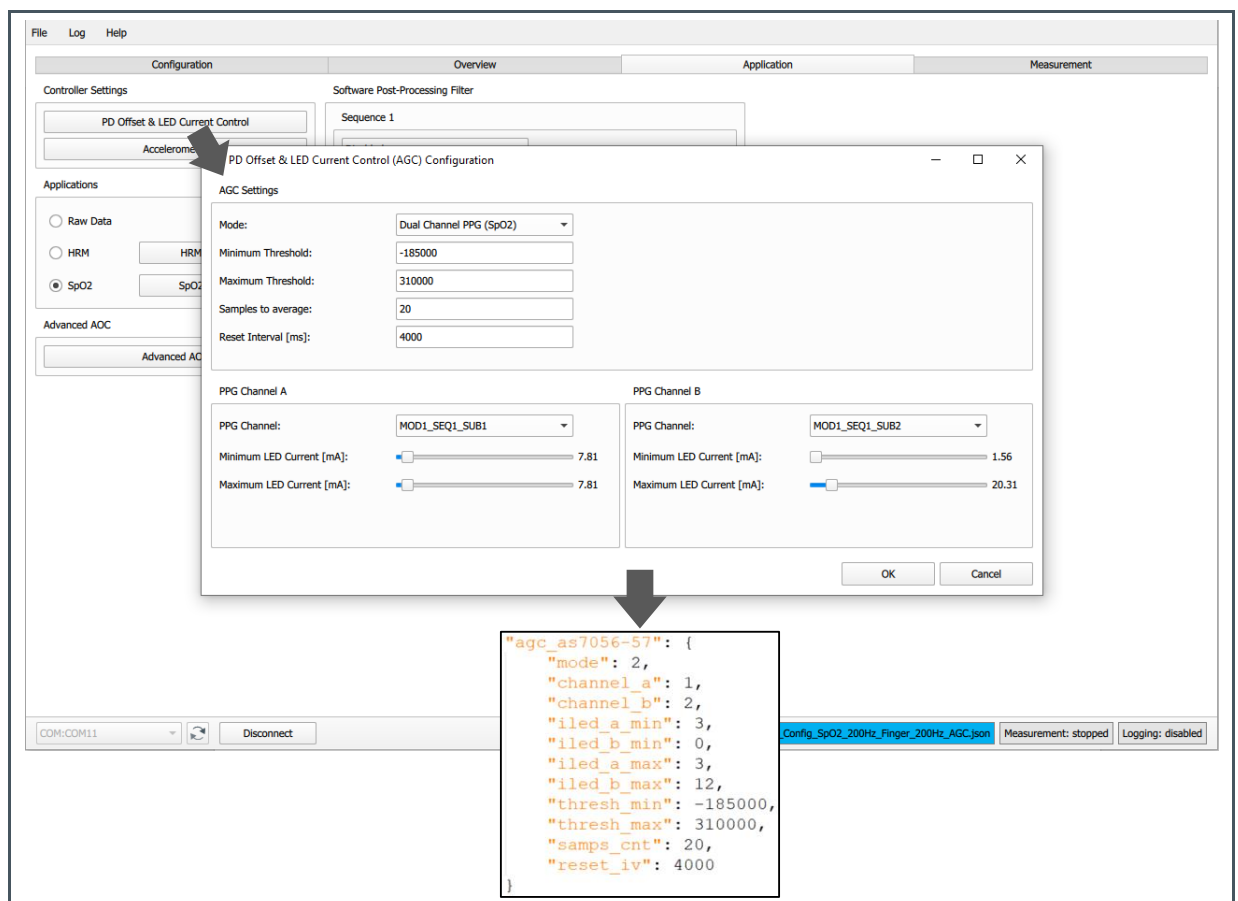
6.3.1 Controller Settings

In this block, the AGC and Accelerometer configuration can be operated. The minimum/maximum ADC count, minimum/maximum LED current, and accelerometer sampling frequency must be set for HRM and SpO₂ applications.

PD Offset & LED Current Control (AGC) Configuration

The configuration of the software-based automatic gain control (AGC) could be set in the PD Offset & LED current section.

Figure 146:
AGC Configuration



The “PD offset & LED current” control is an algorithmic approach to increase the signal quality of the PPG signals. The algorithm continuously monitors PPGn outputs and, if necessary, reconfigures the AS7056/57 while measuring to ensure ideal conditions.



Information

Please do not use enabled AGC and enabled AOC together.

AGC Settings:

Mode: AGC Algorithms to be applied to one PPG channel (HRM fast) or Dual channel PPG (SpO₂) or the AGC algorithm can be disabled.

Minimum Threshold: The minimum threshold of the controlled signal in ADC counts.

Maximum Threshold: The maximum threshold of the controlled signal in ADC counts.

Samples to average: The number of samples to the average, for calculating the signal mean.

Reset Interval [ms]: The interval, in milliseconds, to reset temporary AGC parameters like min/max peak signals.

PPG Channel A & B:

PPG Channel A: The PPG Signal is assigned to the first AGC channel and it is required for one and dual-channel AGC.

PPG Channel B: The PPG Signal is assigned to the second AGC channel and it is required for dual-channel AGC.

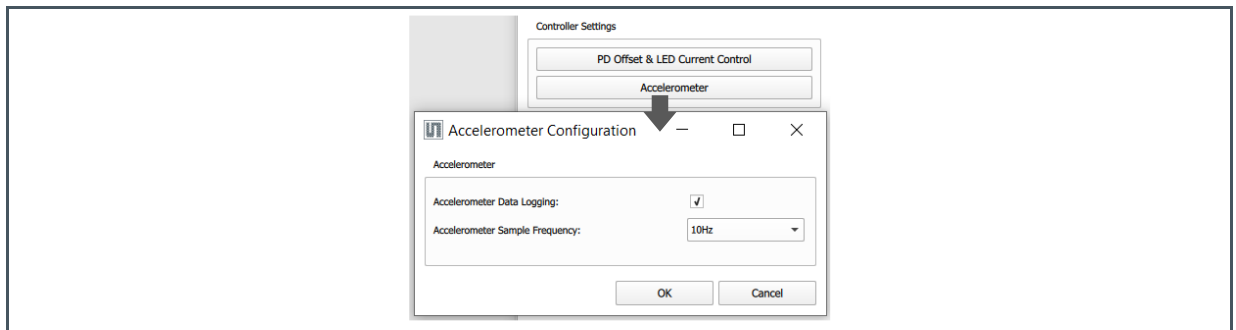
PPG Channel Selection: The PPG channel from MOD_x_SEQ_y_SUB_z can be selected, where x = 1 & 2, y = 1 & 2, and z = 1...8.

Minimum and maximum LED current [mA]: This sets the range in which the LED current can move if the PPG channel A or A/B are enabled. If the PPG channel A or A/B is disabled, the LED will have a constant LED current set in the LED Driver Configuration submenu.

Accelerometer

The accelerometer is also activated when the HRM algorithm is selected. However, the accelerometer can be enabled and disabled via the user interface.

Figure 147:
Accelerometer

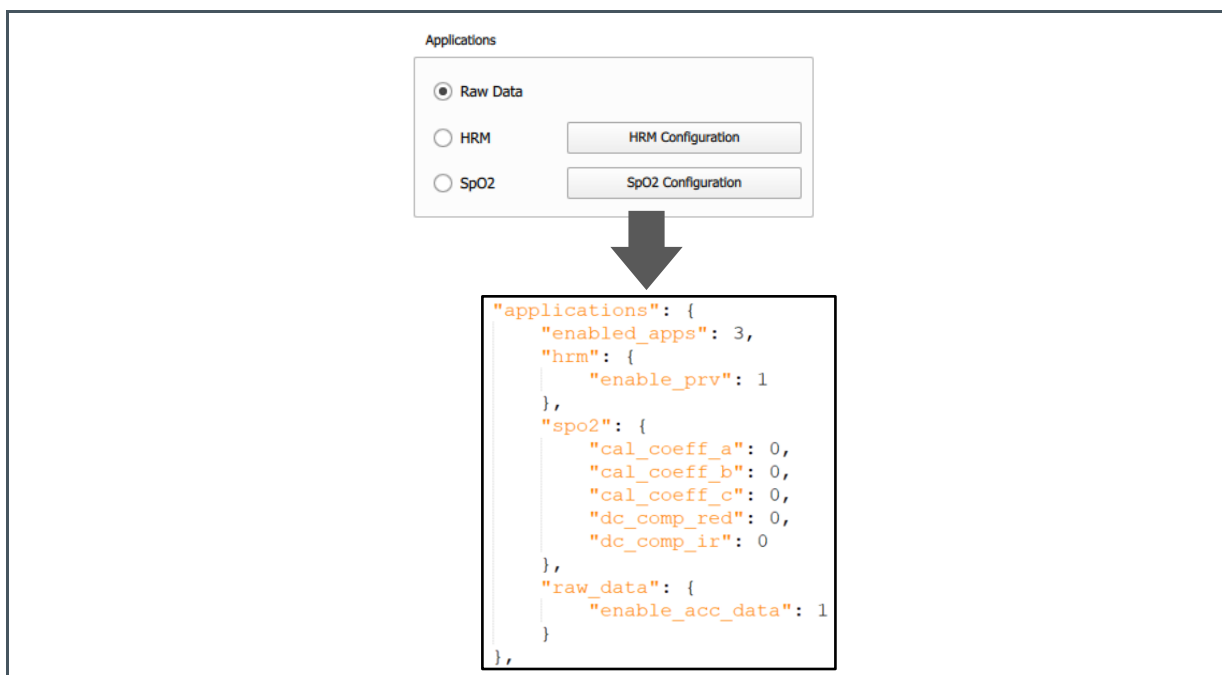


The accelerometer is used for displacement information. The HRM application includes this accelerometer data and makes the accelerometer useful and practical.

6.3.2 Applications

The available applications, also known as bio algorithms, could be configured in the Application section.

Figure 148:
Application Selection

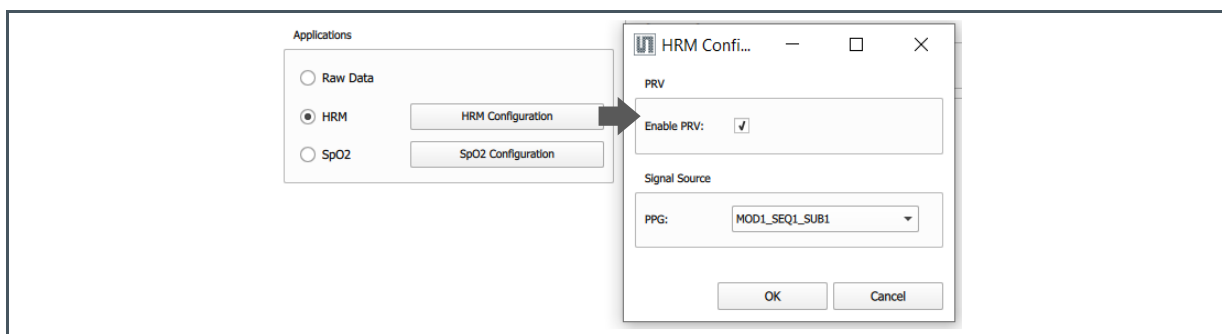


This setting is mainly used to select and change the HRM and SpO₂ applications configuration blocks and parameters.

HRM Configuration

The Signal Routing window, which is a PPG signal source configuration, is used to select the PPG, as shown in Figure 149.

Figure 149:
HRM Configuration

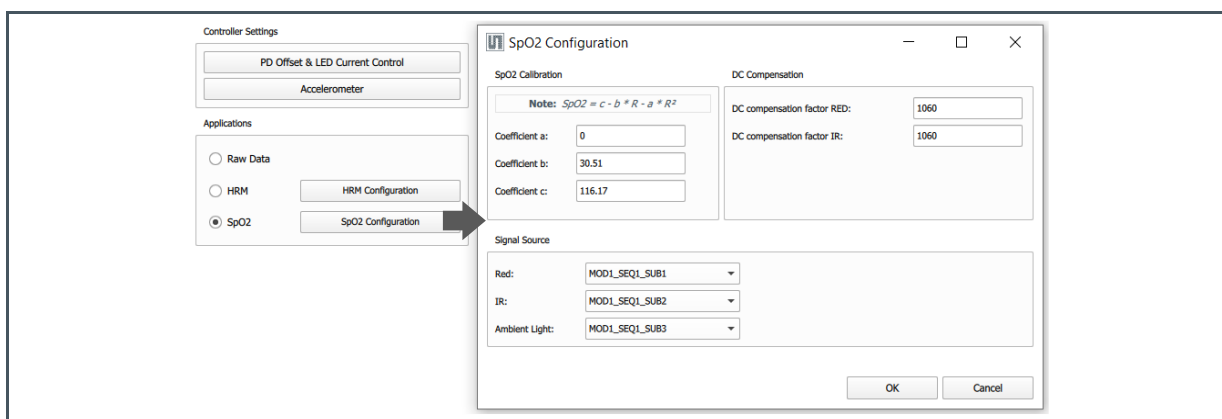


- Pulse Rate Variability (PRV) can be enabled for HRM.
- The PPG channel from MOD_x_SEQ_y_SUB_z can be selected, where x = 1 & 2, y = 1 & 2, and z = 1...8.

SpO₂ Configuration

“SpO₂” is for configuring the SpO₂ Parameters. Enter the parameters associated with the specific settings of the AS7056/57 in the SpO₂ configuration settings. Due to production and assembly tolerances, it is recommended that the below-mentioned factors need to be determined for the device under development; after the sensor has been integrated and the optical configuration is finalized.

Figure 150:
SpO₂ Configuration



SpO₂ Calibration:

The R-value, calculated by the firmware, is used to calculate the oxygen saturation. Different models are proposed in the literature for this purpose. For the AS7056/57, both the quadratic (Equation 3) and the linear model (Equation 4) are implemented:

Equation 3:

$$SpO_2 = c - b * R - a * R^2$$

Equation 4:

$$SpO_2 = -b \cdot R + c$$

DC Compensation:

The resulting factors for the DC compensation, photodiode offset factor RED, and photodiode offset factor IR for the evaluation kit, depend on the different settings of the ADC range, the PD offset range and the PPG clock frequency. These factors, associated with the specific settings of the AS7056/57, must then be entered in the SpO₂ configuration settings.

Signal Source:

The Signal source is for PPG signals for Red, IR, and Ambient Light. The PPG channel from MODx_SEQy_SUBz can be selected where x = 1 & 2, y = 1 & 2, and z = 1...8.

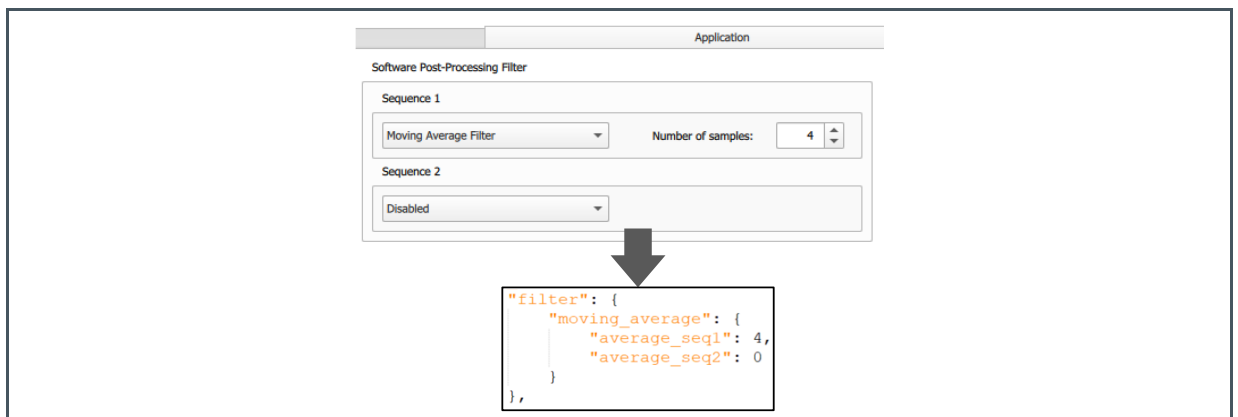
Raw Data

Instead of HRM and SpO₂, Raw data can be selected. The results of an analog to digital converter's (ADC) conversion are represented as a digital number with varying resolution on the bit length and settings used.

Software Post-Processing Filter

A post-processing filter has been implemented to enhance the quality of a signal and reduce the noise levels. A moving average filter, FIR filter, and IIR filter are implemented.

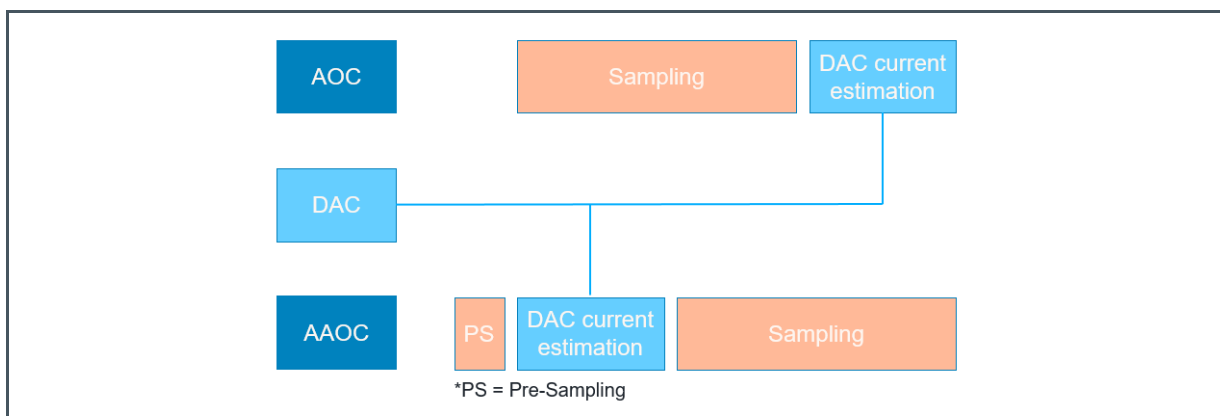
Figure 151:
Software Post-Processing Filter



Advanced AOC

The Feature *Optimal Advanced AOC* (**A**utomatic **O**ffset **C**ontrol) is also called AAOC and is the further development of the previous AOC hardware algorithm. The AAOC is developed to ensure that the PPG signals are placed in the not saturated area of the ADC range.

Figure 152:
AOC vs AAOC



The difference between AAOC and AOC is the point of time of the estimation from the PPG signal value and regulation of the DAC current. While using the AOC, the PPG signal is measured and after that measurement the value of the signal is used to control the DAC current and adjust the Photodiode offset current. Because of this behavior of the AOC, it is possible for a rapid and sharp rise or fall in the PPG signal to result in a saturated signal for at least one or more samples until the DAC current is set to the correct value.

The newly developed AAOC hardware algorithm uses a different behavior to estimate the value of the PPG signal and control the DAC current. The AAOC does a very short pre-measurement to estimate the latest value of the PPG signal, calculates the offset current and adjusts the DAC to use the new value. After the pre-measurement, a single sampling (measurement) is carried out and the measured value is processed as usual. With this new method, it is possible to measure PPG signals without loss of information (saturation), although there is a large influence on the signal, such as a rapidly changing ambient light during movement. If the impact on the signal is greater than the regulation limits, which depend on the sensor configuration (DAC-FSR, ADC-FSR, LED, etc.) the signal will be saturated.

Figure 153:
Overview of the Two Photodiodes Offset Control

	AOC	AAOC
Useable with all sampling/measurement methods	Yes (except AAOC/SAR sampling)	No

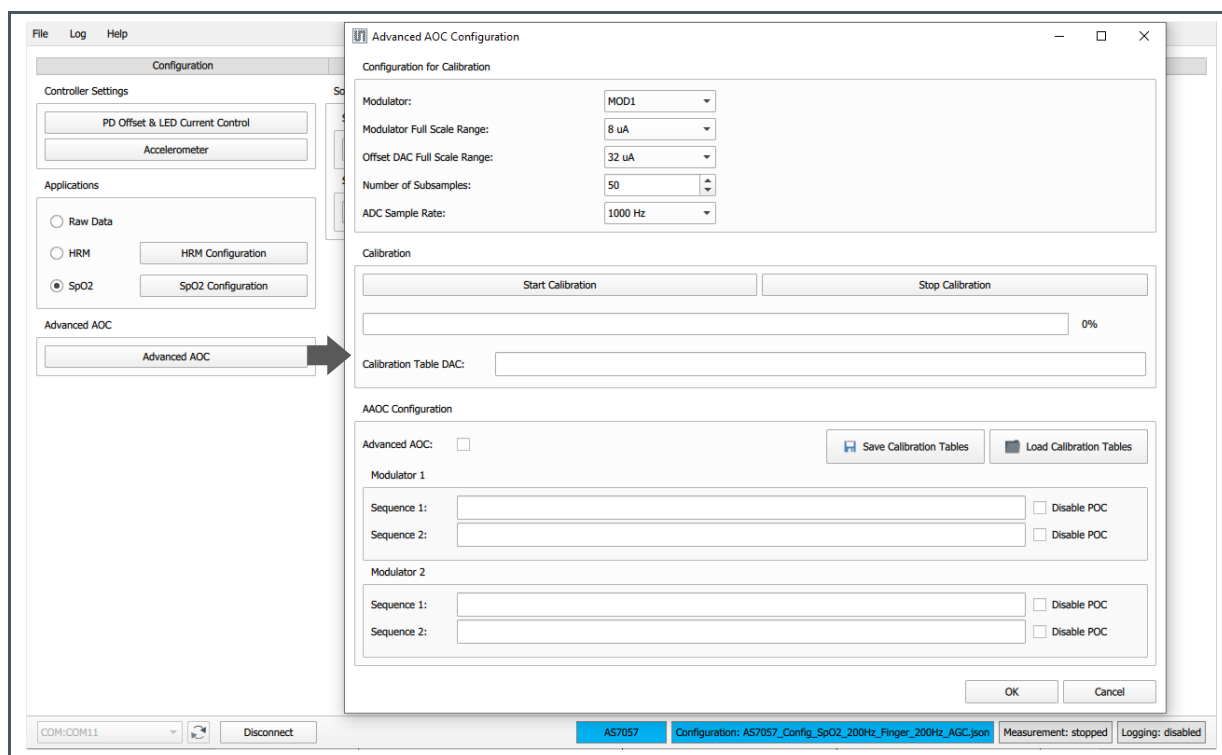
	AOC	AAOC
Saturated signal possible	Yes (fast and high signal changes)	No (except, the technical limits are exceeded)
Selectable per Subsample	Yes	Yes



Information

It is not possible to use the AAOC and AOC for the same subsample.

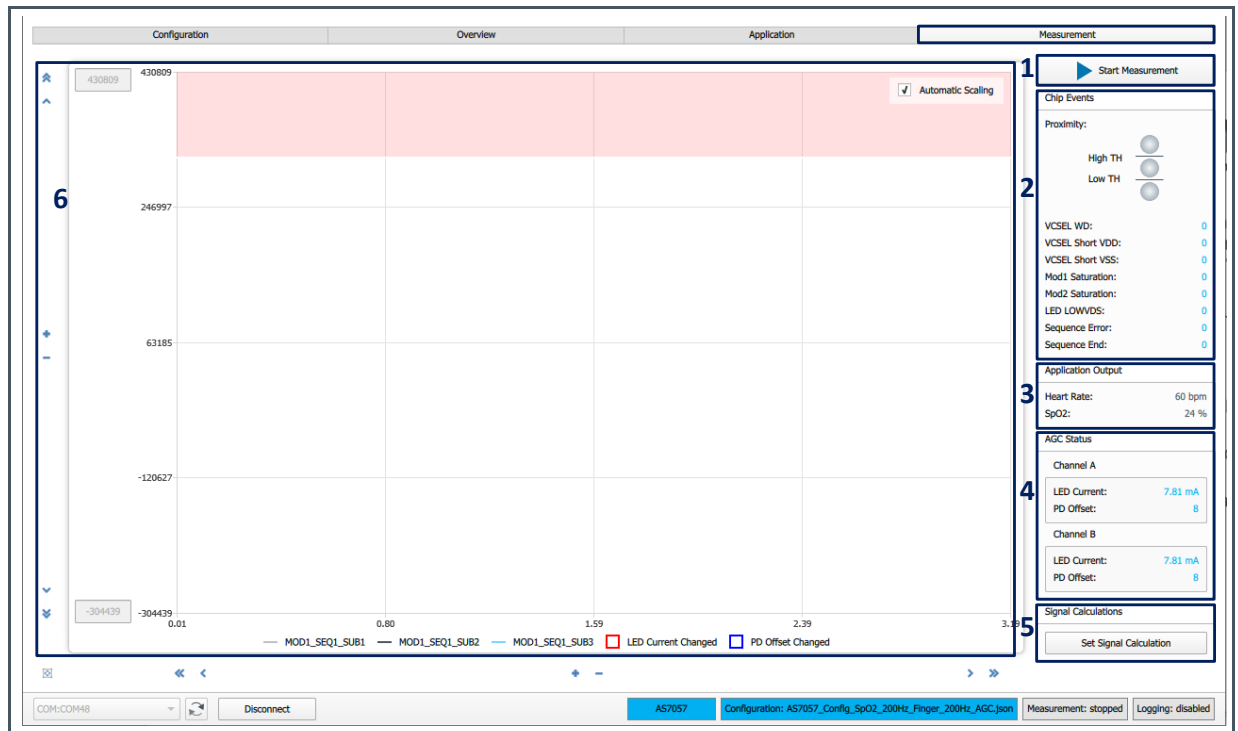
Figure 154:
Advanced AOC



6.4 Measurement Tab

Selecting the Measurement tab (Figure 155) gives access to start and stop the continuous measurement, visualize the chip events, check the application output, and the AGC Status. The PPG ADC count can also be displayed here.

Figure 155:
Measurement Tab Overview

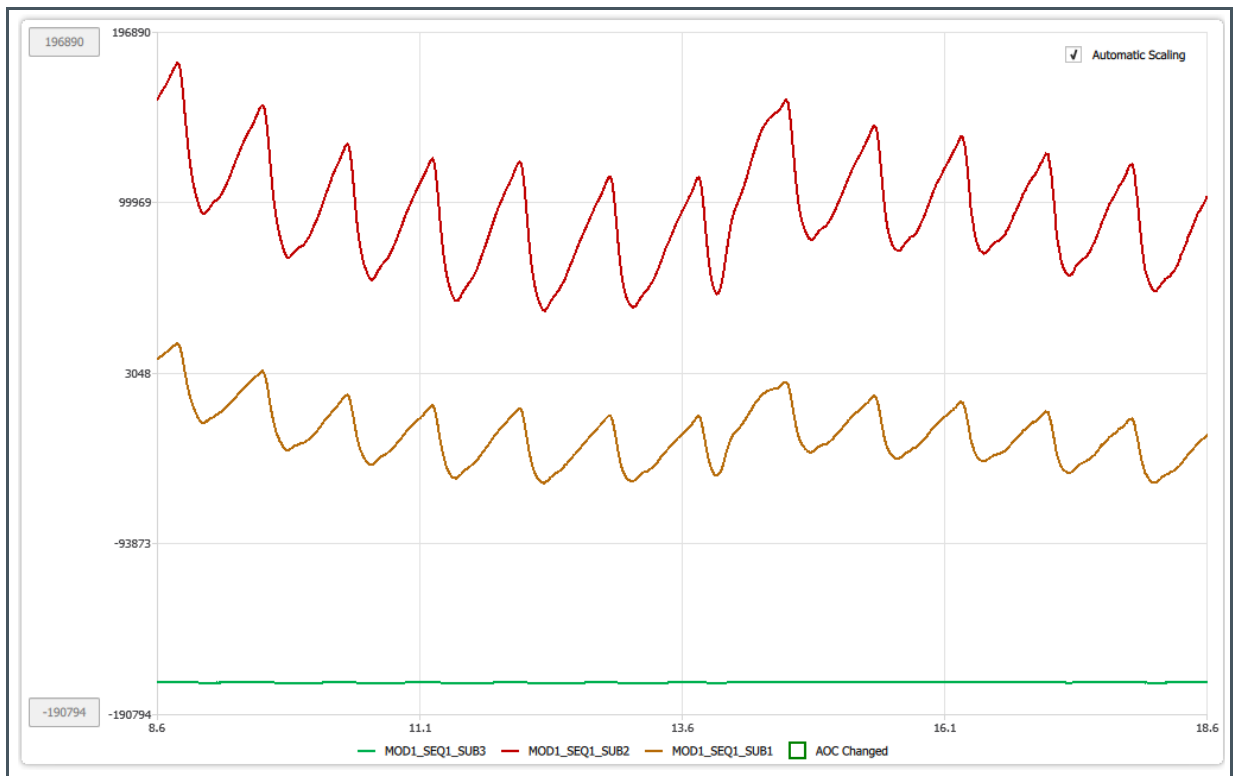


- 1 **Measurement Button:** One button for starting and stopping the measurement
 - “Start Measurement”
 - “Stop Measurement”
- 2 **Chip Event:** Shows the current values’ Proximity, VCSEL/LED, and MOD1 and MOD2. Sequence conditions can also be observed
- 3 **Application Output:** Shows the current Heart Rate & SpO₂ values
- 4 **AGC Status:** The AGC status for the LED current and PD offset. It shows the change in information of the PD Offsets for channels A and B (at min, max, increased, decreased).
- 5 **Signal Calculation:** This allows the user to add a new signal, which is a calculation of two signals to the plot areas.
- 6 **Plot Area:** This displays the signals up to three plots.

Plotting

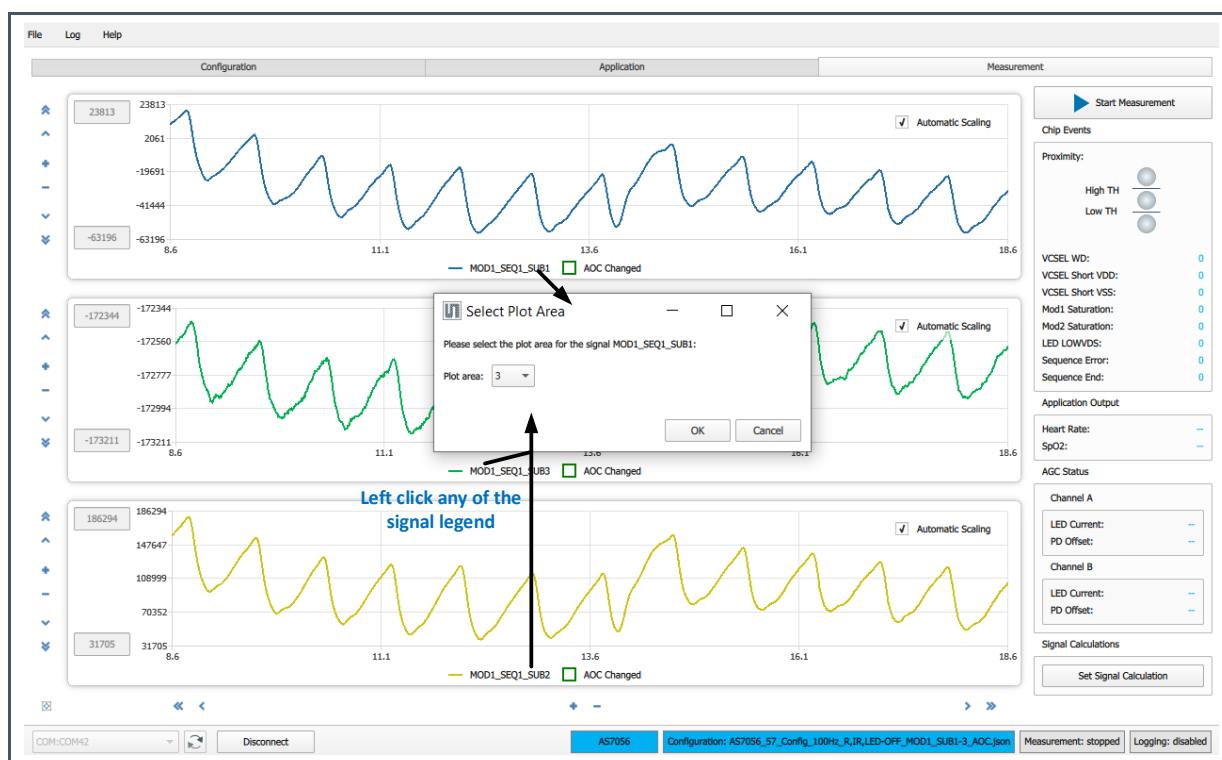
The User Interface provides some functions and signals which are displayed in the plot area.

Figure 156:
Plotting



When “Automatic Scaling” is enabled, the display sets the minimum and maximum of the y-axis automatically. The minimum and maximum of the x-axis are set automatically.

Figure 157:
Selection of Plotting Area

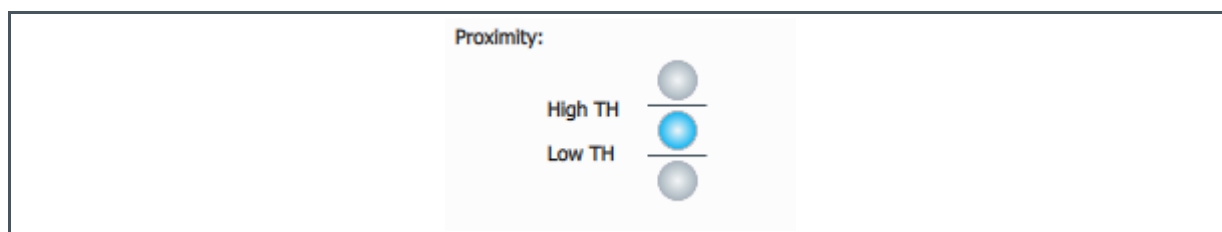


The display area can plot a maximum of three plots for a signal. By default, the plot area displays all the signals into one plot. By left-clicking on the channel name in the legend, a dialog box will open, to represent the signal into another plot area as shown in Figure 157. The plot area contains some functions for automatic scaling, manual scaling, and zooming.

Proximity

The chip event displays the result of the proximity as a traffic light

Figure 158:
Proximity Indication

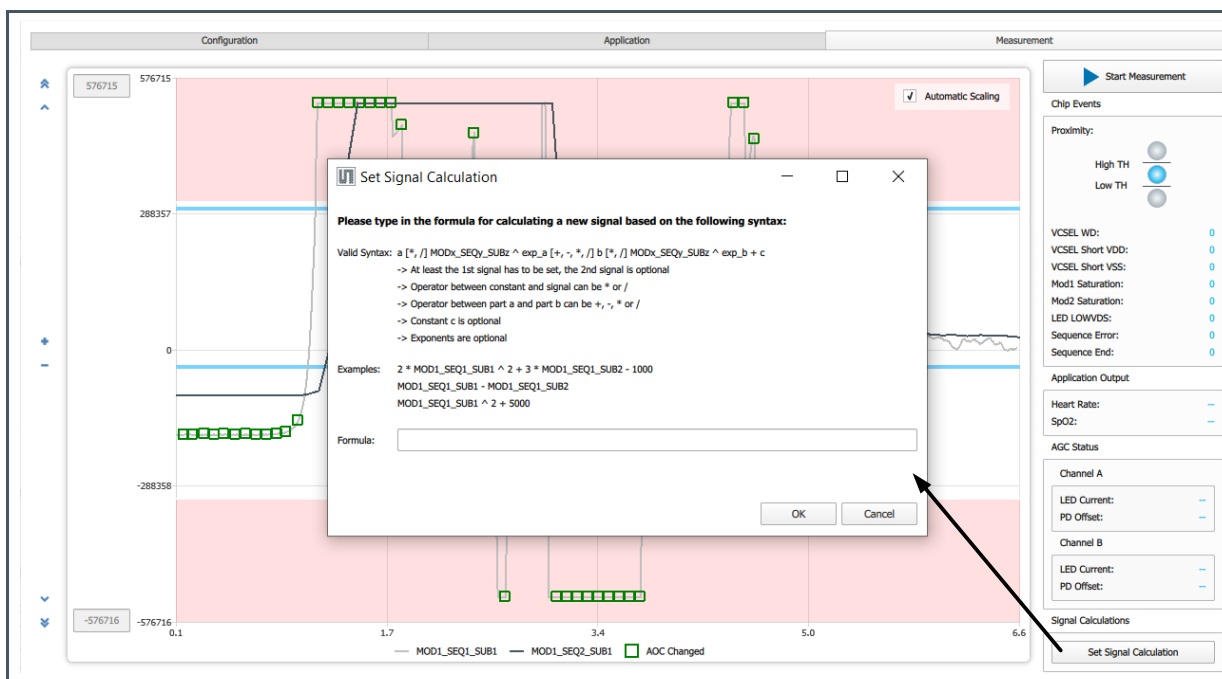


Depending on the proximity state, the blue color will be shown in the bubble (in the middle, above, or below the bubble according to the proximity threshold).

Signal Calculation

It is possible to combine up to two PPG signals with a mathematical formula. The formula can be typed in and activated by the Set Signal Calculation dialog. The new calculated signal will be shown together in the plot area, but they can be shifted to another plot area.

Figure 159:
Signal Calculation Tab

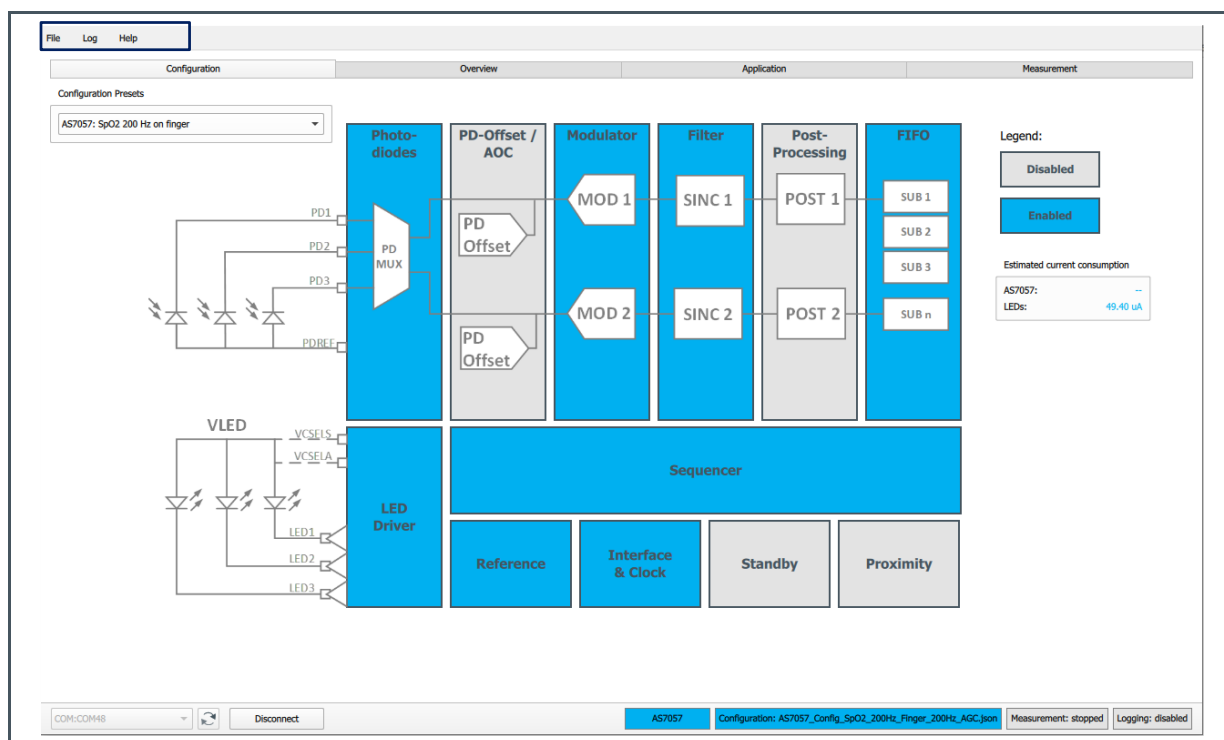


When a new formula is being inserted, the combined series signal will be added to the chart view. The combined signal will be calculated with the upcoming measurement.

6.5 Menu Bar of the User Interface

This section contains options to the register map list, load and save a configuration file, save the log file with the timestamp, and update the firmware.

Figure 160:
Menu Bar

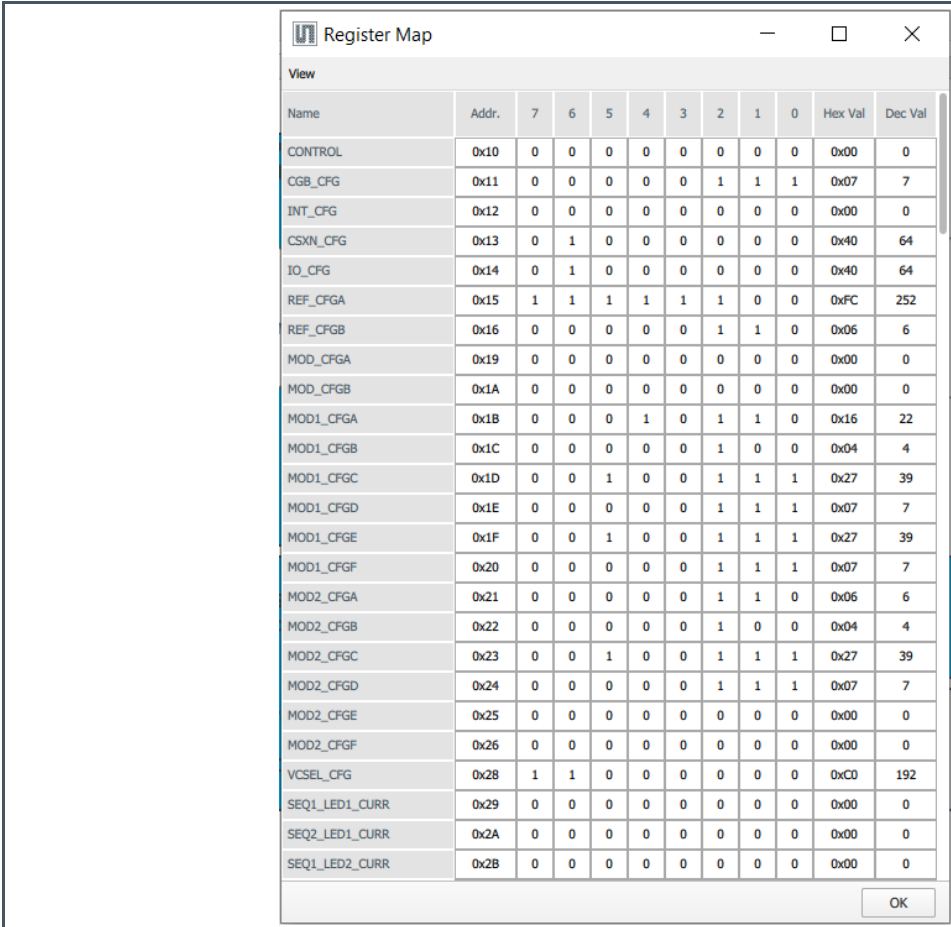


6.5.1 Register Map

The “Register Map” window is used to view/change the contents of the complete set of the AS7056/57 user register. To open it, click on the file, then “→ Register Map” menu.

Changing a register value can be done either by modifying its value in the relevant “Value” field or by toggling a bit by clicking on the relevant bit cell. Changing a value in the register map will not update the current selection in the configuration windows of the GUI. Also, a change in any of the configuration windows will not trigger an automatic update of the already opened register map window. To update the values, click on the refresh button marked under the ‘View’ pane on the “Register Map” in Figure 161 below.

Figure 161:
Register Map List



Register Map											
View											
Name	Addr.	7	6	5	4	3	2	1	0	Hex Val	Dec Val
CONTROL	0x10	0	0	0	0	0	0	0	0	0x00	0
CGB_CFG	0x11	0	0	0	0	0	1	1	1	0x07	7
INT_CFG	0x12	0	0	0	0	0	0	0	0	0x00	0
CSXN_CFG	0x13	0	1	0	0	0	0	0	0	0x40	64
IO_CFG	0x14	0	1	0	0	0	0	0	0	0x40	64
REF_CFGA	0x15	1	1	1	1	1	1	0	0	0xFC	252
REF_CFGB	0x16	0	0	0	0	0	1	1	0	0x06	6
MOD_CFGA	0x19	0	0	0	0	0	0	0	0	0x00	0
MOD_CFGB	0x1A	0	0	0	0	0	0	0	0	0x00	0
MOD1_CFGA	0x1B	0	0	0	1	0	1	1	0	0x16	22
MOD1_CFGB	0x1C	0	0	0	0	0	1	0	0	0x04	4
MOD1_CFGC	0x1D	0	0	1	0	0	1	1	1	0x27	39
MOD1_CFGD	0x1E	0	0	0	0	0	1	1	1	0x07	7
MOD1_CFGE	0x1F	0	0	1	0	0	1	1	1	0x27	39
MOD1_CFGF	0x20	0	0	0	0	0	1	1	1	0x07	7
MOD2_CFGA	0x21	0	0	0	0	0	1	1	0	0x06	6
MOD2_CFGB	0x22	0	0	0	0	0	1	0	0	0x04	4
MOD2_CFGC	0x23	0	0	1	0	0	1	1	1	0x27	39
MOD2_CFGD	0x24	0	0	0	0	0	1	1	1	0x07	7
MOD2_CFGE	0x25	0	0	0	0	0	0	0	0	0x00	0
MOD2_CFGF	0x26	0	0	0	0	0	0	0	0	0x00	0
VCSEL_CFG	0x28	1	1	0	0	0	0	0	0	0xC0	192
SEQ1_LED1_CURR	0x29	0	0	0	0	0	0	0	0	0x00	0
SEQ2_LED1_CURR	0x2A	0	0	0	0	0	0	0	0	0x00	0
SEQ1_LED2_CURR	0x2B	0	0	0	0	0	0	0	0	0x00	0

6.5.2 Saving Current Configuration Settings to a File

The current configuration settings can be exported to a file. To do this, click on the “File → Save Configuration” menu. This will open the “Save Configuration File” dialog box shown in Figure 162. Enter the file name and choose the file location, then click “Save”.

6.5.3 Loading Configuration Settings from a File

To load a previously exported configuration setting, click on the “File → Load Configuration” menu. This will open the “Select Configuration File” dialog box. Select the configuration file from which to load the settings, and click the “Open” button.

The settings imported from the file can be reviewed in the relevant configuration windows.

If the GUI is connected to the board, the newly imported settings will be applied immediately; otherwise, upon successful connection to the board.

Figure 162:
Save and Load Configuration Menu
Entries

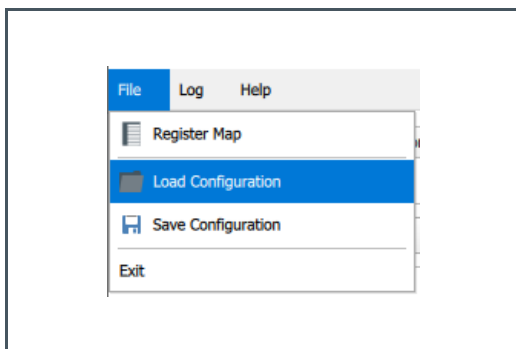
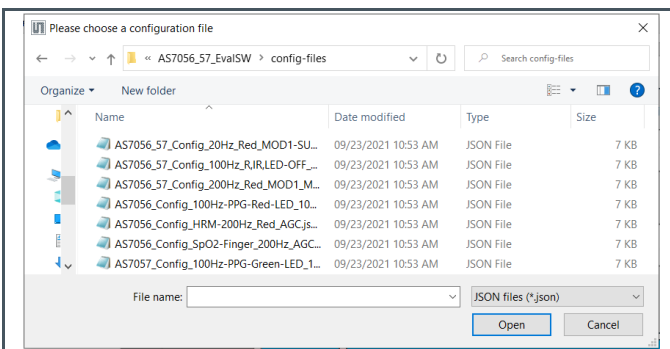


Figure 163:
Save Configuration File Dialog



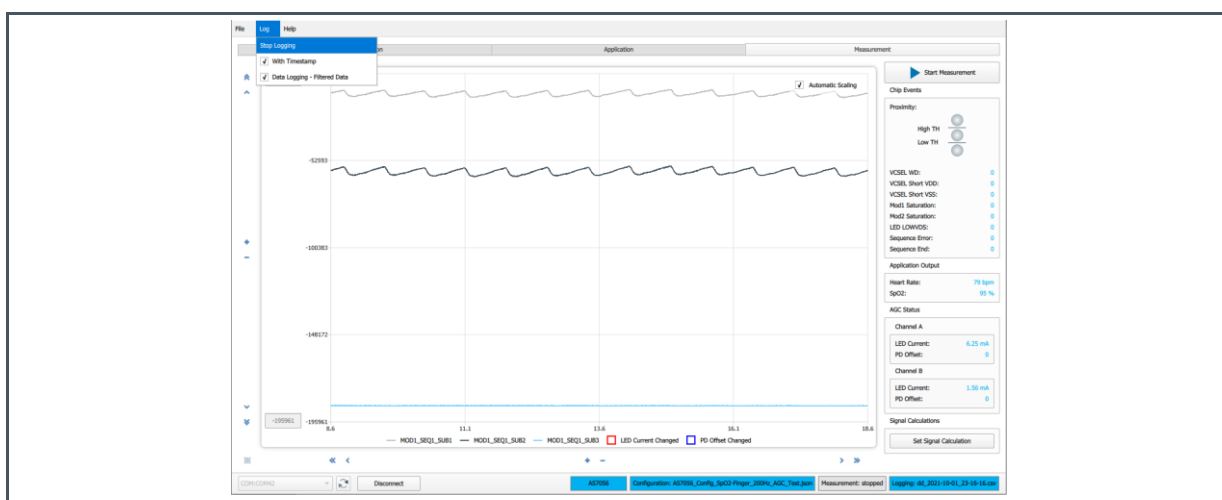
6.5.4 Raw Data Logging and Exporting

Before starting the measurement by clicking the “Start measurement”, click the “Log → Start Logging” to save the raw data from the AS7056/57. A pop-up window will appear to select the file location and save the CSV file. When a measurement is stopped with the “Stop measurement” button, click “Log → Stop Logging” afterward.

With Timestamp: If the timestamp is activated, the timestamp will be added to the selected log file name, when the measurement starts.

Data Logging – Filtered Data: If logging of the filter data is enabled, a separate log file with the name of the selected file and “_filtered” will be created.

Figure 164:
Logging of Raw Data



6.5.5 About Dialog

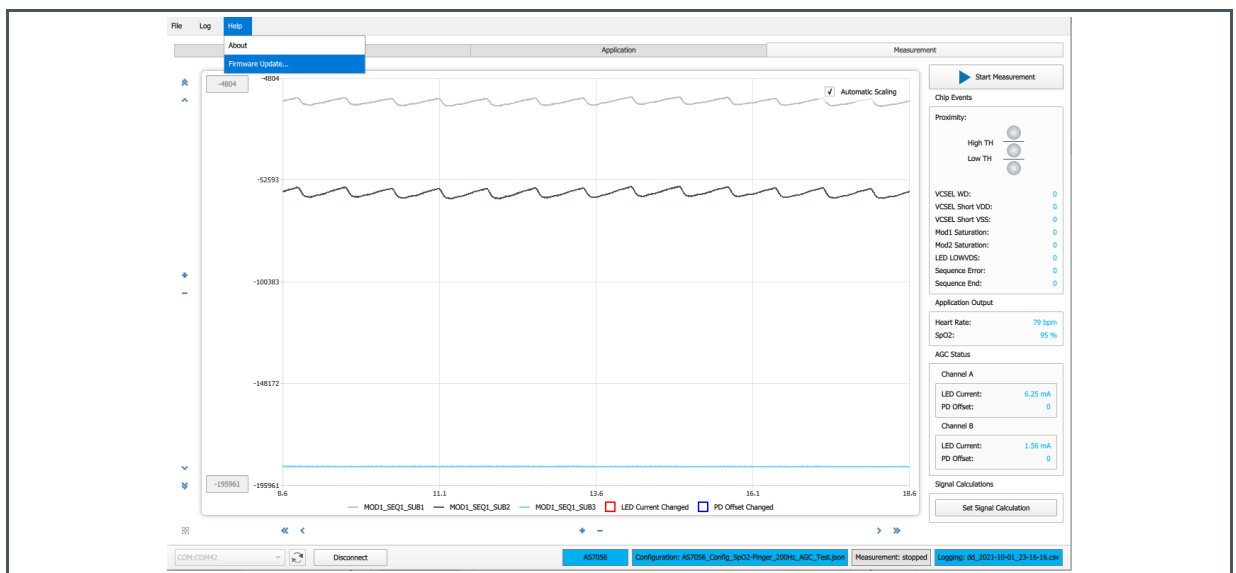
The dialog box is showing software name, software version number, firmware version number, python package version number, and copyright plus company link with technical support link.

6.5.6 Firmware Update

There is an option to upgrade the firmware (FW). To achieve this, perform the following steps:

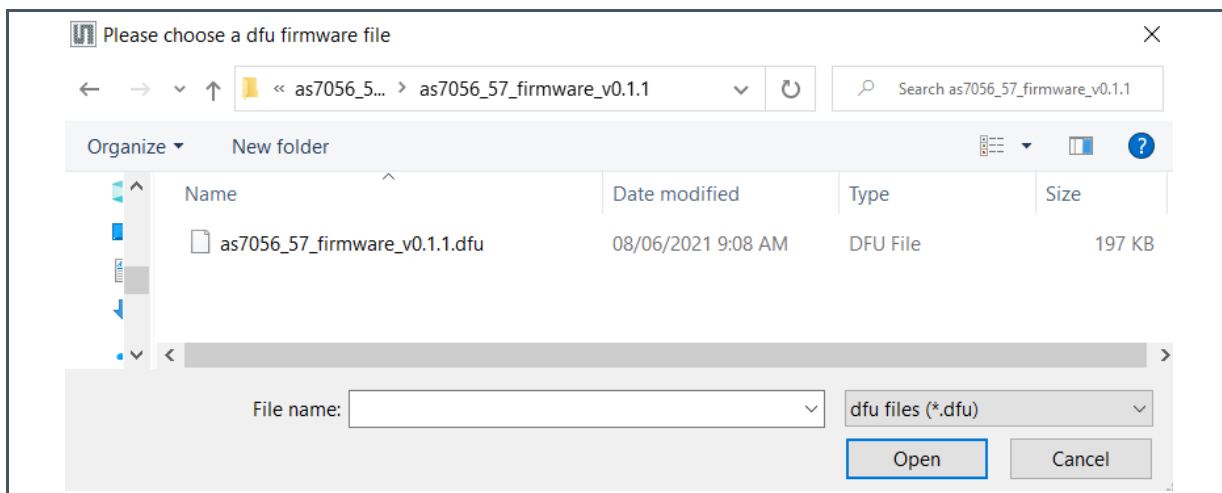
1. Click on the “Help” button and in the drop-down menu click on “Firmware Update...”.

Figure 165:
Firmware Update



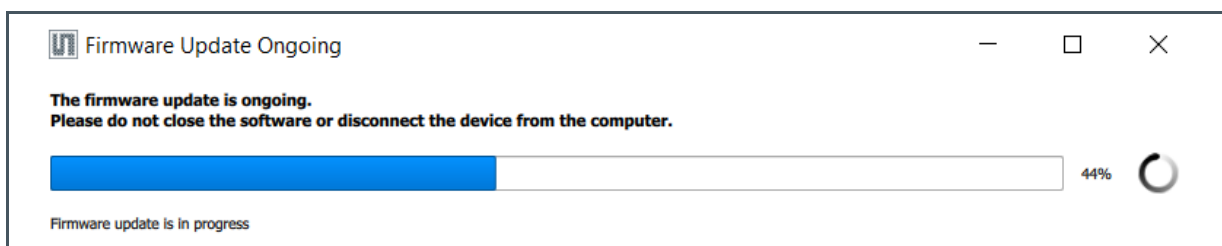
2. In the pop-up window, choose the DFU firmware file used for the AS7056/57 kit.

Figure 166:
Choose the Right DFU File



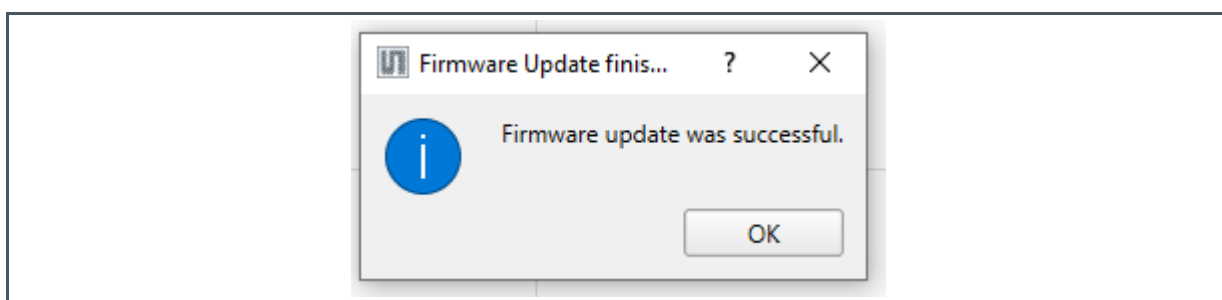
3. Afterward, a pop-up window will open to indicate the update is in progress.

Figure 167:
Firmware Upgrade Stage



4. After the update is complete, a pop-up message will appear in the GUI to indicate the update was successful.

Figure 168:
Successful Firmware Update



7 Revision Information

Changes from previous version to current revision v1-00	Page
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Initial version

- Page and figure numbers for the previous version may differ from page and figure numbers in the current revision.
- Correction of typographical errors is not explicitly mentioned.

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